KGS IAS

72nd BPSC PRELIMS

प्रहार Test Series

Test- 05

Indian Geography 02

Solution

English

Khan Global Studies

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BPSC PRELIMS TEST - 5 (SOLUTION)

Q 1. Ans. B

Explanation:

- **Statement 1 is correct:**
  - ✓ Land heats faster than ocean in summer
  - ✓ Creates thermal low over Indian subcontinent (Thar Desert region)
  - ✓ Ocean remains relatively high pressure
  - ✓ Strong pressure gradient develops (Ocean → Land)
  - ✓ Moist winds move toward India → initiates monsoon circulation
- **Statement 2 is correct:**
  - ✓ Inter-Tropical Convergence Zone shifts northward in summer
  - ✓ Moves up to 20°N–25°N over India
  - ✓ Forms Monsoon Trough (over Indo-Gangetic plains)
  - ✓ Acts as convergence zone attracting moist winds
  - ✓ Controls spatial distribution of rainfall
- **Statement 3 is incorrect:**
  - ✓ Summer condition → Low pressure over land (not high pressure)
  - ✓ High pressure over land occurs in winter season
  - ✓ Winter high pressure → drives dry Northeast Monsoon winds
  - ✓ Hence statement contradicts monsoon mechanism
- **Statement 4 is correct:**
  - ✓ Tibetan Plateau heats intensely in summer
  - ✓ Creates strong convection and upper-air high pressure
  - ✓ Leads to formation of Tropical Easterly Jet (TEJ)
  - ✓ Strengthens monsoon circulation
  - ✓ Supports northward shift of ITCZ

Additional Information

- **Bihar: Monsoon Dependence**
  - ✓ Primary source: Bay of Bengal branch
  - ✓ Direction: Enters from east and southeast
- **Flood Mechanism in Bihar**
  - ✓ Condition: Monsoon trough shifts northward
  - ✓ Impact area: Northern Bihar
  - ✓ Regions: West Champaran, Sitamarhi, Kishanganj
  - ✓ Cause: Orographic rainfall near Himalayas
  - ✓ Rivers affected: Kosi, Gandak, Bagmati
  - ✓ Result: Frequent floods
- **Drought Mechanism in Bihar**
- Condition: Monsoon trough shifts northward
- Effect: Break in monsoon
- Impact: Dry spells, drought-like conditions
- Vulnerable areas: South Bihar
- Regions: Gaya, Aurangabad

Q 2. Ans. D

Explanation:

- **Heat Wave:**
  - ✓ Period: Abnormally high temperatures
  - ✓ Season: March to June (India)
  - ✓ **Heat Wave: Necessary Conditions**
    - ▪ Surface heating: Intense insolation
    - ▪ Atmospheric condition: Stability
    - ▪ Cloud cover: Suppressed
    - ▪ Moisture: Very low
    - ▪ Upper air: Anti-cyclonic circulation present
  - ✓ **Role of Anti-Cyclone**
    - ▪ Mechanism: Subsidence of air
    - ▪ Effect: Adiabatic compression → Warming
    - ▪ Outcome: Heat trapped near surface
    - ▪ Convection: Suppressed
  - ✓ **Absence of Anti-Cyclone: Result**
    - ▪ Effect: Rising hot air → Cooling → Condensation
    - ▪ Outcome: Thunderstorms
    - ▪ Examples:
    - ▪ Nor'westers (Kalbaisakhi): Eastern India
    - ▪ Mango Showers: South India
  - ✓ **Cloud and Moisture Impact**
    - ▪ Clouds: Reflect solar radiation (high albedo)
    - ▪ Moisture: Energy used in evaporation (latent heat)
    - ▪ Result: Reduced temperature rise
  - ✓ **Heat Dome Mechanism**
    - ▪ Cause: Strong anti-cyclonic high pressure
    - ▪ Process: Descending air warms adiabatically

- Effect: Acts as lid → Prevents convection
- Outcome: Persistent heat wave

✓ **Advection of Hot Air**
- Process: Horizontal wind movement
- Wind type: Loo
- Origin: Balochistan and Thar Desert
- Region affected: Indo-Gangetic plains

✓ El Niño Linkage
- Effect: Alters Walker Circulation
- Impact: Delays monsoon onset
- Result: Prolonged dry and hot conditions

✓ IMD Heat Wave Criteria
- **Plains:**
  - ◊ Base temp: ≥ 40°C
  - ◊ Heat wave: +4.5°C to 6.4°C above normal
  - ◊ Severe: > 6.4°C above normal
- **Coastal:**
  - ◊ Base temp: ≥ 37°C
  - ◊ Heat wave: +4.5°C to 6.4°C
  - ◊ Severe: > 6.4°C
- **Hilly:**
  - ◊ Base temp: ≥ 30°C
  - ◊ Heat wave: +4.5°C to 6.4°C
  - ◊ Severe: > 6.4°C

✓ **Bihar: Vulnerability**
- High-risk regions: Gaya, Aurangabad, Rohtas, Kaimur
- **Reason:**
  - ◊ Proximity: Chota Nagpur Plateau
  - ◊ Wind exposure: Loo from Uttar Pradesh

Q 3. Ans. D

Explanation:

- **Precipitation: Classification Basis**

✓ Criteria: Physical state + droplet size + temperature profile

✓ Forms of Precipitation
- Drizzle: Droplet size < 0.5 mm
- Rain: Droplet size 0.5–5 mm
- Sleet: Rain passes through subfreezing layer near surface → Ice pellets
- Hail: Solid ice layers formed in thunderstorms

✓ Hail Formation
- Cloud type: Cumulonimbus
- Process: Strong updrafts → Repeated lifting → Concentric ice layers
- Condition: Violent instability

✓ Sleet Formation
- Condition: Temperature inversion
- Upper layer: Warm
- Near surface: Subfreezing
- Process: Rain freezes into ice pellets

✓ Drizzle vs Rain
- Drizzle: Very fine, floats
- Rain: Larger drops, fall rapidly

✓ Cloud Source Dynamics
- Drizzle: Stratus / Stratocumulus
- Motion: Weak updrafts, stable

✓ Rain: Nimbostratus / Cumulus
- Motion: Moderate updrafts

✓ Hail: Cumulonimbus
- Motion: Violent updrafts (>100 km/h)

✓ Precipitation Formation Processes
- Tropics (India): Collision-Coalescence process
- Mid/high latitudes: Bergeron-Findeisen process

✓ Measurement
- Instrument: Rain Gauge / Ombrometer
- IMD standard: Funnel area 200 cm^2

✓ **Bihar: Pre-Monsoon Hailstorms**
- Season: March–May
- Event: Olavrishti (hailstorms)
- Impact: Crop damage
- Regions: North Bihar
- Crops affected:
- Muzaffarpur Litchi
- Bhagalpur Zardalu Mango

✓ **Bihar: Winter Precipitation**
- Type: Light rain and drizzle
- Season: December–February
- Source: Western Disturbances
- Origin: Mediterranean Sea

Q 4. Ans. B

Explanation:

- **Wind Power in India: Current Leader (2025–26)**

✓ Rank 1: Gujarat

✓ Installed capacity: >14–15 GW

- **Reasons:**

✓ Coastline: ~1600 km

✓ Regions: Kutch

✓ Land: Vast, unobstructed

- ✓ Policy: Strong state RE push (onshore + offshore)
- State Rankings: Wind Power
  - ✓ Rank 1: Gujarat → Kutch Wind Farm
  - ✓ Rank 2: Tamil Nadu → Muppandal (Kanyakumari)
  - ✓ Rank 3: Karnataka → Chitradurga, Belgaum
  - ✓ Rank 4: Maharashtra → Vankusawade (Satara)
  - ✓ Rank 5: Rajasthan → Jaisalmer Wind Park
- Additional Factor
  - ✓ Tamil Nadu:
    - Earlier leader
    - Current: 2nd (>12 GW)
    - Notable: Largest onshore wind farm (Muppandal)
  - ✓ Karnataka:
    - Strong in solar (Pavagada Solar Park)
    - Wind rank: 3rd
    - Focus: Wind-Solar Hybrid
  - ✓ Andhra Pradesh:
    - Wind rank: ~6th
    - Focus: Pumped Storage Projects (PSPs)
- **National Wind Energy Status**
  - ✓ Total capacity: >56 GW
  - ✓ Global rank: 4th largest producer
  - ✓ FY 2025–26 addition: 6.05 GW (highest ever)
- Policy Target (Panchamrit Goals)
    - Total non-fossil capacity target: 500 GW by 2030
    - Wind target: >100 GW
- Geographical Pattern
    - Concentration: Western + Southern states
- Bihar: Wind Energy Status
  - ✓ Installed capacity: ~0
  - ✓ Reason:
    - Landlocked
    - Low wind velocity
    - Indo-Gangetic plains
  - ✓ **Bihar: Alternative Renewable Focus**
    - Agency: BREDA
    - Focus areas:
    - Solar energy
    - Floating solar: Darbhanga, Supaul
    - Bio-energy: Ethanol, compressed biogas

Q 5. Ans. B

Explanation:

- **Statement 1 is incorrect:** Natural Gas → Conventional
  - ✓ A fossil fuel used for decades.
  - ✓ Part of traditional energy sources along with:
    - Coal
    - Petroleum
    - Natural Gas
  - ✓ Therefore not non-conventional.
- **Statement 2 is correct:** Nuclear Energy → Non-Conventional
  - ✓ Considered a modern technological energy source.
  - ✓ Uses uranium/thorium for power generation.
  - ✓ Classified in India under non-conventional energy sources.
- **Statement 3 is correct:** Urban Solid Waste → Non-Conventional
  - ✓ Waste-to-Energy (WtE) is a modern energy technology.
  - ✓ Converts municipal solid waste into electricity.
  - ✓ Falls under non-conventional/renewable energy category.
- **Statement 4 is incorrect:** Firewood & Dung Cakes → Conventional (Correctly classified as conventional)
  - ✓ Traditional rural energy sources used for centuries.
  - ✓ Though biomass-based, they are not modern technologies.
  - ✓ Hence treated as conventional sources.
- **Conventional Sources**
  - ✓ Coal
  - ✓ Petroleum
  - ✓ Natural Gas
  - ✓ Firewood
  - ✓ Dung Cakes
  - ✓ Hydroelectricity (traditional large dams)
- **Non-Conventional Sources**
  - ✓ Solar Energy
  - ✓ Wind Energy
  - ✓ Nuclear Energy
  - ✓ Biogas
  - ✓ Tidal Energy
  - ✓ Geothermal Energy
  - ✓ Waste-to-Energy

Additional information:

- India's First Uranium Mine:
  - ✓ Jaduguda (Jharkhand)
- Example of Waste-to-Energy Plant:

- Timarpur-Okhla Plant (Delhi)
- Policy Link:
  - PM Ujjwala Yojana promotes LPG use to reduce dependence on firewood and dung.

Q 6. Ans. A

Explanation:

- **Statement 1 is correct:**
  - 1G biofuels use food crops grown on arable land.
  - Examples: sugarcane, corn, wheat, soybean, palm oil.
  - Feature: Direct use of food-based biomass.
  - Key issue: Food vs fuel conflict.
- **Statement 2 is incorrect:**
  - Biodiesel is NOT produced by fermentation.
  - Correct process: Transesterification (chemical reaction).
  - Feedstock: Vegetable oils (soybean, palm, jatropha) or animal fats.
  - Fermentation applies to bioethanol, not biodiesel.

Key Facts:

- **Bioethanol:**
  - Produced by fermentation of sugars/starch
  - Sources: sugarcane, corn, wheat
- **Biodiesel:**
  - Produced by transesterification
  - Sources: vegetable oils and animal fats

Q 7. Ans. B

Explanation:

- **Hydrogen "Colors": Concept Basis**
  - Basis: Carbon footprint + production energy source
  - Note: Hydrogen itself is colorless
  - **Green Hydrogen**
    - Source: Water (H_2O)
    - Process: Electrolysis
    - Energy: Renewable (solar, wind)
    - Emission: Zero carbon
    - Nature: Clean hydrogen
  - **Blue Hydrogen**
    - Source: Natural gas (CH_4)
    - Process: Steam Methane Reforming (SMR)
    - Carbon handling: CO_2 captured via CCS
    - Emission: Low / controlled
  - **Grey Hydrogen**
    - Source: Natural gas (CH_4)
    - Process: SMR
    - Carbon handling: No CO_2 capture
    - Emission: High (CO_2 released)
    - Note: Most common global form
  - **Turquoise Hydrogen**
    - Source: Methane (CH_4)
    - Process: Methane pyrolysis
    - Output: Hydrogen + solid carbon
    - Emission: No CO_2 release (solid carbon instead)
  - **Pink Hydrogen**
    - Source: Water (H_2O)
    - Process: Electrolysis
    - Energy: Nuclear power
    - Emission: Zero
    - Note: Clean but non-renewable energy source
- **Additional Information**
  - **National Green Hydrogen Mission (NGHM)**
    - Launch: 2023 (MNRE)
    - Objective: India as global Green Hydrogen hub
  - **Target (2030):**
  - Production capacity: 5 MMT per annum
  - Scheme: SIGHT (financial incentives)

Q 8. Ans. D

Explanation:

- **Statement 1 is correct**
  - Bhilai Steel Plant was set up in Durg district, Chhattisgarh (earlier Madhya Pradesh).
  - Technical + financial assistance: USSR (Soviet Union).
  - It is one of the three major Indo-foreign collaboration steel plants of the 1950s.
- **Statement 2 is correct**
  - Commissioning year: 4 February 1959
  - First President of India: Dr Rajendra Prasad
  - He inaugurated the first blast furnace of Bhilai Steel Plant.
- **Key Facts:**
  - Bhilai Steel Plant:
    - Location: Durg, Chhattisgarh
    - Parent PSU: SAIL (Steel Authority of India Limited)

- Established under: Second Five-Year Plan (1956–61)
- ✓ Other PSU steel plants (common exam trap):
  - Bhilai → USSR
  - Rourkela → West Germany
  - Durgapur → United Kingdom

Q 9. Ans. D

Explanation:

- **Seshachalam Hills: Location & Features**
  - ✓ Location: Southern Andhra Pradesh
  - ✓ Region: Eastern Ghats
  - ✓ Nature: Rugged, dissected ranges
  - ✓ Protected Status
    - Year: 2010
    - Designation: Biosphere Reserve
  - ✓ Drainage System
    - Main river: Pennar
    - Tributaries: Pennar tributaries
    - Effect: Supports local micro-climate
  - ✓ Flagship Species
    - Key species: Red Sanders (Pterocarpus santalinus)
    - Status: Endemic (only natural habitat globally)
    - Threat: Heavy illegal smuggling
    - Other fauna: Slender Loris (nocturnal primate)
- **Red Sanders: Key Facts**
  - ✓ Product: Valued red hardwood
  - ✓ Use: Traditional medicine (East Asia)
  - ✓ Additional use: Shamisen musical instrument
  - ✓ IUCN status: Endangered
  - ✓ CITES: Appendix II
- **Additional Information**
  - ✓ Silent Valley NP
    - Location: Nilgiri Hills, Kerala
    - River: Kunthipuzha (→ Bharathapuzha basin)
    - Flagship species: Lion-tailed Macaque
    - Not associated: Pennar basin / Red Sanders
  - ✓ Eravikulam NP
    - Location: Kerala (Kannan Devan Hills)
    - Ecosystem: Shola-grassland
    - Flagship species: Nilgiri Tahr
    - Special feature: Neelakurinji bloom (12-year cycle)
  - ✓ Nilgiri Biosphere Reserve
    - Location: Tamil Nadu + Kerala + Karnataka
    - Region: Western Ghats
    - Not part of: Eastern Ghats / Pennar basin
- **Bihar: Conservation**
  - ✓ Biosphere reserves: None (UNESCO MAB)
  - ✓ Major protected area:
    - Valmiki National Park & Tiger Reserve (West Champaran)
  - ✓ Aquatic species: Gangetic Dolphin
  - ✓ Location: Vikramshila Gangetic Dolphin Sanctuary
  - ✓ Other species: Gharial
  - ✓ River: Gandak

Q 10. Ans. D

Explanation:

- **Himalayas:**
  - ✓ Type: Young fold mountains
  - ✓ Era: Cenozoic (Tertiary Period)
  - ✓ Cause: Indo-Australian Plate collision with Eurasian Plate
  - ✓ Start of collision: ~50 million years ago (Eocene)
  - ✓ Great Himalayas (Himadri):
    - Epoch: Oligocene (34–23 MYA)
    - Stage: First uplift
  - ✓ Lesser Himalayas (Himachal):
    - Epoch: Miocene (23–5 MYA)
    - Stage: Second uplift
  - ✓ Shiwalik Range:
    - Epoch: Pliocene (5–2.6 MYA)
    - Stage: Third uplift
    - Origin: Riverine sediment accumulation
  - ✓ Tertiary Rock System: Oil & Coal
    - Epochs: Eocene
    - Resources:
      - ◊ Crude oil
      - ◊ Natural gas
      - ◊ Tertiary coal
    - Regions:
      - ◊ Assam (Digboi, Naharkatiya)
      - ◊ Gujarat
- Geological Time Scale (Tertiary Period)
  - ✓ Paleocene (66–56 MYA):
    - Event: Deccan Trap cooling
  - ✓ Eocene (56–34 MYA):

- Event: Assam oil + Tertiary coal formation
- Oligocene (34–23 MYA):
  - Event: Great Himalayas uplift
- Miocene (23–5 MYA):
  - Event: Lesser Himalayas uplift
- Pliocene (5–2.6 MYA):
  - Event: Shiwaliks uplift

- **Correct Himalayan order:**
  - Oligocene → Miocene → Pliocene
- **Additional Information**
  - **Northern Plains Formation**
    - Process: Foredeep formation south of Himalayas
    - Filling period: Pleistocene + Holocene
    - Material: Alluvium (Ganga + tributaries)
    - Bihar Landmass
      - ◊ Type: Alluvial plain
      - ◊ Coverage: >95% area
      - ◊ Feature: Flat terrain
      - ◊ Bedrock: Buried peninsular rocks

Q 11. Ans. B

Explanation:

- **Isostatic Rebound**
  - Event: Melting of Pleistocene ice sheets (~11,700 years ago)
  - Effect: Removal of heavy load
  - Response: Crust rises upward
  - Process: Isostatic rebound (post-glacial rebound)
- **Key Facts**
  - Isostatic Depression:
    - Cause: Addition of mass (sediments/ ice)
    - Effect: Crust sinks
    - Examples:
      - ◊ Ganga-Brahmaputra Delta
      - ◊ Mississippi Delta
  - Emergent Coastlines
    - Cause: Land uplift after rebound
    - Features: Raised beaches, marine terraces
    - Regions: Fennoscandia, Hudson Bay
  - Modern Relevance
    - Cause: Glacier melting (climate change)
    - Effect: Ongoing isostatic rebound
    - Impact: Local seismic activity (glacial earthquakes)
  - **Theories of Isostasy**
    - Airy Model:
      - ◊ Density: Constant
      - ◊ Thickness: Variable
      - ◊ Concept: Deep mountain roots
    - Pratt Model:
      - ◊ Density: Variable
      - ◊ Depth: Constant
      - ◊ Concept: Lighter rocks form higher regions
  - **Bihar: Geological Context**
    - No glacial rebound: Not glaciated region
    - Isostatic Depression:
      - ◊ Cause: Heavy alluvial deposition
      - ◊ Rivers: Ganga, Kosi, Gandak
      - ◊ Effect: Crustal subsidence
    - Tectonic Interaction
      - ◊ Himalayas: Uplift (erosion unloading)
      - ◊ Bihar plains: Subsidence (sediment loading)
    - Seismic Implication
      - ◊ Cause: Stress along faults (Himalayan Frontal Thrust)
      - ◊ Zone: Seismic Zone V
      - ◊ Regions: Sitamarhi, Madhubani, Darbhanga, Supaul

Q 12. Ans. D

Explanation:

- **Statement 1 is correct:**
  - Gutenberg discontinuity: Depth ~2900 km (~1800 miles)
  - Location: Inside Earth
  - Feature: Abrupt change in seismic wave behavior
- **Statement 2 is correct:**
  - P-waves: Velocity decreases at boundary
  - S-waves: Disappear completely
  - Reason:
    - S-waves cannot travel through liquids
  - Inference:
    - Above boundary: Solid mantle
  - Below boundary: Liquid outer core
- **Statement 3 correct:**
  - Outer core: Hotter than mantle (~700°C more)
  - Nature: Liquid, high iron content

- ✓ Density: Higher than mantle
- ✓ Boundary name: Core–Mantle Boundary (CMB)

Q 13. Ans. A

Explanation:

- **Gaddi Community:**
  - ✓ Type: Semi-nomadic, transhumant tribe
  - ✓ Region: Dhauladhar Range (Himachal Pradesh), parts of Jammu & Kashmir
  - ✓ Occupation: Shepherds
  - ✓ Livelihood: Sheep and goat rearing
  - ✓ Migration:
    - Summer: High-altitude pastures
    - Winter: Lower hills
  - ✓ Gaddi Handicrafts & Products
  - ✓ **Gardu:**
    - Type: Blanket
    - Pattern: Black-white square checks
    - Fabric: Patti (45×60 cm)
    - Method: Two pieces joined, handwoven
  - ✓ **Gardi:**
    - Type: Smaller blanket
    - Feature: Lighter than Gardu
    - Pattern: Same as Gardu
    - Use: Children, travel warmth
  - ✓ **Patti:**
    - Type: Fabric
    - Color: Single (off-white/black)
    - Shape: Narrow, long
    - Use:
      - ◊ Chola/Cholu
      - ◊ Suthan (pyjama)
      - ◊ Kurti (shirt)
      - ◊ Topi (cap)
  - ✓ **Dodh:**
    - Type: Blanket
    - Color: Single (white/gray)
    - Size/Weight: Similar to Gardu
    - Decoration: Colored threads
  - ✓ **Thalch:**
    - Type: Rope
    - Material: Goat hair
    - Length: 8–10 m
    - Use: Carrying loads, drying wool
  - ✓ **Khalri:**
    - Type: Bag
    - Material: Sheep/goat skin
    - Process: Skin removed, dried, stitched
    - Feature: Hair removed
  - ✓ **Thobi (Sella):**
    - Type: Carpet
    - Material: Goat hair
    - Texture: Rough
    - Feature: Durable
    - Size: Variable

Q 14. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Majuli is purely a region of fluvial geomorphology.
  - ✓ Formed by river processes: sediment deposition + channel shifting
  - ✓ River system: Brahmaputra
  - ✓ Type: Riverine island (world's largest)
- **Statement 2 is incorrect:**
  - ✓ Southern boundary: Main Brahmaputra channel
  - ✓ Northern boundary: Kherkutia Xuti (anabranch of Brahmaputra)
  - ✓ Subansiri River: Joins Kherkutia Xuti on north
  - ✓ Conclusion: Not bounded entirely by Subansiri

Additional Information:

- **Majuli:**
  - ✓ Location: Assam
  - ✓ Type: Fluvial (riverine) island
  - ✓ Formation: Braided channel dynamics + heavy sediment load
  - ✓ **Administrative status:**
    - District: First island district in India (2016)
  - ✓ **Cultural Facts**
    - Neo-Vaishnavism center: Srimanta Sankardeva
    - Institutions: Sattras
    - Art forms: Sattriya dance, Bhaona, mask-making

Q 15. Ans. B

Explanation:

- **Chitrakote Waterfall location:**
  - ✓ River: Indravati
  - ✓ State: Chhattisgarh
  - ✓ Reason:

- Waterfall formed on Indravati River
    - Type: Widest waterfall in India
    - Shape: Horseshoe
    - Nickname: "Niagara Falls of India"
- **Additional Information**
  - ✓ Mahanadi:
  - ✓ Reason:
    - Major river of Chhattisgarh but not linked to Chitrakote
    - Associated feature: Hirakud Dam
  - ✓ Godavari:
  - ✓ Reason:
    - Parent river
    - Indravati is tributary of Godavari
    - Trap: Basin-level confusion
  - ✓ Narmada:
  - ✓ Reason:
    - Forms Dhuandhar Falls (not Chitrakote)
    - Flows in rift valley

Q 16. Ans. A

Explanation:

- **Rourkela Steel Plant → Germany**
  - ✓ Reason:
    - Established with West German collaboration
  - ✓ Location: Odisha
- **Bhilai Steel Plant → USSR**
  - ✓ Location: Chhattisgarh
- **Durgapur Steel Plant → United Kingdom**
  - ✓ Location: West Bengal
- **Bokaro Steel Plant → USSR**
  - ✓ Location: Jharkhand
  - ✓ Year: 1964 (later addition)

Additional Information

- **Core steel plants:**
  - ✓ Rourkela → West Germany
  - ✓ Bhilai → USSR
  - ✓ Durgapur → UK
  - ✓ **Bokaro → USSR**
- **Bihar Context**
  - ✓ Pre-2000:
  - ✓ Bokaro part of Bihar
  - ✓ Post-2000:
  - ✓ Bokaro in Jharkhand
  - ✓ Loss: Steel + coal + heavy industry
- **Current Role of Bihar**
  - ✓ Type: Consumption hub
  - ✓ Supply: SAIL network
  - ✓ Cities: Patna, Muzaffarpur
  - ✓ Use: Infrastructure (bridges, metro)

Q 17. Ans. A

Explanation:

- Karur → Tamil Nadu
  - ✓ Reason: Major textile and handloom hub
- Ilkal → Karnataka
  - ✓ Reason: Ilkal saree
  - ✓ Feature: Tope Teni technique, Kasuti embroidery, red silk pallu
- Pochampally → Telangana
  - ✓ Reason: Ikat weaving
  - ✓ Feature: Tie-dye before weaving, geometric patterns
- Chanderi → Madhya Pradesh
  - ✓ Reason: Fine, sheer fabric
  - ✓ Feature: Silk-cotton blend, gold zari
  - ✓ Use: Sarees, kurtas

Additional Information

- **GI Protection:**
  - ✓ Ilkal Saree
  - ✓ Pochampally Ikat
  - ✓ Chanderi Fabric
  - ✓ Law: GI Act, 1999
- **Pochampally (Current Affairs):**
  - ✓ Recognition: UNWTO Best Tourism Village
- **Bihar Linkages**
  - ✓ Bhagalpur:
  - ✓ Title: Silk City
  - ✓ Product: Tussar (non-mulberry silk)
- Nalanda (Baswanbigha):
  - ✓ Product: Bawan Buti saree
  - ✓ Feature: 52 motifs
- Institutional Support:
  - ✓ Upendra Maharathi Shilp Anusandhan Sansthan, Patna

Q 18. Ans. D

Explanation:

- **Chennai: "Detroit of India"**
  - ✓ Reason: Largest automobile manufacturing hub
  - ✓ Key Features
    - Industry share: ~30% of India's automobile production

- Export hub: Major contributor to auto exports
- Industrial zones: Sriperumbudur, Oragadam
- ✓ Port Advantage
- ✓ Ports:
  - Chennai Port
  - Kamarajar Port
  - Benefit: Easy export of vehicles
- **Other Important City**
  - ✓ Surat: Diamond City
    - Fact: >90% of world diamond cutting and polishing
  - ✓ Mysuru: Sandalwood City
    - Fact: Sandalwood oil, soap production
  - ✓ Nashik: Wine Capital of India
    - Fact: ~50% of India's wineries
  - ✓ Bhubaneswar: Temple City of India
    - Fact: ~7000 temples historically
  - ✓ Kollam: Cashew Capital of the World
    - Fact: Major global cashew processing hub

Q 19. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Nuclear Power Corporation of India Limited (NPCIL): Public Sector Enterprise
  - ✓ Administrative control: Department of Atomic Energy
  - ✓ Role: Nuclear power generation in India
- **Statement 2 is correct:**
  - ✓ Nuclear Power Corporation of India Limited: Holds equity participation in
  - ✓ Bharatiya Nabhikiya Vidyut Nigam Limited (BHAVINI)
  - ✓ Nature: Partial ownership (financial support)

Key Facts

- **BHAVINI:**
  - ✓ Type: PSU under Department of Atomic Energy
  - ✓ Role: Fast Breeder Reactors
- **Nuclear Programme (India):**
  - ✓ **Stage 1:**
    - Handled by NPCIL
    - Reactor type: PHWR (Uranium-based)
  - ✓ **Stage 2:**
    - Handled by BHAVINI
    - Reactor: Prototype Fast Breeder Reactor (Kalpakkam)
  - ✓ **Stage 3:**
    - Handled by Bhabha Atomic Research Centre
    - Focus: Thorium-based reactors

Q 20. Ans. D

Explanation:

- **Uranium: Discovery**
  - ✓ Martin Klaproth: 1789
  - ✓ Country: Germany
  - ✓ Mineral: Pitchblende
  - ✓ Named after: Uranus
- **Uranium: Nature**
  - ✓ Heavy metal
  - ✓ High energy density
  - ✓ Used in nuclear energy: >60 years
- **Occurrence**
  - ✓ Average crustal concentration: 2–4 ppm
  - ✓ Abundance: Similar to tin, tungsten, molybdenum
  - ✓ Also present in: Seawater
- **Major Uranium Deposits (India)**
  - ✓ Singhbhum Shear Zone: Jharkhand
  - ✓ Cuddapah Basin: Andhra Pradesh, Telangana
  - ✓ Mahadek Basin: Meghalaya
  - ✓ Delhi Supergroup: Rajasthan
  - ✓ Bhima Basin: Karnataka
- **Resource Distribution (% U_3O_8)**
  - ✓ Andhra Pradesh: ~49%
  - ✓ Jharkhand: ~26%
  - ✓ Meghalaya: ~9%
  - ✓ Others: Remaining share
- **Key Facts**
  - ✓ Largest reserves: Cuddapah Basin
  - ✓ Oldest mining belt: Singhbhum
- **Deposit types:**
  - ✓ Vein type: Singhbhum, Bhima
  - ✓ Sandstone type: Mahadek
  - ✓ Strata-bound: Cuddapah

Q 21. Ans. B

Explanation:

- **Coal India Limited (CIL)**
  - ✓ HQ: Kolkata
  - ✓ Reason: Legacy of Damodar Valley coalfields + colonial trade center

- ✓ Status: Maharatna
- ✓ Ministry: Coal

- **GAIL (India) Limited**
  - ✓ HQ: New Delhi
  - ✓ Reason: Centralized gas pipeline administration
  - ✓ Status: Maharatna
  - ✓ Ministry: Petroleum and Natural Gas
- **Rashtriya Ispat Nigam Limited (RINL)**
  - ✓ HQ: Visakhapatnam
  - ✓ Plant: Visakhapatnam Steel Plant
  - ✓ Feature: First shore-based steel plant in India
  - ✓ Advantage: Import coal + export steel via port
  - ✓ Status: Navratna
  - ✓ Ministry: Steel
- **Hindustan Aeronautics Limited (HAL)**
  - ✓ HQ: Bengaluru
  - ✓ Established: 1940
  - ✓ Reason: Inland location for defence manufacturing
  - ✓ Status: Maharatna
  - ✓ Ministry: Defence

Additional Information

- Bihar Context
  - ✓ No Maharatna/Navratna HQ
  - ✓ Major PSU presence: NTPC plants
  - ✓ Locations: Barh, Kahalgaon, Nabinagar
- Hindustan Urvarak & Rasayan Limited (HURL)
  - ✓ Type: Joint venture PSU
  - ✓ Promoters: CIL, NTPC, IOCL
  - ✓ Plant: Barauni Fertiliser revived

NTPC (National Thermal Power Corporation) → New Delhi
ONGC (Oil and Natural Gas Corporation) → Dehradun
IOCL (Indian Oil Corporation Ltd.) → New Delhi
BPCL (Bharat Petroleum Corporation Ltd.) → Mumbai
HPCL (Hindustan Petroleum Corporation Ltd.) → Mumbai
SAIL (Steel Authority of India Ltd.) → New Delhi
BHEL (Bharat Heavy Electricals Ltd.) → New Delhi
NALCO (National Aluminium Company Ltd.) → Bhubaneswar
CIL (Coal India Ltd.) → Kolkata
GAIL (Gas Authority of India Ltd.) → New Delhi
RINL (Rashtriya Ispat Nigam Ltd.) → Visakhapatnam
HAL (Hindustan Aeronautics Ltd.) → Bengaluru

Q 22. Ans. C

Explanation:

- **Water Act, 1974**
  - ✓ CPCB established
  - ✓ Objective: Control water pollution
  - ✓ Focus: Streams and wells cleanliness
- **Air Act, 1981**
  - ✓ CPCB mandate expanded
  - ✓ Added: Air pollution control
- **Environment Protection Act, 1986**
  - ✓ Umbrella environmental law
  - ✓ CPCB: Technical advisory role
- **Forest Conservation Act, 1980**
  - ✓ Focus: Forest land diversion
  - ✓ No role in CPCB formation
  - ✓ Key Supporting Facts
- **CPCB**
  - ✓ Type: Statutory body
  - ✓ Year: 1974
  - ✓ Mandate evolution
  - ✓ 1974: Water pollution
  - ✓ 1981: Air pollution added
  - ✓ 1986: Advisory role strengthened
- **Pollutants:**
  - ✓ PM10, PM2.5, NO_2, SO_2, CO, O_3, NH_3, Pb
- **Bihar Context**
  - ✓ State body: Bihar State Pollution Control Board
  - ✓ Formed under: Water Act, 1974
- **High AQI cause:**
  - ✓ Indo-Gangetic trough
  - ✓ Cold air inversion
  - ✓ Stubble smoke + local emissions
  - ✓ Ganga Monitoring
  - ✓ Parameters: BOD, Fecal Coliform
  - ✓ Stretch: Buxar to Bhagalpur

Q 23. Ans. D

Explanation:

- **Statement 1 is correct:**
  - ✓ National Mineral Development Corporation (NMDC) established: 1958
  - ✓ Status: Navratna PSU
  - ✓ Administrative ministry: Ministry of Steel
- **Statement 2 is correct:**
  - ✓ NMDC: Largest iron ore producer in India
  - ✓ Production: Tens millions of tonnes annually

- **Statement 3 is correct:**
  - ✓ Majhgawan mine: Panna, Madhya Pradesh
  - ✓ Operator: NMDC
  - ✓ Status: Only fully mechanised diamond mine in India
- **Key Facts**
  - ✓ Major mining regions
    - Bailadila: Chhattisgarh
    - Donimalai: Karnataka
  - ✓ Bailadila ore
    - Type: High-grade hematite
    - Iron content: >65%
    - Export: Japan, South Korea
  - ✓ Corporate update
    - NMDC Steel Limited (NSL): Separate entity
    - Plant: Nagarnar, Chhattisgarh
  - ✓ PSU–Mineral Mapping
    - NMDC: Iron ore, diamond
    - SAIL: Steel production
    - NALCO: Aluminium
    - HCL: Copper
    - UCIL: Uranium
- **Bihar Context**
  - ✓ No major iron ore reserves
  - ✓ Lost Singhbhum belt after 2000
  - ✓ Key minerals
    - Limestone: Rohtas, Kaimur
    - Pyrite: Amjhore
    - Sand: Riverine
  - ✓ Exploration
    - Magnetite traces: Jamui, Rohtas

Q 24. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Gulf of Mannar Biosphere Reserve
  - ✓ Area: ~10,500 km^2
  - ✓ Established: 1989
  - ✓ UNESCO MAB: 2001
  - ✓ Islands: 21
- **Statement 2 is correct:**
  - ✓ Location: Laccadive Sea, Indian Ocean
  - ✓ Between: Southeast coast of India and northwest coast of Sri Lanka
  - ✓ Includes: 21 islands
  - ✓ Status: First Marine Biosphere Reserve in South and Southeast Asia
  - ✓ Includes: Marine National Park
  - ✓ Feature: High marine biodiversity
- **Key Facts**
  - ✓ Boundaries
    - India side: Rameswaram
    - Sri Lanka side: Mannar Island
    - Adam's Bridge (Ram Setu): Shallow chain
  - ✓ Rivers draining
    - Tambraparni: India
    - Malvathu Oya: Sri Lanka
  - ✓ Ecosystem
    - Coral reefs
    - Seagrass beds
    - Mangroves
  - ✓ Flagship species
    - Dugong
  - ✓ UNESCO timeline
    - First India BR: Nilgiri Biosphere Reserve (2000)
    - Next: Gulf of Mannar, Sundarbans Biosphere Reserve (2001)
- **Bihar Context**
  - ✓ No Biosphere Reserve
  - ✓ Aquatic flagship: Gangetic Dolphin
  - ✓ Sanctuary: Vikramshila (Bhagalpur)

Q 25. Ans. C

Explanation:

- According to the National Wildlife Action Plan (2002–2016), issued by the Ministry of Environment, Forest and Climate Change, areas within 10 km of national parks and wildlife sanctuaries are recommended to be designated as Eco-Sensitive Zones (ESZs) or eco-fragile zones.
- **The following table compiles the list of activities under the ESZ Notification that are permitted, regulated or prohibited.**

Prohibited Activities	Regulated Activities/ Business Activities	Permitted Activities
Commercial Mining	Felling Of Trees	Ongoing agriculture and horticulture practices by local communities

Setting Saw Mill	Establishment of Hotels and Resorts	Rain Water Harvesting
Industries causing water, air, soil, noise pollution	Erection of electrical cables	Organic Farming
Commercial use of firewood		Use of renewable energy and adoption of green technologies for all activities
Establishment of major hydroelectric projects	Drastic Change of Agricultural Systems	
Use of Hazardous Substances for any activity	Commercial use of natural water resources, including groundwater harvesting	
Undertaking tourism Activities like over-flying the National Park Area, hot air balloons etc.	Signboards and Hoardings, widening of roads	
	Movement of Vehicular traffic at night	
	Introduction of Exotic Species	
	Pollution caused due vehicles, discharge of effluents and solid waste in the environment (water bodies, terrestrial areas, air etc.)	

Q 26. Ans. B

Explanation:

- **Sundarbans National Park**
  - ✓ Type: Mangrove forest ecosystem
  - ✓ Formation: Ganga + Brahmaputra + Meghna delta
  - ✓ Location: Bay of Bengal
  - ✓ Status: Largest continuous mangrove forest in world
  - ✓ **UNESCO Status**
    - Indian part: Sundarbans National Park
    - World Heritage Site: 1987
    - Bangladesh part: 1997
  - ✓ **Biodiversity**
    - Flagship species: Royal Bengal Tiger
    - Unique: Only mangrove habitat with significant tiger population
  - ✓ **Ramsar Status**
    - Year: 2019
    - Area: ~4230 sq km
    - India's largest Ramsar site
  - ✓ **Mangrove Adaptations**
    - Pneumatophores: Aerial roots for oxygen
    - Vivipary: Seeds germinate on parent plant

Additional Information

- **Manas National Park**
  - ✓ Type: Terrestrial Himalayan foothill
  - ✓ UNESCO: 1985
  - ✓ River: Manas
- **Dibru-Saikhowa Biosphere Reserve**
  - ✓ Type: Inland riverine floodplain
  - ✓ Rivers: Brahmaputra, Lohit
  - ✓ Famous: Feral horses
- **Simlipal Biosphere Reserve**
  - ✓ Type: Inland deciduous forest
  - ✓ Famous: Melanistic tigers
- **Transboundary Linkage**
  - ✓ **Sundarbans → Bangladesh counterpart**
- **Parallel (Bihar)**
  - ✓ Valmiki Tiger Reserve → Connected with
  - ✓ Chitwan National Park
  - ✓ Hydrological Link
  - ✓ Ganga through Bihar → sediment to Sundarbans
  - ✓ **Aquatic Fauna Contrast**
    - Sundarbans: Saltwater crocodile
    - Bihar rivers: Gharial

Q 27. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Sarhuli Mandar: Sacred groves linked to the Sarhul festival
  - ✓ Local term: Sarna Sthal
  - ✓ Dominant tree: Sal
- **Statement 2 is correct:**
  - ✓ Sacred groves: Sarnas
  - ✓ Region: Chotanagpur plateau (Bihar, Jharkhand)

Additional Information

- **Origin**
  - ✓ Approx period: ~600 BCE
  - ✓ Associated tribe: Munda
- **Village structure**
  - ✓ Each village: One Sarna (sacred grove)
  - ✓ Priest titles: Pahan, Deuri, Naik, Kelo
- **Cultural feature**
  - ✓ Dhumkuria: Bachelors' dormitory
- **Religious significance**
  - ✓ Nature worship
  - ✓ Sal tree: Sacred focus
  - ✓ Sarhul festival: Spring celebration linked to forests

Q 28. Ans. D

Explanation:

- **Archean rocks are found in the Karnataka Plateau (Dharwar craton).**
  - ✓ The Karnataka Plateau hosts the famous Dharwar craton, one of the oldest geological formations in India.
  - ✓ It consists of highly metamorphosed gneisses and schists, forming part of the Archean basement complex.
- **Archean rocks are present in Tamil Nadu.**
  - ✓ Large parts of Tamil Nadu expose ancient crystalline rocks (gneissic complex), belonging to the Archean system.
  - ✓ These rocks form the stable southern shield region of India.
- **Archean rocks occur in Madhya Pradesh and Chhattisgarh.**
  - ✓ Despite being famous for younger formations like the Vindhyans, regions such as Bundelkhand and Balaghat in Madhya Pradesh and adjoining Chhattisgarh expose Archean basement rocks.
  - ✓ These form the underlying shield beneath later sedimentary layers.
- **Archean rocks are found in the Chotanagpur Plateau region.**
  - ✓ The Chotanagpur Plateau (mainly in Jharkhand and nearby areas) is a major exposure zone of Archean rocks.
  - ✓ It is rich in mineral resources due to the associated Dharwar formations.

The 5 Major Archean Cratons of India

- Dharwar Craton: Karnataka Plateau
- Bastar Craton: Chhattisgarh and parts of MP/Odisha.
- Singhbhum Craton: Chotanagpur Plateau (Jharkhand/Odisha).
- Bundelkhand Craton: MP and southern UP.
- Aravalli Craton: Rajasthan.

Q 29. Ans. C

Explanation:

- **Assertion (A) is correct:**
  - ✓ Mohorovičić Discontinuity (Moho):
    - ▪ Boundary between Earth's crust and mantle.
    - ▪ Identified by sudden increase in seismic wave velocity.
  - ✓ Reason:
    - ▪ Mantle rocks are denser than crustal rocks.
    - ▪ P-waves and S-waves travel faster in materials that are denser and more rigid/incompressible.
- **Reason (R) is incorrect:**
  - ✓ Moho does not occur at uniform depth globally.
  - ✓ Depth varies significantly depending on crust type:
    - ▪ Oceanic crust: ~ 5–10 km deep
    - ▪ Continental crust: ~ 35–40 km deep
    - ▪ Mountain regions (e.g., Himalayas): up to 70 km or more

Additional information:

Major Internal Discontinuities (Top → Bottom)

- Conrad Discontinuity
  - ✓ Between upper crust (SIAL) and lower crust (SIMA)
- Mohorovičić (Moho)
  - ✓ Between crust and mantle
- Repetti Discontinuity
  - ✓ Between upper mantle and lower mantle
- Gutenberg Discontinuity
  - ✓ Between mantle and outer core
  - ✓ S-waves disappear → outer core is liquid
- Lehmann Discontinuity
  - ✓ Between outer core and inner core
  - ✓ Inner core is solid

Q 30. Ans. A

Explanation:

- **Statement 1 is correct:**

- ✓ Bhils: Largest Scheduled Tribe in India (Census 2011)
- ✓ Population: ~17 million
- ✓ Share: ~38% of total ST population

- **Statement 2 is incorrect:**
  - ✓ Largest Bhil population: Madhya Pradesh
  - ✓ High concentration: Jhabua, Alirajpur
  - ✓ Chhattisgarh: Dominated by Gond tribe, not Bhils

Additional Information:

- Distribution: Gujarat, Rajasthan, Maharashtra, Madhya Pradesh
- Etymology: "Billu" (bow)
- Known for: Archery, guerrilla warfare
- Art: Pithora painting
- Festival: Baneshwar Fair (Dungarpur, Rajasthan)
- Historical link: Allied with Maharana Pratap at Battle of Haldighati

Q 31. Ans. C

Explanation:

- **Bhotia: High-altitude pastoralists**
  - ✓ Region: Uttarakhand, Himachal Pradesh, Sikkim
  - ✓ Practice: Transhumance (Bugyals)
- **Shompen: Endemic to Great Nicobar Island**
  - ✓ Habitat: Tropical evergreen rainforest
  - ✓ Type: Semi-nomadic, isolated
- **Toda: Located in Nilgiri Hills**
  - ✓ Economy: Pastoral (buffalo-based)
  - ✓ Known for: Dogles huts, Toda embroidery (GI tag)
- **Tharu: Terai belt (marshy plains)**
  - ✓ States: Uttar Pradesh, Bihar, Uttarakhand
  - ✓ Economy: Settled agriculture

Additional Information:

- PVTGs: Shompen, Toda
- Tharu (Bihar): West Champaran (Terai)
- Bhotia vs Tharu: Mountain pastoral vs plain agriculture
- Andaman–Nicobar distinction:
- Shompen: Nicobar
- Jarawa/Sentinelese: Andaman

Q 32. Ans. D

Explanation:

Vembanad Lake: Longest lake in India

- Type: Brackish water lagoon
- Location: Kerala backwaters
- Separated from sea: Barrier island and sand spits
- Parallel to: Arabian Sea (Laccadive Sea)

Additional Information

- **Ashtamudi Lake:**
  - ✓ Type: Estuarine lake
  - ✓ Feature: Eight-branched (palm-shaped)
  - ✓ Not defined by barrier island
- **Sasthamkotta Lake:**
  - ✓ Type: Freshwater lake
  - ✓ Location: Inland Kerala
  - ✓ No connection to sea
- **Vellayani Lake:**
  - ✓ Type: Freshwater lake
  - ✓ Location: Thiruvananthapuram (inland)
  - ✓ No coastal features
- **Key Facts:**
  - ✓ Vembanad-Kol Wetland: Ramsar Site
  - ✓ 2nd largest wetland in India
  - ✓ Rivers draining: Pamba, Achankovil, Meenachil
  - ✓ Event: Nehru Trophy Boat Race (Punnamada)
  - ✓ Contains: Willingdon Island (Kochi Port)
- **Related Coastal Lakes:**
  - ✓ Chilika: Odisha: Largest brackish lagoon in Asia
  - ✓ Pulicat: AP–TN: Barrier island (Sriharikota)
  - ✓ Ashtamudi: Kerala: Estuarine
- **Bihar Link:**
  - ✓ Kanwar Lake: Largest freshwater oxbow lake in Asia
  - ✓ Type: Fluvial origin (Burhi Gandak)
  - ✓ Status: Ramsar Site

Q 33. Ans. C

Explanation:

- **Sambhar Lake: Largest inland salt lake in India**
  - ✓ Location: Arid Rajasthan
- **Pangong Tso: High-altitude endorheic lake**
  - ✓ Region: Trans-Himalayan Ladakh
- **Lonar Lake: Meteorite crater lake**
  - ✓ Formed in Deccan basalt
- **Nalsarovar Lake: Seasonal wetland and bird sanctuary**
  - ✓ Location: Lowland plains of Gujarat

Additional Information:

- **Geomorphology:**

- ✓ Sambhar: Aeolian depression, highly saline
- ✓ Pangong Tso: Tectonic/glacial, brackish, freezes in winter
- ✓ Lonar: Meteorite impact, saline + alkaline
- ✓ Nalsarovar: Relict sea, freshwater to brackish
- **Ramsar Sites:**
  - ✓ Sambhar, Lonar, Nalsarovar: Ramsar wetlands
- **Other lake:**
  - ✓ Wular Lake: Largest freshwater lake in India
- **Special facts:**
  - ✓ Pangong Tso: Divided by LAC (India–China)
  - ✓ Lonar: Pink water due to Haloarchaea microbes

Bihar context:

- Lakes: Fluvial origin
- Kanwar Lake: Largest freshwater oxbow lake in Asia, Ramsar Site
- Other wetlands: Kusheshwar Asthan, Goga Bil

Q 34. Ans. C

Explanation:

- **Footloose Industry:**
  - ✓ Industry independent of raw material, market, and location constraints
  - ✓ Characteristics:
    - No dependence on bulky raw materials
    - Transport cost: negligible
    - Products: high-value, low-weight
    - Labour: small, highly skilled
    - Pollution: minimal
    - Location factor: connectivity (internet, power, transport)
  - ✓ Examples of Footloose Industries:
    - Information Technology (IT)
    - Electronics assembly
    - Diamond cutting
  - ✓ Key Concept:
    - Footloose: "can locate anywhere"
    - Priority: infrastructure over resources
- **Bihar Context:**
  - ✓ Heavy industry disadvantage: far from the Chota Nagpur Plateau
  - ✓ Advantage: suitable for footloose industries
  - ✓ **Focus sectors:**
    - IT and ITeS hubs: Patna
    - Textiles and garments: labour-based
    - Ethanol and light manufacturing
  - ✓ **Policy Link:**
    - Industrial promotion: IT parks, textile policy
    - Driver: skilled youth + low dependence on raw materials

Q 35. Ans. C

Explanation:

- **Atomic Energy Commission:**
  - ✓ Established: 1948
  - ✓ Role: Policy-making body for the nuclear programme
- **Atomic Energy Establishment Trombay:**
  - ✓ Established: 3 January 1954
  - ✓ Purpose: Nuclear R&D hub
- **Bhabha Atomic Research Centre:**
  - ✓ Renamed from AEET: 22 January 1967
  - ✓ Reason: In memory of Homi Jehangir Bhabha

Key Facts:

- **Administrative control: Department of Atomic Energy (DAE)**
  - ✓ Major reactors at BARC:
  - ✓ Apsara: 1956, first in Asia
  - ✓ CIRUS: 1960, Canada collaboration
  - ✓ Dhruva: 1985, indigenous
- **Programme link:**
  - ✓ Three-stage nuclear programme: Thorium utilisation
- **Important Distinction:**
  - ✓ Establishment: 1954
  - ✓ Renaming: 1967

Q 36. Ans. A

Explanation:

- **Rajasthan Atomic Power Station:**
  - ✓ Location: Rajasthan
  - ✓ Near: Rana Pratap Sagar Dam
- **Kaiga Generating Station:**
  - ✓ Location: Karnataka
  - ✓ River: Kali River
- **Kakrapar Atomic Power Station:**
  - ✓ Location: Gujarat
  - ✓ River: Tapi River
- **Narora Atomic Power Station:**
  - ✓ Location: Uttar Pradesh

- ✓ River: Ganga

Key Facts:

- Operator: Nuclear Power Corporation of India Limited
- Reactor type:
  - ✓ Most plants: PHWR (indigenous)
- Exceptions:
  - ✓ Kudankulam Nuclear Power Plant: VVER (Russia)
  - ✓ Tarapur Atomic Power Station: BWR (first plant, 1969)
- Special facts:
  - ✓ Narora: Only plant in Gangetic alluvial plain

Bihar Context:

- No operational nuclear plant
- Power supply: Eastern Regional Grid
- Sources: Narora, Kakrapar, Rawatbhata
- Constraint: Cooling water availability

Q 37. Ans. A

Explanation:

- **BCC Belt: Ballari–Chitradurga–Chikkamagaluru belt**
  - ✓ Major mineral: Iron ore (hematite + magnetite)
  - ✓ **Includes:**
    - ▪ Bababudan Hills: First discovery of iron ore
    - ▪ Kudremukh: Large magnetite deposits

Additional Information

- **Bauxite:**
  - ✓ Present in Karnataka
- **Manganese:**
  - ✓ Occurs with iron
  - ✓ Main deposits: Sandur (Ballari), Shivamogga
- **Copper:**
  - ✓ Minor deposits in Chitradurga (Ingaldhal)
  - ✓ Major belts: Jharkhand, MP, Rajasthan

Key Facts:

- Type: High-grade iron ore belt
- Use: Steel industry
- **Industrial linkage:**
  - ✓ Feeds plants like JSW Steel (Vijayanagar)
  - ✓ Export: Ports of Mangaluru, Goa

Bihar Context:

- Iron ore loss: After 2000 bifurcation (Jharkhand)
- New findings:
  - ✓ Jamui: Magnetite potential
- Dependence:
  - ✓ Imports iron ore from Jharkhand, Odisha
  - ✓ Steel units: Bihta, Patna, Begusarai

Q 38. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Atmosphere largely transparent to short-wave insolation
  - ✓ Lower troposphere not heated directly by solar radiation
  - ✓ Heating mechanism: Earth's surface → conduction, convection, long-wave radiation
  - ✓ Pre-monsoon thermal low: due to intense surface heating over NW India
- **Statement 2 is correct:**
  - ✓ Active monsoon: thick cloud cover present
  - ✓ Clouds absorb and re-radiate long-wave terrestrial radiation
  - ✓ Effect: heat trapped at night
  - ✓ Result: higher minimum (night) temperatures over Central India

Key Facts:

- **Indian Context:**
  - ✓ Thermal low region: Northwest India
- **Monsoon system:**
  - ✓ Monsoon Trough controls active/break phases
- **Bihar Link:**
  - ✓ Pre-monsoon: Loo winds (surface heating)
  - ✓ Monsoon: Cloud cover → warm, humid nights
  - ✓ Climate type: Continental (high temperature range)

Q 39. Ans. A

Explanation:

- **Aravalli Range: Alignment SW → NE**
  - ✓ Arabian Sea monsoon: Same direction
  - ✓ Result: Winds flow parallel, no obstruction
  - ✓ No uplift → no condensation → low rainfall
- **Key Facts:**
  - ✓ **Monsoon branches:**
    - ▪ Arabian Sea branch
    - ▪ Bay of Bengal branch
  - ✓ **Orography impact:**

- Western Ghats: Perpendicular → heavy rain
- Khasi Hills: Funnel effect → extreme rain
- Aravalli Range: Parallel → negligible rain

✓ **Additional factor:**
- Mid-tropospheric subsidence → dry conditions in Thar

- **Bihar Contrast:**
  - ✓ Rain source: Bay of Bengal branch
  - ✓ Barrier: Himalayas (perpendicular)
- **Rainfall pattern:**
  - ✓ North Bihar > South Bihar
  - ✓ East Bihar > West Bihar
  - ✓ High rainfall zone: Kishanganj

Q 40. Ans. B

Explanation:

- **Statement 1 is correct:**
  - ✓ Specific heat capacity of water > land: ~5 times
  - ✓ Arabian Sea heats slowly → moderates temperature
  - ✓ Mumbai: maritime influence → no extreme pre-monsoon heat
- **Statement 2 is incorrect:**
  - ✓ Nagpur: inland continental location
  - ✓ High diurnal range: hot days + rapid night cooling
  - ✓ Mumbai: low diurnal range due to humidity + cloud cover

Key Facts:

- Continentality: Distance from sea ↑ → temperature extremes ↑
- Diurnal range:
  - ✓ Coastal: Low
  - ✓ Inland: High
- **Humidity effect:**
  - ✓ Water vapour traps long-wave radiation → warm nights
- **Land–Sea Breeze:**
  - ✓ Day: Sea → Land (cooling)
  - ✓ Night: Land → Sea

Bihar Link:

- Gaya: extreme continental climate
- High diurnal + annual range
- East Bihar (e.g., Kishanganj): slightly moderated climate
- West/South Bihar: more extreme temperatures

Q 41. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Bihar lies entirely north of the Tropic of Cancer (≈ 24°20'N to 27°31'N)
  - ✓ Latitude: Subtropical zone
  - ✓ Climate type: Warm temperate/subtropical continental
  - ✓ Result:
    - High annual temperature range
    - Very hot summers
    - Cold winters
    - Strong continentality due to inland position
- **Statement 2 is incorrect:**
  - ✓ During Winter Solstice (~21 December)
  - ✓ Sun is over Tropic of Capricorn
  - ✓ Northern Hemisphere receives low-angle sunlight
  - ✓ Latitude effect in winter:
    - Higher latitude → shorter day length
    - Lower latitude → longer day length
  - ✓ Bihar (higher latitude in India)
  - ✓ has shorter daylight than
  - ✓ Kerala (lower latitude, closer to the equator)
- **Statement 3 is correct:**
  - ✓ Tropical wet evergreen forests require:
    - Temperature: High year-round
    - Rainfall: > 200 cm annually
    - No long dry season
  - ✓ Subtropical climate of Gangetic Plains:
    - Strong winter season
    - Prolonged dry period before the monsoon
- **Result:**
  - ✓ Evergreen forests cannot sustain
  - ✓ Deciduous forests dominate (leaf shedding during the dry season)

Key Facts:

- **Latitude zones in India:**
  - ✓ South of Tropic of Cancer: Tropical climate (equable)
  - ✓ North of Tropic of Cancer: Subtropical climate (seasonal extremes)
- **Forest control factors:**
  - ✓ Evergreen: constant heat + rainfall
  - ✓ Deciduous: seasonal dryness

Bihar Link:

- **Bihar climate:**
  - ✓ Subtropical monsoon type
  - ✓ Strong seasonal contrast
  - ✓ Supports moist + dry deciduous forests
- **Vegetation pattern:**
  - ✓ North Bihar: higher rainfall → moist deciduous (Sal dominance)
  - ✓ South Bihar: lower rainfall → dry deciduous (Mahua, Khair, Amaltas)

Q 42. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Santali is the most widely spoken Munda language (Austro-Asiatic family)
  - ✓ Script: Ol Chiki
  - ✓ Creator: Pandit Raghunath Murmu (1925)
  - ✓ Purpose: Promote Santali literature and cultural identity
- **Statement 2 is correct:**
  - ✓ Ol Chiki script
  - ✓ Developed in 1925
  - ✓ By Pandit Raghunath Murmu
  - ✓ Used specifically for the Santali language
  - ✓ Important for the linguistic identity of the Santhal community
- **Statement 3 is incorrect:**
  - ✓ **Santhal religion is NOT monotheistic in strict sense**
  - ✓ **Correct structure:**
    - ▪ Supreme deity: Thakur Ji / Marang Buru
    - ▪ Core belief system: Animism + nature worship
    - ▪ Spirits worshipped: Bongas (forest, hill, household spirits)
    - ▪ Sacred place: Jaherthan (sacred grove)
    - ▪ Practices: Rituals + sacrifices + ancestral spirit worship
    - ▪ Witchcraft belief: Traditionally present
    - ▪ Not rejection of spirits
    - ▪ Not pure monotheism
    - ▪ Functionally animistic + polytheistic system
- **Key Facts:**
  - ✓ Santhal religion features:
    - ▪ Nature-based worship system
    - ▪ Ancestor spirit worship
    - ▪ Sacred grove centered rituals
  - ✓ **Constitutional recognition:**
    - ▪ Santali included in Eighth Schedule via 92nd Amendment Act, 2003
- **Bihar Link:**
  - ✓ Bihar Santhal distribution:
    - ▪ Concentrated in eastern & south-eastern districts
    - ▪ Key districts: Purnea, Katihar, Saharsa, Banka, Jamui
    - ▪ Linguistic overlap: Maithili + Surjapuri + Santali transition zone
  - ✓ Historical context:
    - ▪ Santhal Pargana (now in Jharkhand) originally part of unified Bihar
    - ▪ Linked to Santhal Hool (1855) anti-colonial uprising

Q 43. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ **Western Disturbances:**
    - ▪ Extratropical storms originating near the Mediterranean region
    - ▪ Travel eastward via the westerly jet stream
    - ▪ Bring winter rainfall (Dec–Feb) to North India
    - ▪ Crucial for Rabi crops
    - ▪ Major benefit: Wheat germination and early growth
- **Statement 2 is incorrect:**
  - ✓ Western Disturbances occur in winter (Dec–Feb), not monsoon season
  - ✓ Kharif crops (rice, cotton, maize) depend on Southwest Monsoon (June–September), not winter rain
  - ✓ Correct classification:
    - ▪ Rabi season: Western Disturbances
    - ▪ Kharif season: Southwest Monsoon
  - ✓ **No direct link between Western Disturbances and Kharif sowing**

Key Facts:

- **Crop season linkage:**
  - ✓ Rabi: Winter sowing (Oct–Dec), harvest (Apr–Jun)
  - ✓ Kharif: Monsoon sowing (Jun–Jul), harvest (Sep–Oct)
  - ✓ Zaid: Summer season crops (Mar–May)

- **Western Disturbance origin:**
  - ✓ Mediterranean region
  - ✓ Moves via westerly jet stream

Bihar Link:

- Bihar:
  - ✓ Receives weakened Western Disturbances after west-to-east moisture loss
  - ✓ Benefits: Light winter rain for wheat, mustard, maize
- **Impacts:**
  - ✓ Cold waves (Shitlahar) after passage
  - ✓ Dense fog in winter (Jan) affecting transport

Q 44. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Bhadla Solar Park:
    - Location: Jodhpur district, Rajasthan
    - Terrain: Arid desert, high solar insolation
    - Capacity: ~2245 MW (2.25 GW)
    - Status: Largest operational solar park in India and among the largest in the world
    - Advantage: Minimal cloud cover + flat sandy land
- **Statement 2 is incorrect:**
  - ✓ Kaimur Plateau has solar potential but:
    - High population density nearby
    - Fertile Gangetic plains nearby
    - Land acquisition constraints
    - Bihar solar development focuses on:
    - Floating solar plants
    - Small/medium ground projects
- **Statement 3 is correct:**
  - ✓ Pavagada Solar Park:
    - Capacity: ~2050 MW
    - Location: Tumakuru district, Karnataka
    - Known as: Shakti Sthala
    - Model: Land lease system
    - **Key feature:**
      - ◊ ~13,000 acres leased from farmers
      - ◊ Farmers retain ownership
      - ◊ Fixed annual income + inflation-linked returns
      - ◊ Avoids large-scale land acquisition conflict
- **Key Facts:**
  - ✓ Solar park classification:
    - Ultra Mega Solar Park: ≥ 500 MW
  - ✓ Major parks:
    - Bhadla: largest (Rajasthan)
    - Pavagada: farmer lease model (Karnataka)
    - Rewa: supplies Delhi Metro (Madhya Pradesh)
    - Kurnool: early large-scale park (Andhra Pradesh)
- **Bihar Link:**
  - ✓ **Bihar:**
    - **Limited feasibility for large solar parks due to:**
    - **Dense population**
    - **Fertile alluvial plains**
  - ✓ **Energy strategy:**
    - Floating solar on water bodies
    - Small ground-mounted projects
  - ✓ **Emerging sites:**
    - Darbhanga (floating solar)
    - Supaul (floating solar)
    - Kajra (Lakhisarai) and Pirpainti (Bhagalpur): repurposed thermal plant land for solar + battery storage

Q 45. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Agra and Darjeeling lie at nearly the same latitude (~27°N)
  - ✓ But climate differs due to altitude:
    - Agra: ~170 m (Gangetic plains)
    - Darjeeling: ~2000+ m (Himalayas)
  - ✓ Key result:
    - Higher altitude → lower temperature
    - Darjeeling is much colder than Agra due to normal lapse rate
- **Statement 2 is incorrect:**
  - ✓ Atmosphere is heated from bottom → terrestrial radiation from Earth's surface
  - ✓ Not by direct absorption of incoming short-wave solar radiation
  - ✓ Why Darjeeling is colder:
    - Thin air at high altitude
    - Low water vapor, CO_2, dust
    - Less trapping of long-wave radiation
    - Rapid heat loss to space
  - ✓ So:
    - High altitude does NOT mean more heat absorption

- It means lower heat retention
- **Key Supporting Facts:**
  - ✓ **Atmospheric heating:**
    - Incoming solar radiation passes through atmosphere
    - Earth absorbs it and re-radiates long-wave heat
    - Lower atmosphere traps this heat
  - ✓ **Lapse rate:**
    - Temperature decreases with altitude in troposphere
- **Bihar Link:**
  - ✓ **Bihar:**
    - Entirely low-lying Gangetic plain state
    - No major altitude influence on climate
    - **Contrast:**
      - ◊ Plains: temperature driven by continentality + latitude
      - ◊ Hill regions (older unified Bihar/ Jharkhand belt): altitude-driven cooling
  - ✓ **Example (historical Bihar geography):**
    - Ranchi / Netarhat region:
      - ◊ Cooler climate due to elevation
      - ◊ Strong lapse-rate effect (absent in modern Bihar plains)

Q 46. Ans. B

Explanation:

Statement 1 is correct:

- **Karewa Formation:**
  - ✓ Meaning: "Elevated table-land"
  - ✓ Composition: Pleistocene (Quaternary) deposits
    - Sand
    - Silt
    - Clay
    - Loam
    - Glacial boulders
  - ✓ Origin: Fluvio-lacustrine (river + lake) environment
  - ✓ Setting: Formed in the structural basin of the Jhelum system

Statement 2 is incorrect:

- Highly porous and unconsolidated sediments
- Rapid percolation of rainwater into subsurface
- Surface remains dry
- No permanent rivers originate on Karewa tops
- Water presence in Kashmir:
  - ✓ Rivers like Jhelum River cut through Karewas in deep valleys
  - ✓ But Karewa plateaus themselves remain dry uplands
- **So:**
  - ✓ Misconception: "Kashmir = water-rich everywhere"
  - ✓ Reality: Karewa surfaces are dry terrace systems

Key Facts:

- **Formation mechanism:**
  - ✓ Tectonic uplift of Pir Panjal blocked drainage
  - ✓ Formation of ancient lake (Pleistocene)
  - ✓ Sediment deposition over time
  - ✓ Later drainage by Jhelum River incision
- **Geology type:**
  - ✓ Lacustrine + glacial sedimentary terraces

Bihar Link:

- **Bihar:**
  - ✓ Composed of Quaternary alluvial plains (Ganga system)
  - ✓ Formation: continuous river deposition (not lake-based terraces like Karewas)
  - ✓ **Contrast:**
    - Kashmir Karewas: lacustrine + glacial terraces (dry uplands)
    - Bihar plains: fluvial alluvium (river-deposited fertile plains)
- **Unique crop:**
  - ✓ Karewas: saffron (dry, cold, well-drained soils)
  - ✓ Bihar wetlands: Makhana (waterlogged chaurs and maans)

Q 47. Ans. B

Explanation:

- **Statement 1 is correct:**
  - ✓ 2G biofuels: Use non-food feedstock
  - ✓ Feedstock: Agricultural residue, forestry waste, bagasse, municipal waste
  - ✓ Objective: Avoid food vs fuel conflict
- **Statement 2 is incorrect:**
  - ✓ 2G ethanol: Requires breakdown of lignocellulosic biomass
  - ✓ Cellulose + hemicellulose bound with lignin
  - ✓ Process: Pre-treatment + enzymatic hydrolysis → simple sugars → fermentation

- **Key Facts:**
  - ✓ **1G biofuels:**
    - Feedstock: Sugarcane, corn, broken rice
    - Process: Direct fermentation of sugars/starch
  - ✓ **2G biofuels:**
    - Feedstock: Stubble, wood waste
    - Limitation: Complex structure →costly processing
  - ✓ **3G biofuels:**
    - Feedstock: Algae
    - Feature: High yield, no arable land
  - ✓ **4G biofuels:**
    - Feedstock: Genetically modified organisms
    - Feature: High carbon capture
- **Policy framework:**
  - ✓ National Policy on Biofuels: Target E20 by 2025-26
  - ✓ Pradhan Mantri JI-VAN Yojana: Funding for 2G ethanol
- **Bihar facts:**
  - ✓ Ethanol Production Promotion Policy: First state policy
  - ✓ Plant: Purnea: First grain-based ethanol plant (2022)
  - ✓ Type: 1G (broken rice, maize)
- **Potential:**
  - ✓ North Bihar: High agri-residue
  - ✓ Regions: Kosi, Seemanchal
  - ✓ Use: 2G ethanol production → stubble management

Q 48. Ans. D

Explanation:

- **Assertion (A) is incorrect:**
  - ✓ ITCZ: Not safe for aviation
  - ✓ Feature: Severe turbulence, icing, lightning, microbursts
  - ✓ Cause: Intense vertical convection
- **Reason (R) is correct:**
  - ✓ ITCZ: Convergence of trade winds
  - ✓ Inter-Tropical Convergence Zone: NE + SE trade winds meet
  - ✓ Effect: Strong uplift of warm moist air
  - ✓ Clouds: Towering cumulonimbus (Cb)
  - ✓ Vertical extent: Can penetrate tropopause
- **Why (R) does not explain (A):**
  - ✓ R explains convection mechanism
  - ✓ A is opposite of reality (ITCZ is hazardous, not safe)
- **Key Facts:**
  - ✓ Doldrums: Calm surface winds but intense vertical instability
  - ✓ Tropopause height:
    - Equator: ~18 km
    - Poles: ~8 km
  - ✓ Cb cloud height: 50,000–60,000 ft
  - ✓ Aircraft cruise: ~35,000 ft
- **ITCZ nature:**
  - ✓ Thermal equator: Migrates seasonally
- **India link:**
  - ✓ ITCZ shift → Monsoon Trough
- **Bihar impact:**
  - ✓ North shift: Heavy rainfall in Himalayan foothills
  - ✓ Flood-prone regions: Kosi, Seemanchal, Champaran

Q 49. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Western Disturbances: Extratropical cyclones from Mediterranean
  - ✓ First indicator: Cirrus clouds in upper troposphere
  - ✓ Feature: Thin, wispy, high-altitude clouds
- **Statement 2 is incorrect:**
  - ✓ Cirrus clouds: Made of ice crystals
  - ✓ But: Do not produce precipitation reaching ground
  - ✓ Snowfall source: Nimbostratus clouds (low–mid level, thick)
- **Key Facts:**
  - ✓ Cloud sequence (frontal system):
  - ✓ Cirrus → Cirrostratus → Altostratus → Nimbostratus
  - ✓ **Cloud characteristics:**
    - Cirrus: High altitude, no rainfall
    - Cirrostratus: Halo around sun/moon
    - Altostratus: Grey sky
    - Nimbostratus: Continuous rain/snow
  - ✓ **Western Disturbances:**
    - Type: Temperate cyclones
    - Driver: Temperature contrast (frontogenesis)
    - Steering: Subtropical Westerly Jet Stream

- ✓ **Cloud taxonomy:**
  - Cumulus: Fair weather
  - Cumulonimbus: Thunderstorm, hail
  - Nimbostratus: Steady precipitation
- **Bihar relevance:**
  - ✓ Halo sign: Cirrostratus → approaching disturbance
  - ✓ Effect: Winter rain or cold wave (Shitlahar)
  - ✓ **Risk:**
    - Localized cumulonimbus → hailstorms
    - Impact: Damage to Rabi crops (wheat, mustard)

Q 50. Ans. B

Explanation:

- **Offshore Trough:**
  - ✓ Shallow elongated low-pressure zone
  - ✓ Location: Parallel to west coast (Gujarat → Kerala)
  - ✓ Season: Southwest Monsoon
  - ✓ Impact on rainfall:
    - Strong + near coast → Heavy rainfall
    - Weak / shifts seaward → Reduced rainfall
- **Additional Information**
  - ✓ Mascarene High:
    - Location: Near Madagascar
    - Role: Drives SW Monsoon (macro-scale)
    - Not coastal / not parallel to India
  - ✓ **Equatorial Trough:**
    - Global low-pressure belt near equator
    - Associated with Inter-Tropical Convergence Zone
    - Not localized along west coast
  - ✓ **Subtropical Ridge:**
    - High-pressure belt (~30° latitude)
    - Effect: Subsidence → Dry conditions
- **Key Facts:**
  - ✓ **Orographic synergy:**
    - Offshore Trough + Western Ghats
    - Moist winds forced upward → Heavy rain
  - ✓ **Important distinction:**
    - Offshore Trough: Arabian Sea branch (west coast)
    - Monsoon Trough: Bay of Bengal branch (north India)
- **Bihar Link:**
  - ✓ Rainfall driver: Monsoon Trough
  - ✓ Normal position: Rajasthan → Bay of Bengal
  - ✓ Shift north (Break Monsoon):
    - Central India: Rainfall decreases
    - North Bihar: Heavy rainfall
  - ✓ Flood cause:
    - Rivers: Kosi, Gandak, Bagmati
    - Regions: West Champaran, Sitamarhi, Kishanganj

Q 51. Ans. B

Explanation:

Bharat Heavy Electricals Limited (BHEL):

- Established: 1964
- Type: Central Public Sector Enterprise (Maharatna status granted in 2013)
- Administrative Ministry: Ministry of Heavy Industries
- Core Role: Design, manufacture, install, and service heavy electrical equipment
- Major Products: Turbines, Boilers, Generators, Transformers, Switchgear
- Primary Use: Equipment supply for thermal, hydro, and nuclear power plants
- Important Clarification:
  - ✓ Does not generate electricity
  - ✓ Does not mine fuel
  - ✓ Supplies machinery to power producers such as NTPC and NHPC
- Financial role
  - ✓ Power project financing institutions:
    - IREDA — Indian Renewable Energy Development Agency
    - PFC — Power Finance Corporation
    - REC — Rural Electrification Corporation
- Regulatory role — Incorrect association with BHEL
  - ✓ Grid frequency control: Grid Controller of India Limited (Grid-India)
  - ✓ Electricity tariff regulation: Central Electricity Regulatory Commission (CERC)
- Coal mining role — Incorrect association with BHEL
  - ✓ Coal mining and distribution: Coal India Limited (CIL)
  - ✓ Administrative ministry: Ministry of Coal

Value Addition:

- Maharatna Status (2013): Allows greater financial autonomy and large investment capacity.
- Historical Origin: Expansion of heavy industries under the Second Five-Year Plan (1956–61) based on the Mahalanobis Model.
- Bihar Context: Equipment supplied to major projects:
  - ✓ Kanti Thermal Power Station (Muzaffarpur)
  - ✓ Nabinagar Super Thermal Power Project (Aurangabad)

Q 52. Ans. B

Explanation:

Classical Theory of Industrial Location (Alfred Weber):

- Key principle: Industries minimise transport cost.
- Weight-losing raw materials: Raw materials lose significant weight during processing.
- Result: It becomes cheaper to locate the factory near raw material sources rather than transporting bulky raw materials over long distances.

Examples of weight-losing industries:

- Iron and steel industry
- Copper smelting
- Cement industry

Reason:

- Large quantities of raw material are needed.
- Finished products weigh less than raw materials.
- Transporting raw materials is costlier than transporting finished goods.

Factory located close to the consumer market:

- This applies to weight-gaining industries.
- These industries produce goods heavier than raw materials.
- Example: Beverage and soft drink industries.

Factory located at or near raw materials:

- Applies to weight-losing industries such as iron and steel.
- Raw materials like iron ore and coal are bulky and heavy.
- Hence, industries are established near mining regions.

Industries can be set up anywhere:

- Heavy industries are strongly influenced by transport costs.
- Location decisions depend heavily on raw material availability.

Must be located only at coastal ports:

- Coastal location is beneficial for export-oriented industries, not mandatory.
- Many heavy industries are inland near mineral belts, not ports.

Additional High-Value Facts

Weight-Losing Industries — Key Feature

- Raw materials lose weight during production.
- Location preference: Raw material regions

Indian Examples:

- Iron and Steel Plants:
  - ✓ Jamshedpur
  - ✓ Rourkela
  - ✓ Bhilai
- Located near iron ore and coal belts.

Weight-Gaining Industries:

- The final product is heavier than the raw materials.
- Location preference: Market areas

Examples:

- Automobile assembly
- Beverage manufacturing

Q 53. Ans. A

Explanation:

Bailadila and Dalli-Rajhara mines:

- Bailadila mines
  - ✓ Location: Dantewada district, Chhattisgarh
  - ✓ Ore type: High-grade hematite iron ore
  - ✓ Name origin: "Bailadila" means hump of an ox, describing the hill shape
  - ✓ Industrial importance: Large-scale production and export-quality ore
- Dalli-Rajhara mines
  - ✓ Location: Balod district (earlier Durg), Chhattisgarh
  - ✓ Industrial link: Captive iron ore mines for the Bhilai Steel Plant
  - ✓ Operational control: Steel Authority of India Limited (SAIL)
  - ✓ Importance: Ensures steady supply of iron ore to one of India's major steel plants

Bauxite:

- Major bauxite regions in India:
  - ✓ Kalahandi and Koraput (Odisha)
  - ✓ Lohardaga (Jharkhand)

Manganese:

- Major manganese belts:
  - ✓ Balaghat (Madhya Pradesh)

- ✓ Nagpur–Bhandara (Maharashtra)
- ✓ Kendujhar–Sundergarh (Odisha)
- These areas dominate manganese production, not Bailadila.

Copper:

- Major copper-producing areas:
  - ✓ Malanjkhand (Balaghat, Madhya Pradesh)
  - ✓ Khetri-Singhana belt (Rajasthan)
  - ✓ Singhbhum (Jharkhand)
- Bailadila and Dalli-Rajhara are not copper zones.

Additional High-Value Facts

Geological Association

- Iron ore in this region belongs mainly to the Dharwar rock system.
- Dharwar rocks are:
  - ✓ Very ancient
  - ✓ Rich in metallic minerals such as iron, manganese, and gold

Export Route — Bailadila Iron Ore

- High-grade iron ore from Bailadila is exported to:
  - ✓ Japan
  - ✓ South Korea

Transport system:

- Ore is transported via a Kothavalasa–Kirandul (KK) railway line.
- Destination port: Visakhapatnam Port (Andhra Pradesh)

Industrial Location Link:

- The iron ore belt covering Chhattisgarh, Jharkhand, and Odisha influenced the location of major steel plants:

Key steel plants in this belt:

- Bhilai Steel Plant (Chhattisgarh)
- Bokaro Steel Plant (Jharkhand)
- Rourkela Steel Plant (Odisha)

Reason for clustering:

- Availability of iron ore
- Availability of coal nearby
- Reduced transportation costs for heavy raw materials

Q 54. Ans. A

Explanation:

Statement 1 is correct: It is part of the Nanda Devi Biosphere Reserve and is a UNESCO World Heritage Site.

- Nanda Devi National Park
  - ✓ Location: Uttarakhand (Chamoli district)
  - ✓ Protected status: Part of Nanda Devi Biosphere Reserve
  - ✓ UNESCO recognition: Included in the Nanda Devi and Valley of Flowers World Heritage Site
- It is one of India's most important high-altitude protected areas.

Statement 2 is correct: It is known for alpine meadows, high-altitude lakes and rich biodiversity.

- The park lies in the Greater Himalayas, resulting in typical high-altitude Himalayan ecosystems.
- Major landscape features:
  - ✓ Alpine meadows (locally called Bugyals)
  - ✓ Glacial valleys
  - ✓ High-altitude lakes
  - ✓ Snow-covered peaks

Important wildlife found:

- Snow Leopard
- Himalayan Musk Deer
- Bharal (Blue Sheep)
- Himalayan Tahr

Statement 3 is incorrect: It is famous for mangrove forests and coral reef ecosystems.

- This statement is incorrect because mangroves and coral reefs are coastal ecosystems, not Himalayan ecosystems.
- Nanda Devi National Park is located in a mountainous region, far from coastal areas.
- Mangrove forests are typically found in regions such as:
  - ✓ Sundarbans
  - ✓ Mahanadi Delta
  - ✓ Godavari and Krishna deltas
- Coral reefs occur mainly in:
  - ✓ Gulf of Mannar
  - ✓ Lakshadweep
  - ✓ Andaman and Nicobar Islands

Additional High-Value Facts:

Nanda Devi Biosphere Reserve: Key Features

- Established: 1988
- Earlier Named: Sanjay Gandhi National Park
- Core components:
  - ✓ Nanda Devi National Park
  - ✓ Valley of Flowers National Park
- Known for high-altitude biodiversity and fragile Himalayan ecosystems.
- UNESCO MAB (Man and Biosphere) Programme recognition: 2004.

Valley of Flowers:

- Famous for seasonal alpine flowers.
- Located near the Badrinath region.
- One of the most biologically rich alpine zones in India.

Q 55. Ans. B

Explanation:

- **NTPC's present strategy:**
  - ✓ Continues coal-based thermal power generation to meet base-load electricity demand.
  - ✓ Simultaneously expands renewable energy capacity (solar, wind, green hydrogen).
  - ✓ Follows a gradual energy transition, not an abrupt shutdown of thermal plants.
- **Key policy direction:**
  - ✓ Coal plants are modernised or replaced with more efficient technology.
  - ✓ Renewable capacity is being scaled up aggressively through subsidiary expansion and new projects.

Additional High-Value Facts:

- Renewable Energy Target:
  - ✓ Around 60 GW of renewable capacity is planned by 2032.
- Energy Mix Concept:
  - ✓ Thermal power: Provides base-load (continuous electricity).
  - ✓ Solar & wind: Intermittent sources, depend on weather and daylight.
- Bihar-Relevant NTPC Plants (Exam Useful):
  - ✓ Kahalgaon Super Thermal Power Station
  - ✓ Barh Super Thermal Power Station
  - ✓ Nabinagar Power Generating Company
  - ✓ Barauni Thermal Power Station

Q 56. Ans. B

Explanation:

- **Assertion (A) — Correct**
  - ✓ Biogas plants:
    - Divert organic waste (agricultural residue, dung, municipal waste) from landfills.
    - Generate renewable energy (methane-rich biogas).
  - ✓ Hence, they support waste reduction + clean energy production.
- **Reason (R) — Correct**
  - ✓ Anaerobic digestion:
    - Breakdown of organic waste without oxygen.
    - Produces methane-rich biogas usable as fuel.
- **Relationship — Not a correct explanation**
  - ✓ (R) explains how biogas is produced (scientific process).
  - ✓ But landfill reduction happens because waste is collected and diverted to plants (logistical action).
  - ✓ Therefore, (R) does not fully explain (A).

Quick High-Yield Additions:

- Biogas Composition
  - ✓ Methane: 50–65%
  - ✓ Carbon dioxide: 35–45%
- Important Indian Schemes
  - ✓ GOBARdhan Scheme
    - Converts cattle dung and organic waste into biogas and manure.
    - Ministry of Jal Shakti (under Swachh Bharat Mission Grameen), with multi-ministerial support
  - ✓ SATAT Scheme
    - Promotes Compressed Bio-Gas (CBG) as a transport fuel.
    - Ministry of Petroleum & Natural Gas (MoPNG)
- By-product
  - ✓ Digestate (slurry):
    - Rich in NPK nutrients
    - Used as organic fertiliser

Q 57. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Pawan Hans provides Medivac (Medical Evacuation) helicopter services for ONGC offshore operations, especially in the Bombay High region.
  - ✓ These helicopters remain on 24/7 standby to evacuate injured or critically ill personnel from offshore rigs to mainland hospitals.
- **Statement 2 is incorrect:**
  - ✓ Pawan Hans does not transport only personnel.
  - ✓ It also carries:
    - Essential supplies
    - Drilling tools

    - Lightweight equipment
  - ✓ Cargo transport is a core operational function in offshore oil exploration logistics.

Quick Value-Addition:

- Pawan Hans Limited
  - ✓ Established: 1985
  - ✓ Primary purpose: Helicopter support for oil exploration and remote area connectivity.
  - ✓ Historically operated under the Ministry of Civil Aviation.
- Major Operational Roles
  - ✓ Offshore oil support (especially Bombay High)
  - ✓ Inter-island connectivity:
    - Andaman & Nicobar
    - Lakshadweep
  - ✓ Heli-services in difficult terrains:
    - North-East India
    - Himalayan pilgrimage routes (Kedarnath, Amarnath)
- Bombay High — Key Fact
  - ✓ India's largest offshore oilfield
  - ✓ Located about 160 km off Mumbai in the Arabian Sea
  - ✓ Discovered in 1974 by ONGC

Q 58. Ans. C

Explanation:

- **Statement 1 is correct**
  - ✓ Mangrove plants in the Sundarbans develop pneumatophores (breathing roots).
  - ✓ Function:
    - Grow upward above waterlogged soil
    - Help in oxygen exchange in anaerobic (oxygen-poor) muddy soils.
- **Statement 2 is correct:**
  - ✓ The Sundarbans is a dynamic tidal delta.
  - ✓ Landforms are continuously modified due to:
    - Strong tidal action
    - Sediment movement from rivers and sea
  - ✓ Results:
    - Continuous erosion and deposition
    - Formation and disappearance of islands.
- **Statement 3 is incorrect:**
  - ✓ The eastward shifting of the Ganga has reduced, not increased, freshwater supply to the Indian Sundarbans.
  - ✓ This has led to:
    - Higher salinity levels
    - Stress on mangrove vegetation
    - Ecological degradation risks.

Additional information:

- Sundarbans — Key Identity
  - ✓ World's largest mangrove forest
  - ✓ Located in the Ganga–Brahmaputra delta
  - ✓ Named after the Sundari tree (Heritiera fomes).
- Major Global Designations
  - ✓ UNESCO World Heritage Site (1987)
  - ✓ Biosphere Reserve (1989)
  - ✓ Ramsar Site (2019)
  - ✓ Tiger Reserve — habitat of the Royal Bengal Tiger.
- Ecological Speciality
  - ✓ Only mangrove forest globally with a wild tiger population.
  - ✓ Highly sensitive to:
    - Sea-level rise
    - Salinity changes
    - Cyclones.

Q 59. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Murlen National Park
    - Location: Champhai district, Mizoram
    - Situated near the Indo–Myanmar border (close to the Chin Hills).
  - ✓ Important habitat for Vavu (Hume's Pheasant) — the state bird of Mizoram.
- **Statement 2 is incorrect:**
  - ✓ The park does not have tropical deciduous forests.
  - ✓ Dominant vegetation type:
    - Sub-tropical evergreen and semi-evergreen forests
  - ✓ Reason:
    - High altitude (≈1500–1900 m)
    - Heavy monsoon rainfall supporting evergreen vegetation.

Additional information:

- Mizoram — National Parks
  - ✓ Murlen National Park — Champhai district
  - ✓ Phawngpui National Park (Blue Mountain NP) — southern Mizoram
- Key Wildlife in Murlen

- ✓ Hume's Pheasant (Vavu) — State bird
- ✓ Hoolock Gibbon — India's only ape
- ✓ Himalayan Black Bear
- ✓ Serow — State animal of Mizoram

Q 60. Ans. C

Explanation:

- **Statement C is incorrect:**
  - ✓ Nilgiri Tahr is endemic to the Western Ghats, not Central India.
  - ✓ Bandhavgarh National Park (Madhya Pradesh, Vindhya Range) has:
  - ✓ Tropical moist deciduous forests
  - ✓ High Royal Bengal Tiger density
  - ✓ It does not host Nilgiri Tahr.
- **Statement A is correct:**
  - ✓ Eravikulam National Park (Kerala) protects the largest wild population of Nilgiri Tahr.
  - ✓ Located near Munnar, it is the most important conservation site for the species.
- **Statement C is correct:**
  - ✓ Mukurthi National Park (Tamil Nadu) was established mainly to conserve Nilgiri Tahr.
  - ✓ It forms part of the Nilgiri Biosphere Reserve.
- **Statement D is correct:**
  - ✓ Nilgiri Tahr habitat:
    - ▪ High-altitude grasslands
    - ▪ Shola forests
    - ▪ Elevation generally above 1200–2000 m
  - ✓ Found only in the Western Ghats.

Additional information:

- Nilgiri Tahr
  - ✓ Scientific name: Nilgiritragus hylocrius
  - ✓ State animal: Tamil Nadu
  - ✓ IUCN Status: Endangered
  - ✓ Wildlife Protection Act: Schedule I species
- Major Protected Areas
  - ✓ Eravikulam National Park — Kerala (largest population)
  - ✓ Mukurthi National Park — Tamil Nadu
  - ✓ Silent Valley landscape — Western Ghats
- Geographical info:
  - ✓ Palghat Gap divides Nilgiri Tahr populations:
  - ✓ North: Nilgiri Hills
  - ✓ South: Anamalai & Cardamom Hills
- Associated Famous Plant
  - ✓ Neelakurinji flower
    - ▪ Blooms once every 12 years
    - ▪ Seen widely in the Eravikulam region

Q 61. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Seshachalam Hills are the primary natural habitat of Red Sanders (Pterocarpus santalinus).
  - ✓ Red Sanders is:
    - ▪ Endemic to the Eastern Ghats of Andhra Pradesh
    - ▪ Found mainly in the Seshachalam hill ranges
  - ✓ This makes the region globally important for rare timber conservation.
- **Statement 2 is incorrect:**
  - ✓ Seshachalam Hills do not connect the Eastern and Western Ghats.
  - ✓ The Eastern Ghats and Western Ghats meet in the Nilgiri Hills region, not in Seshachalam.
  - ✓ Hence, the statement misidentifies the geographical junction point.

Additional information:

- Seshachalam Biosphere Reserve
  - ✓ Declared: 2010
  - ✓ Location: Andhra Pradesh
  - ✓ Districts: Tirupati and YSR Kadapa
  - ✓ Includes:
    - ▪ Sri Venkateswara National Park
    - ▪ Sri Venkateswara Wildlife Sanctuary
- Red Sanders — Key Facts
  - ✓ Scientific name: Pterocarpus santalinus
  - ✓ IUCN Status: Endangered
  - ✓ CITES: Appendix II
  - ✓ Uses:
    - ▪ Premium furniture
    - ▪ Musical instruments
    - ▪ Traditional medicine
- Geographical Trap (Frequently Asked)
  - ✓ Eastern & Western Ghats meet at: Nilgiri Hills
  - ✓ Not at: Seshachalam Hills
- Cultural Link
  - ✓ Tirumala Venkateswara Temple is located on the Seshachalam Hills, making the region culturally significant.

Q 62. Ans. A

Explanation:

- **Statement A is correct:**
  - ✓ The Inter-Tropical Convergence Zone (ITCZ):
    - Is a low-pressure belt near the equator.
    - Formed where the Northeast and Southeast trade winds converge.
  - ✓ During Northern Hemisphere winter (December):
    - Sun's vertical rays shift toward the Tropic of Capricorn.
    - Maximum heating shifts to the Southern Hemisphere.
    - As a result, the ITCZ shifts south of the equator.

Additional information:

- ITCZ — Key Identity
  - ✓ Also called Doldrums (low wind zone).
  - ✓ Associated with:
    - Rising air currents
    - Heavy convectional rainfall
    - Thunderstorms
- Seasonal Position
  - ✓ January: South of the equator
  - ✓ July: North of the equator (over the Indian subcontinent)
- Related fact:
  - ✓ Summer: ITCZ shifts north →Southwest Monsoon begins.
  - ✓ Winter: ITCZ shifts south →Northeast Monsoon dominates.

Q 63. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ The Himalayan mountain system formed through multiple orogenic (mountain-building) phases, not a single event.
  - ✓ The first major uplift occurred during the Eocene–Oligocene period (≈50–30 million years ago).
  - ✓ This phase formed the Greater Himalayas (Himadri) — the highest and oldest Himalayan range.
- **Statement 2 is incorrect:**
  - ✓ The Shiwalik Range (Outer Himalayas) did not form simultaneously with the Greater Himalayas.
  - ✓ It formed much later, during the Pliocene–Pleistocene period (≈5–2 million years ago).
  - ✓ Composition:
    - Unconsolidated sediments (sand, gravel, conglomerates)
    - Deposited by rivers from the older Himalayan ranges
    - Later folded into hills.

Additional information:

Three Phases of Himalayan Formation (Very Important)

- Phase 1 — Greater Himalayas (Himadri)
  - ✓ Time: Eocene–Oligocene
  - ✓ Highest peaks formed.
- Phase 2 — Lesser Himalayas (Himachal)
  - ✓ Time: Miocene
  - ✓ Middle mountain ranges developed.
- Phase 3 — Shiwaliks (Outer Himalayas)
  - ✓ Time: Pliocene–Pleistocene
  - ✓ Youngest and most fragile hills.
- Tethys Sea:
  - ✓ Himalayan rocks formed from sediments deposited in the Tethys geosyncline.
- Most fragile Himalayan zone:
  - ✓ Shiwaliks
  - ✓ Highly prone to:
    - Landslides
    - Soil erosion
    - Earthquake damage.
- Syntaxial Bends (Highly Tested)
  - ✓ Western: Near Nanga Parbat (Indus Gorge)
  - ✓ Eastern: Near Namcha Barwa (Brahmaputra turn)

Q 64. Ans. B

Explanation:

- **Pair 1 — Correct**
- Orang National Park: Assam
  - ✓ Located on the northern bank of the Brahmaputra River.
  - ✓ Situated in the Darrang and Sonitpur districts of Assam.
  - ✓ Often called "Mini Kaziranga" due to its similar landscape and fauna.
- **Pair 2 — Incorrect**
- Gir National Park:
  - ✓ Gir National Park is located in Gujarat, not Rajasthan.

- ✓ It is the only natural habitat of the Asiatic Lion.
- ✓ Rajasthan parks include:
  - Ranthambore
  - Sariska
  - Mukundra Hills
- **Pair 3 — Incorrect**
- Kaziranga National Park:
  - ✓ Kaziranga is located in Assam, not West Bengal.
  - ✓ Districts: Golaghat and Nagaon.
  - ✓ Famous for:
    - Great One-Horned Rhinoceros
    - UNESCO World Heritage Site.
- **Pair 4 — Correct**
- Periyar National Park: Kerala
  - ✓ Located in the Cardamom Hills (Western Ghats).
  - ✓ Situated in the Idukki and Pathanamthitta districts.
  - ✓ Centred around Periyar Lake.

Additional information:

Orang National Park

- State: Assam
- Nickname: Mini Kaziranga
- Key species:
  - ✓ One-horned rhinoceros
  - ✓ Tiger
  - ✓ Elephant

Kaziranga National Park

- State: Assam
- UNESCO World Heritage Site
- Located between:
  - ✓ Brahmaputra River (north)
  - ✓ Karbi Anglong Hills (south)

Periyar National Park

- State: Kerala
- Famous feature:
  - ✓ Artificial Periyar Lake
  - ✓ Formed by the Mullaperiyar Dam (1895)
- Also, an Elephant Reserve.

Q 65. Ans. D

Explanation:

- **High Isolation in Ladakh:**
  - ✓ High altitude (≈3000 m and above) → Atmosphere becomes thin.
  - ✓ Low humidity (very little water vapour).
  - ✓ Very little cloud cover and dust due to rain-shadow conditions.
  - ✓ Result:
    - Minimal scattering and absorption of solar radiation.
    - Strong direct isolation reaches the surface.

Additional information:

Rain Shadow Effect:

- Caused by Great Himalayas blocking monsoon winds.
- Leads to:
  - ✓ Very low rainfall (<10 cm annually).
  - ✓ Dry, dust-free atmosphere.
  - ✓ Enhanced solar radiation intensity.

Ladakh Paradox:

- Very strong sunshine but cold climate.
- Possible to experience:
  - ✓ Sunstroke in sunlight
  - ✓ Freezing in shade
- Due to thin air and strong radiation loss at night.

Trans-Himalayan Location Facts:

- Ladakh lies between:
  - ✓ Karakoram Range (north)
  - ✓ Zaskar Range (south)
- Indus River flows between Ladakh and Zaskar ranges.

Q 66. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Monazite is a phosphate mineral containing:
    - Rare Earth Elements (REEs)
    - Thorium Oxide (ThO_2)
  - ✓ Typical composition in Indian monazite:
    - ≈55–60% Rare Earth Oxides (REO)
    - ≈8–10% Thorium Oxide
  - ✓ Hence, the mineral is strategically important for rare earth and nuclear energy resources.
- **Statement 2 is incorrect:**
  - ✓ The state-wise order of monazite reserves is wrong.
  - ✓ Correct order (largest to smallest reserves):
    - Andhra Pradesh — Highest reserves
    - Odisha — Second
    - Tamil Nadu — Third

Additional information:

Monazite — Key Facts

- Found mainly as placer deposits (heavy mineral sands).

- Occurs along coastal beaches.
- Usually associated with:
  - ✓ Ilmenite
  - ✓ Rutile
  - ✓ Zircon
  - ✓ Garnet

Strategic Importance
- Contains Thorium, important for:
  - ✓ India's three-stage nuclear programme
  - ✓ Future Thorium-based reactors (AHWR concept).
- Mining and processing are controlled mainly by:
  - ✓ IREL (India) Limited
  - ✓ Under the Department of Atomic Energy (DAE).
- Major Monazite-bearing Coastal States:
  - ✓ Andhra Pradesh
  - ✓ Odisha
  - ✓ Tamil Nadu
  - ✓ Kerala

Q 67. Ans. B

Explanation:

- **Correct Epoch: Miocene**
  - ✓ The second major orogenic phase of Himalayan formation occurred during the Miocene epoch (≈20–10 million years ago).
  - ✓ This phase resulted in the uplift of the Lesser Himalayas (Himachal range).
  - ✓ Continuous tectonic compression from the northward-moving Indian Plate caused rocks south of the Greater Himalayas to fold and rise.
- **Eocene:**
  - ✓ Associated with the first major uplift.
  - ✓ Formed the Greater Himalayas (Himadri).
- **Pliocene:**
  - ✓ Linked with the third phase of uplift.
  - ✓ Responsible for forming the Shiwalik (Outer Himalayas).
- **Holocene:**
  - ✓ Current geological epoch.
  - ✓ Characterised mainly by alluvial deposition, not mountain uplift.

Additional information:

Important Structural Boundaries:
- Main Central Thrust (MCT)
  - ✓ Separates Greater Himalayas and the Lesser Himalayas.
- Main Boundary Thrust (MBT)
  - ✓ Separates Lesser Himalayas and the Shiwaliks.

Key Ranges of the Lesser Himalayas
- Pir Panjal: Longest range
- Dhauladhar Range
- Mahabharat Range
- Mussoorie Range

Q 68. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Stalagmites are depositional features found in limestone (karst) caves.
  - ✓ Formation process:
    - ▪ Water containing dissolved calcium carbonate ($CaCO_3$) drips onto the cave floor.
    - ▪ Evaporation and CO_2 release cause $CaCO_3$ to precipitate.
    - ▪ This builds cone-shaped deposits upward from the cave floor.
- **Statement 2 is correct:**
  - ✓ Borra Caves
    - ▪ Location: Ananthagiri Hills, Araku Valley, Andhra Pradesh.
    - ▪ One of the largest limestone cave systems in India.
    - ▪ Formed by chemical weathering and erosion by the Gosthani River.
- **Statement 3 is incorrect:**
  - ✓ The Aravalli Range is not a major karst cave region.
  - ✓ Reasons:
    - ▪ Dominated by metamorphic rocks (quartzite, schist, marble).
    - ▪ Dry climate limits chemical weathering.
  - ✓ India's largest limestone cave systems are mainly found in:
    - ▪ Meghalaya (Khasi–Jaintia Hills)
    - ▪ Due to heavy rainfall + thick limestone beds.

Additional information:
- Major Limestone Cave Regions in India
  - ✓ Meghalaya — Longest cave systems (e.g., Krem Liat Prah)
  - ✓ Borra Caves — Andhra Pradesh
  - ✓ Kutumsar Cave — Kanger Valley NP, Chhattisgarh

- ✓ Gupteshwar Cave — Odisha
- ✓ Rohtas Plateau — Bihar (minor karst features)

Q 69. Ans. B

Explanation:

- **Assertion (A) is true:**
  - ✓ Rainfall in the Northern Plains during the Southwest Monsoon decreases from east to west.
  - ✓ Typical trend:
    - Kolkata → ~150 cm
    - Patna → ~100 cm
    - Prayagraj → ~90 cm
    - Delhi → ~60 cm
  - ✓ Cause:
    - Gradual loss of moisture as monsoon winds move inland.
- **Reason (R) is true:**
  - ✓ The Indo-Gangetic Plain shows a gentle rise in elevation from east to west.
  - ✓ Elevation pattern:
    - Bengal Delta → near sea level
    - Punjab–Haryana Plains → about 250–300 m.
- **(R) is the not correct explanation of (A):**
  - ✓ Rainfall decreases mainly due to:
    - Moisture depletion as winds move farther from the Bay of Bengal.
  - ✓ The gentle westward rise of land is:
    - Too gradual to act as a significant rainfall barrier.
  - ✓ Therefore:
    - (R) is true but does not explain (A).

Additional information:

Main Cause of East-to-West Rainfall Decrease

- Distance from sea (continentality effect)
- Progressive moisture exhaustion of monsoon winds.

Important Supporting Concepts

- Bay of Bengal Branch
  - ✓ Moves along the Ganga plains.
  - ✓ Loses moisture continuously inland.
- Arakan Yoma Role
  - ✓ Deflects monsoon winds westward into India.
  - ✓ Ambala Water Divide
  - ✓ Divides drainage of:
    - Indus system (west)
    - Ganga system (east)
- Isohyets
  - ✓ Lines connecting equal rainfall points.
  - ✓ Show a clear decrease westward in the Northern Plains.

Q 70. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Sasthamkotta Lake
    - Location: Kollam district, Kerala
    - Largest natural freshwater lake in Kerala
    - Designated as a Ramsar Wetland Site in 2002.
  - ✓ It serves as the major drinking water source for the Kollam district.
- **Statement 2 is correct:**
  - ✓ The lake has remarkably pure water, maintained naturally by:
    - Chaoborus larvae (phantom midges)
- These larvae:
  - Feed on bacteria, algae, and organic matter
  - Help maintain water clarity and quality
  - Reduce the chances of algal blooms.

Additional information:

Kerala — Ramsar Sites

- Vembanad-Kol Wetland
  - ✓ Largest lake in Kerala
  - ✓ Brackish water lagoon
- Ashtamudi Wetland
  - ✓ Estuarine lagoon
  - ✓ Famous for eight arms/channels
- Sasthamkotta Lake
  - ✓ Only freshwater Ramsar site among Kerala's major lakes.

Hydrological Features

- No major surface rivers feed the lake.
- Water source:
  - ✓ Rainfall
  - ✓ Underground springs

Q 71. Ans. B

Explanation:

- **Madden–Julian Oscillation (MJO):**
  - ✓ A travelling atmospheric disturbance in the tropics.

- ✓ Moves eastward along the equator.
- ✓ Completes one cycle in about 30–60 days.
- ✓ Consists of:
  - ▪ Active phase → heavy clouds and rainfall
  - ▪ Suppressed phase → reduced rainfall.
- **Impact on Indian Monsoon:**
  - ✓ Active MJO over the Indian Ocean
    - ▪ Strengthens the Southwest Monsoon
    - ▪ Causes wet spells in India.
  - ✓ Suppressed phase over the Indian Ocean
    - ▪ Leads to break monsoon (dry spells).

Additional information:

MJO vs ENSO (Very Important)

- MJO
  - ✓ Intra-seasonal phenomenon
  - ✓ Duration: 30–60 days
  - ✓ Moves across equatorial regions.
- ENSO (El Niño/La Niña)
  - ✓ Inter-annual phenomenon
  - ✓ Lasts months to years
  - ✓ Mostly over the Pacific Ocean.

Role in Cyclone Formation

- Active MJO phase:
  - ✓ Increases cloud formation
  - ✓ Enhances cyclone development over:
    - ▪ Bay of Bengal
    - ▪ Arabian Sea.

Q 72. Ans. C

Explanation:

- **Correct Mechanism — ITCZ**
  - ✓ The Inter-Tropical Convergence Zone (ITCZ) shifts north and south following the apparent movement of the Sun.
  - ✓ It crosses the equator twice a year:
    - ▪ During the Vernal Equinox (March)
    - ▪ During the Autumnal Equinox (September)
  - ✓ Each passage produces:
    - ▪ Strong convection
    - ▪ Heavy rainfall
  - ✓ This creates the double peak (biannual maxima) rainfall pattern in equatorial regions.

Additional information:

- Regions in India Showing Biannual Rainfall Pattern
  - ✓ Andaman & Nicobar Islands
  - ✓ Lakshadweep Islands
  - ✓ Southern tip of peninsular India
  - ✓ Parts of coastal Tamil Nadu and Kerala

Key Term — Zenithal Rainfall

- Occurs when:
  - ✓ Sun is directly overhead.
  - ✓ ITCZ passes across the region.
- Produces:
  - ✓ Intense convectional rainfall.

Seasonal Link

- March–April: First rainfall peak
- September–October: Second rainfall peak
- Controlled by ITCZ migration.

Q 73. Ans. B

Explanation:

Mechanical Exfoliation:

- Occurs due to thermal expansion and contraction of rock surfaces.
- Typical in semi-arid regions with a large diurnal temperature range.
- Day: Outer rock layer heats and expands.
- Night: Outer layer cools and contracts.
- Repeated stress causes outer layers to peel off like onion skins.
- Produces rounded granitic domes and boulders seen in:
  - ▪ Hampi (Karnataka)
  - ▪ Parts of Telangana and Andhra Pradesh.
- **Frost Action:**
  - ✓ Requires freezing temperatures (<0°C).
  - ✓ The Deccan Plateau has a warm tropical climate, not freezing winters.
  - ✓ Frost weathering is typical of the Himalayan regions, not peninsular India.
- **Chemical Carbonation:**
  - ✓ Mainly affects limestone and dolomite (karst regions).
  - ✓ Deccan Plateau rocks are granite and gneiss, resistant to carbonation.
- **River Erosion:**
  - ✓ Rivers form valleys and gorges, not widespread onion-shaped domes.
  - ✓ Exfoliation acts over large exposed rock surfaces, not just riverbeds.

Additional information:

- Key Landform Term:
  - ✓ Tors → Isolated piles of rounded granite boulders.

- Common Rock Type Involved:
  - ✓ Granite (hard, poor conductor of heat).
- Typical Climate Required:
  - ✓ Semi-arid climate
  - ✓ Large day–night temperature variation

Q 74. Ans. B

Explanation:

Molasse:
- Refers to thick terrestrial sediments deposited in a foreland basin.
- Composed of gravel, sand, clay, and conglomerates.
- Formed from the erosion of newly uplifted mountains.
- In the Himalayas:
  - ✓ Rivers eroded the Greater and Lesser Himalayas.
  - ✓ Deposited sediments in the foredeep south of the mountains.
  - ✓ These sediments were later folded during the Pliocene–Pleistocene.
- Result: Formation of the Shiwalik (Outer Himalaya).
- **Flysch:**
  - ✓ Forms in deep marine environments.
  - ✓ Deposited before or during early mountain building, not in outer foothills.
- **Ophiolite:**
  - ✓ Represents oceanic crust fragments thrust onto continents.
  - ✓ Found in suture zones, not river-deposited foreland sediments.
- **Batholith:**
  - ✓ A massive intrusive igneous rock body formed from magma.
  - ✓ Not related to sedimentary river deposits.

Additional information:
- Shiwalik Composition
  - ✓ Mainly unconsolidated molasse deposits.
  - ✓ Highly prone to:
    - ▪ Landslides
    - ▪ Soil erosion
- Associated Landforms
  - ✓ Duns (Doons):
    - ▪ Flat valleys between the Shiwalik and the Lesser Himalaya
    - ▪ Examples:
      - ◊ Dehradun
      - ◊ Kotli Dun
- Geological Sequence (Very Important)
  - ✓ Greater Himalaya → Lesser Himalaya → Shiwalik (Molasse)

Q 75. Ans. B

Explanation:
- **Statement 1 is correct:**
  - ✓ Northeast Monsoon (Oct–Dec) winds are initially dry continental winds.
  - ✓ Become moist after crossing the Bay of Bengal.
  - ✓ Heavy rainfall was mainly triggered by:
    - ▪ Easterly waves
    - ▪ Low-pressure systems
    - ▪ Tropical cyclones
- **Statement 2 is incorrect:**
  - ✓ El Niño:
    - ▪ Suppresses Southwest Monsoon rainfall.
    - ▪ But generally enhances Northeast Monsoon rainfall.
  - ✓ Reason:
    - ▪ Favours cyclogenesis in the Bay of Bengal.
    - ▪ Leads to above-normal rainfall on:
      - ◊ Tamil Nadu coast
      - ◊ South Coastal Andhra Pradesh.
- **Statement 3 is incorrect:**
  - ✓ Somali Jet (Findlater Jet):
    - ▪ Active during the Southwest Monsoon, not the Northeast Monsoon.
    - ▪ Weakens/disappears during winter.
  - ✓ Northeast Monsoon moisture source:
    - ▪ Northeast Trade Winds crossing the Bay of Bengal, not reversed Somali Jet.

Additional information:
- Core Region of Northeast Monsoon
  - ✓ Tamil Nadu
  - ✓ Puducherry
  - ✓ Coastal Andhra Pradesh
  - ✓ Rayalaseema
  - ✓ Parts of Kerala & South Interior Karnataka
  - ✓ Tamil Nadu receives ~48% annual rainfall from NEM.

Rainfall Drivers During NEM
- Easterly Waves
  - ✓ Move east →west across the Bay of Bengal.
  - ✓ Major cause of rainfall variability.

- Cyclone Formation
  - ✓ The Bay of Bengal remains warm after the SW Monsoon withdrawal.
  - ✓ Peak cyclone season:
    - ▪ October–November

Q 76. Ans. B

Explanation:

- **Nagaland is the Indian state predominantly associated with Sino-Tibetan language speakers.**
  - ✓ Majority of indigenous tribes speak Sino-Tibetan (Tibeto-Burman) languages.
  - ✓ Major tribal languages:
  - ✓ Ao
  - ✓ Angami
  - ✓ Konyak
  - ✓ Sema (Sumi)
  - ✓ Lotha
- Part of the North-East linguistic belt, dominated by Sino-Tibetan languages.
- Official language of Nagaland: English
- Common lingua franca: Nagamese (Assamese-based creole).
- Eighth Schedule Sino-Tibetan languages:
    - ▪ Bodo
    - ▪ Manipuri (Meitei)
- Kerala:
  - ✓ Dominant language: Malayalam
  - ✓ Language family: Dravidian
  - ✓ Typical of South Indian plateau region.
- Gujarat:
  - ✓ Dominant language: Gujarati
  - ✓ Language family: Indo-Aryan (Indo-European)
  - ✓ Part of North-Western Indo-Aryan belt.
- Jharkhand:
  - ✓ Major tribal languages:
    - ▪ Santhali
    - ▪ Mundari
    - ▪ Ho
  - ✓ Language family: Austro-Asiatic (Munda group)
  - ✓ Core region of Austric languages in India.

Additional information:

Four Major Language Families in India

- Indo-European (Indo-Aryan)
  - ✓ Largest share of population
  - ✓ Regions: North, Central, West India
- Dravidian
  - ✓ Southern India
  - ✓ Examples: Tamil, Telugu, Kannada, Malayalam
- Austro-Asiatic (Austric)
  - ✓ Central-Eastern tribal belt
  - ✓ Jharkhand, Odisha, Chhattisgarh
- Sino-Tibetan
  - ✓ Himalayan & North-Eastern India
  - ✓ Nagaland, Manipur, Mizoram, and Arunachal Pradesh.
- Santhali is the only Austro-Asiatic language in the Eighth Schedule and the only scheduled language with its own indigenous script known as Ol Chiki, invented by Pandit Raghunath Murmu.
- Andamanese languages (spoken by the Great Andamanese, Jarawa, and Sentinelese in the Andaman Islands) are considered language isolates or a separate family.

Q 77. Ans. A

Explanation:

- **Bordoishila**
  - ✓ Local name for violent pre-monsoon thunderstorms.
  - ✓ Occurs during April–May (hot weather season).
  - ✓ Also referred to in geography texts as Bardoli Chheerha.
  - ✓ Caused by:
    - ▪ Intense surface heating
    - ▪ Moisture-laden winds from Bay of Bengal
    - ▪ Strong convective instability
- **Weather Characteristics:**
  - ✓ Sudden heavy rainfall
  - ✓ Strong squally winds
  - ✓ Thunder and lightning
  - ✓ Occasional hailstorms
  - ✓ Short duration but high intensity.
- **Agricultural Importance:**
  - ✓ Highly beneficial for:
    - ▪ Tea plantations
    - ▪ Jute cultivation
    - ▪ Early paddy sowing
  - ✓ Provides pre-monsoon soil moisture.

Additional information:

- Regional Barrier Effect

- ✓ Moisture-laden winds get trapped between:
  - Meghalaya Plateau (Garo–Khasi–Jaintia Hills)
  - Eastern Himalayas
- ✓ This funneling effect increases storm intensity in Assam Valley.
- Cultural Geography Link:
  - ✓ In Assamese folklore:
    - Bordoishila is symbolised as a woman traveling to celebrate Rongali Bihu.
    - Storm timing coincides with Rongali Bihu festival (April).

Q 78. Ans. B

Explanation:

- **Winter cooling in North India is mainly due to slanting Sun's rays spreading energy over a larger area and losing heat through a longer atmospheric path.**
  - ✓ Earth's axis tilted 23.5° → Sun shifts southward in December.
  - ✓ Around Winter Solstice (~21 December):
    - Sun overhead at Tropic of Capricorn.
    - North India receives highly oblique (slanting) rays.
- Why slanting rays cause cooling:
  - ✓ Energy Spread Effect
    - Slanting rays cover larger surface area.
    - Heat per unit area decreases.
  - ✓ Atmospheric Path Effect
    - Rays travel through thicker atmospheric layer.
    - Increased absorption, scattering, reflection.
    - Less heat reaches the surface.

Additional information:

- Continentality Effect
  - ✓ North India far from sea → extreme winters.
  - ✓ Coastal areas show moderate temperatures.
- Himalayan Barrier Role
  - ✓ Blocks extreme cold winds from Central Asia.
  - ✓ Without Himalayas → North India would be much colder.
- Winter Temperature Inversion
  - ✓ Rapid night cooling → cold air trapped near surface.
  - ✓ Leads to:
    - Dense fog
    - Smog in Indo-Gangetic plains.
- Western Disturbances
  - ✓ Winter cyclones from Mediterranean region.
  - ✓ Cause:
    - Winter rainfall
    - Snowfall in Himalayas.

Q 79. Ans. A

Explanation:

- **Assertion (A) is True:**
  - ✓ October Heat occurs in North Indian plains after the retreat of the Southwest Monsoon.
  - ✓ Weather becomes hot, humid, and uncomfortable, especially during October.
  - ✓ It represents a transition phase from monsoon to winter.
- **Reason (R) is True and Correctly Explains A:**
  - ✓ Oppressive weather occurs due to three combined factors:
    - Clear skies
      - ◊ Monsoon clouds disappear → direct solar heating increases.
    - High temperature
      - ◊ Strong sunlight heats land rapidly.
    - High humidity
      - ◊ Moisture from monsoon-saturated soil evaporates.
  - ✓ This combination produces stifling, oppressive conditions known as October Heat.

Additional information:

- Seasonal Transition
  - ✓ October–November = Retreating Monsoon period.
  - ✓ Monsoon withdrawal begins from:
    - Northwestern India: September
    - Northern Plains: Mid-October
    - Southern India: Early December
- Diurnal Temperature Range
  - ✓ Days: Hot and humid
  - ✓ Nights: Cooler due to rapid heat loss under clear skies.
- Underlying Mechanism
  - ✓ Southward movement of:
    - Sun
    - ITCZ
  - ✓ Leads to weakening of monsoon trough and formation of high pressure over land.

Q 80. Ans. B

Explanation:

- **Edmund Halley:**
  - ✓ Proposed the thermal theory of the Indian monsoon in 1686.
  - ✓ Explained monsoon as a large-scale land and sea breeze system.
  - ✓ Based on differential heating of land and ocean.
  - ✓ Core mechanism of thermal theory:
    - Summer
      - ◊ Land heats faster than ocean.
      - ◊ Low pressure develops over land.
      - ◊ Winds blow sea to land, forming southwest monsoon.
    - Winter
      - ◊ Land cools faster than ocean.
      - ◊ High pressure develops over land.
      - ◊ Winds blow land to sea, forming northeast monsoon.
- **Alexander von Humboldt:**
  - ✓ Known for physical geography and Humboldt Current.
- **Gilbert Walker:**
  - ✓ Discovered Southern Oscillation.
  - ✓ Linked to ENSO and Walker circulation, not thermal theory.
- **Vilhelm Bjerknes:**
  - ✓ Developed polar front theory.
  - ✓ Related to temperate cyclone formation.

Additional information:

- Dynamic concept, Hermann Flohn, 1951
  - ✓ Monsoon linked to seasonal shift of ITCZ.
- Jet stream theory, M.T. Yin, 1949 and P. Koteswaram, 1952
  - ✓ Monsoon burst linked to jet stream movement.
- Tibetan plateau role
  - ✓ Acts as high-altitude heat source strengthening monsoon.

Q 81. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Bay of Bengal has higher cyclone frequency than Arabian Sea.
  - ✓ About 80 percent of North Indian Ocean cyclones occur here.
  - ✓ Reasons:
    - Warmer sea surface temperatures
    - Freshwater influx from Ganga and Brahmaputra reduces mixing
    - Lower vertical wind shear compared to Arabian Sea.
- **Statement 2 is correct:**
  - ✓ Some cyclones originate as typhoons in the South China Sea.
  - ✓ These systems move west across Indochina, weaken over land.
  - ✓ They re-intensify over Bay of Bengal, forming severe cyclones.
- **Statement 3 is incorrect:**
  - ✓ Uses extreme words like always and never, which are rarely correct.
  - ✓ Arabian Sea cyclones can strike Indian west coast, especially:
    - Gujarat
    - Maharashtra
  - ✓ Recent examples:
    - Cyclone Tauktae
    - Cyclone Nisarga
    - Cyclone Biparjoy.

Additional information:

- Changing cyclone trends
  - ✓ Arabian Sea cyclones are increasing due to:
    - Rising sea surface temperatures
    - Climate change effects.
- Cyclone seasons in North Indian Ocean
  - ✓ Pre-monsoon: April to June
  - ✓ Post-monsoon: October to December, main cyclone season.
- Essential conditions for cyclone formation
  - ✓ Sea temperature above 27°C
  - ✓ Presence of Coriolis force
  - ✓ Low vertical wind shear
  - ✓ Pre-existing low-pressure system
  - ✓ Upper-level divergence.

Q 82. Ans. C

Explanation:

Southwest monsoon direction is produced mainly by the Coriolis force, described by Ferrel's law, acting on southeast trade winds after they cross the equator.

Core mechanism

- During summer, intense heating over India and Tibetan Plateau creates low pressure.

- Southeast Trade Winds from Southern Hemisphere move north to fill this low-pressure area.
- After crossing the equator, winds are deflected to the right in the Northern Hemisphere.
- This deflection converts southeast trades into southwest monsoon winds.

Key governing law:

- Ferrel's law
  - ✓ Moving objects are deflected:
    - Right in Northern Hemisphere
    - Left in Southern Hemisphere
  - ✓ This effect is caused by the Coriolis force due to Earth's rotation.

Q 83. Ans. B

Explanation:

- **Chromite:**
  - ✓ Sukinda Valley, located in Jajpur district, Odisha, holds about 97 to 98 percent of India's chromite reserves.
  - ✓ Chromite is the only commercial ore of chromium.
  - ✓ Major uses:
  - ✓ Production of stainless steel
  - ✓ Manufacture of nichrome
  - ✓ Production of refractory bricks for furnaces.
  - ✓ Mining type:
    - Mostly open-cast mining.
- **Bauxite:**
  - ✓ Major deposits in Kalahandi and Koraput districts of Odisha.
  - ✓ Important deposit area:
    - Panchpatmali hills.
- **Copper:**
  - ✓ Major copper mining regions:
    - Malanjkhand, Madhya Pradesh
    - Khetri, Rajasthan
    - Singhbhum, Jharkhand.
- **Zinc:**
    - Major zinc deposits concentrated in Rajasthan.
    - Important area:
      - ◊ Zawar mines, Udaipur district.

Additional information:

- National importance
  - ✓ Odisha dominates chromite production in India.
  - ✓ Sukinda Valley is among the largest chromite reserves globally.
- Environmental concern
  - ✓ Mining has caused contamination by hexavalent chromium (Cr VI).
  - ✓ Toxic chemical linked to:
    - Water pollution
    - Health hazards.
- Industrial relevance
  - ✓ Chromium provides:
    - Corrosion resistance
    - Hardness to stainless steel.

Q 84. Ans. A

Explanation:

- **Uranium:**
  - ✓ Tummalapalle Mine, located in YSR Kadapa district, Andhra Pradesh, is famous for uranium extraction.
  - ✓ Operated by Uranium Corporation of India Limited (UCIL) under the Department of Atomic Energy (DAE).
  - ✓ Considered one of the largest uranium deposits in India, with reserves more than 1.5 lakh tonnes.
- Bauxite:
  - ✓ Major deposits in Andhra Pradesh located in:
    - Visakhapatnam district
    - East Godavari district
  - ✓ Found mainly in Eastern Ghats region, not Tummalapalle.
- Iron ore:
  - ✓ Major iron ore belts in India:
    - Odisha
    - Jharkhand (Singhbhum)
    - Chhattisgarh (Bailadila)
    - Karnataka (Bellary region).
- Manganese:
  - ✓ Major manganese-producing states:
    - Odisha
    - Maharashtra
    - Madhya Pradesh (Balaghat).

Additional information:

- Geological setting:
  - ✓ Uranium at Tummalapalle occurs in dolostone rocks of the Cuddapah Basin.
- Other important uranium mines in India:
  - ✓ Narwapahar, Jharkhand
  - ✓ Turamdih, Jharkhand
  - ✓ Rohil, Rajasthan.

Q 85. Ans. A

Explanation:

- **Assertion (A) is true:**
  - ✓ Aluminium smelting plants are usually located near major power sources, not at bauxite mines.
  - ✓ Typical locations:
    - Near hydroelectric dams
    - Near pit-head thermal power plants.
  - ✓ Reason: electricity is the most critical requirement.
- **Reason (R) is true and correct explanation of (A)**
  - ✓ Aluminium smelting uses the Hall-Héroult electrolytic process.
  - ✓ This process is extremely power-intensive.
  - ✓ About 13,000 to 15,000 kWh of electricity needed to produce 1 tonne of aluminium.
  - ✓ Electricity cost forms 30 to 40 percent of total production cost.
  - ✓ Therefore, cheap and uninterrupted electricity determines plant location more than raw material proximity.

Additional information:

- Two-stage production process
  - ✓ Refining stage
    - Bauxite converted to alumina.
    - Refineries located near bauxite mines.
  - ✓ Smelting stage
    - Alumina converted to aluminium.
    - Smelters located near power plants.
  - ✓ The Hall-Héroult Process is an electrolytic method for producing aluminum metal from alumina dissolved in molten cryolite.
- Transport logic
  - ✓ Easier to transport alumina than transmit massive electricity over long distances.
- Cryolite role
  - ✓ Mixed with alumina during smelting.
  - ✓ Lowers melting point and improves electrical conductivity.
- **Indian examples:**
  - ✓ NALCO, Angul (Odisha)
    - Uses power from Talcher thermal power plant.
  - ✓ BALCO, Korba (Chhattisgarh)
    - Located near Korba thermal power station.
  - ✓ HINDALCO, Renukoot (Uttar Pradesh)
    - Uses power from Rihand dam and Singrauli thermal plant.
  - ✓ INDAL, Hirakud (Odisha)
    - Uses hydroelectric power from Hirakud dam.

Q 86. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Land has lower specific heat capacity than water.
  - ✓ Therefore, land heats faster than oceans during summer.
  - ✓ This creates high temperature over land, forming a low-pressure area.
  - ✓ This is the basic principle behind summer monsoon formation.
- **Statement 2 is incorrect:**
  - ✓ Intense summer heating causes air to rise, not sink.
  - ✓ Rising air produces a low-pressure zone, not a high-pressure subsidence zone.
  - ✓ Over northwestern India, a thermal low develops, not high pressure.
- **Statement 3 is correct:**
  - ✓ During winter:
    - Land cools faster → high pressure develops over land.
    - Ocean remains warmer → low pressure develops over sea.
  - ✓ Winds move from land to sea, forming the northeast monsoon.

Additional information:

- Specific heat capacity comparison
  - ✓ Water about 4.18 J/g°C
  - ✓ Land about 0.8 to 1.0 J/g°C
  - ✓ Water heats and cools much more slowly than land.
- Monsoon trough
  - ✓ Summer low-pressure belt over northwestern India.
  - ✓ Its shifting controls:
    - Active monsoon periods
    - Break monsoon phases.
- Modern understanding
  - ✓ Differential heating explains basic setup.
  - ✓ Full monsoon system also depends on:
    - ITCZ migration
    - Upper-air circulation
    - Jet streams.

Q 87. Ans. D

Explanation:

- **Statement 1 is correct:**
  - ✓ Koderma–Gaya–Hazaribagh belt is the most famous mica belt in India.
  - ✓ Located along the northern edge of the Chhota Nagpur Plateau.
  - ✓ Historically known for producing high-quality ruby mica.
- **Statement 2 is correct:**
  - ✓ The Jaipur–Ajmer–Bhilwara belt in Rajasthan is a major mica-producing region.
  - ✓ Ajmer serves as an important centre within this belt.
- **Statement 3 is correct:**
  - ✓ The Nellore belt in Andhra Pradesh is one of the largest mica-producing regions in India.
  - ✓ Known for producing greenish mica varieties.
  - ✓ Andhra Pradesh is currently the leading producer of mica in India.

Additional facts:

- Three major mica belts in India
  - ✓ Koderma belt
    - ▪ Jharkhand
    - ▪ Extends into Bihar (Nawada, Jamui).
  - ✓ Rajasthan belt
    - ▪ Jaipur–Ajmer–Bhilwara region.
  - ✓ Nellore belt
    - ▪ Andhra Pradesh
    - ▪ Presently the largest producer.
- Uses of mica:
  - ✓ Electrical insulation.
  - ✓ Electronics and capacitor manufacturing.
  - ✓ Heat-resistant materials.
- Geological occurrence:
  - ✓ Found mainly in pegmatite veins.
  - ✓ Associated with igneous and metamorphic rocks.

Q 88. Ans. B

Explanation:

- **Meghalaya**
  - ✓ Rat-hole mining is mainly practised in Meghalaya, especially in:
    - ▪ Jaintia Hills
    - ▪ Khasi Hills
    - ▪ Garo Hills (limited areas).
  - ✓ It is used because coal seams here are:
    - ▪ Thin
    - ▪ Discontinuous
    - ▪ Difficult for mechanised mining.

Why rat-hole mining is used in Meghalaya:

- Coal type
  - ✓ Meghalaya has tertiary coal deposits.
  - ✓ These seams are usually less than 2 meters thick.
- Mining method
  - ✓ Narrow vertical pits are dug.
  - ✓ Small horizontal tunnels are made to extract coal manually.
  - ✓ Tunnels are very small, resembling rat holes, hence the name.

Additional facts:

- NGT ban (2014)
  - ✓ Rat-hole mining banned by the National Green Tribunal due to:
    - ▪ Frequent accidents and deaths
    - ▪ Severe environmental damage.
- Environmental issue
  - ✓ Causes acid mine drainage:
    - ▪ Sulfur reacts with water and air
    - ▪ Produces sulfuric acid
    - ▪ Pollutes rivers like Kopili and Lukha.

Q 89. Ans. D

Explanation:

- **Pair 1 is correct:** NW-1 : Ganga–Bhagirathi–Hooghly river system
  - ✓ NW-1 runs from Prayagraj to Haldia.
  - ✓ Length about 1,620 km, making it the longest national waterway in India.
  - ✓ Passes through:
    - ▪ Uttar Pradesh
    - ▪ Bihar
    - ▪ Jharkhand
    - ▪ West Bengal.
- **Pair 2 is correct:** NW-2 : Brahmaputra river
  - ✓ NW-2 runs from Dhubri to Sadiya in Assam.
  - ✓ Length about 891 km.
  - ✓ Important transport route for north-eastern India.
- **Pair 3 is correct:** NW-4 : Krishna and Godavari river system

- ✓ NW-4 connects stretches of:
  - Godavari river
  - Krishna river
  - Coastal canal networks.
- ✓ Passes through:
  - Telangana
  - Andhra Pradesh
  - Tamil Nadu
  - Puducherry.

Additional facts:

- NW-1 → Ganga–Bhagirathi–Hooghly
- NW-2 → Brahmaputra
- NW-3 → West Coast Canal (Kerala)
- NW-4 → Krishna–Godavari system
- NW-5 → Brahmani–Mahanadi delta
- NW-6 → Barak river (Assam)
- National Waterways Act, 2016
  - ✓ Declared 111 national waterways in India.
- Inland Waterways Authority of India (IWAI)
  - ✓ Established in 1986.
  - ✓ Headquarters: Noida.

Q 90. Ans. B

Explanation:

- **Subtropical Westerly Jet (STWJ):**
  - ✓ Season: winter and pre-monsoon
  - ✓ Location: south of Himalayas
  - ✓ Effect: acts as upper-air lid
  - ✓ Suppresses: convection and monsoon onset
- **Monsoon Onset:**
  - ✓ Late May–early June: STWJ shifts north of Himalayas
  - ✓ Region: over Tibetan Plateau
  - ✓ Effect: removal of upper-air barrier
  - ✓ Result: sudden monsoon burst
- **Additional Information**
  - ✓ **Tibetan Plateau Role:**
    - Surface: intense heating: low pressure
    - Upper troposphere: strong high pressure
    - Generates: Tropical Easterly Jet (TEJ)
    - Not linked to STWJ formation
  - ✓ **Oceanic Cyclones:**
    - Level: lower troposphere
    - Driver: latent heat
    - Not controlled by jet streams

Q 91. Ans. C

Explanation:

- **Mango Showers: pre-monsoon thunderstorms**
  - ✓ Season: April–May
  - ✓ Region: Kerala, Coastal Karnataka
  - ✓ **Mechanism:**
    - Intense solar heating: strong convection
    - Warm air rises rapidly
    - Moist sea breeze: Arabian Sea enters
    - Formation: cumulonimbus clouds
    - Result: short, intense rainfall
  - ✓ **Agricultural Role:**
    - Helps early ripening of mangoes
- **Additional Information**
  - ✓ **Western Disturbances:**
    - Type: extra-tropical cyclones
    - Origin: northwest
    - Season: winter
    - Region: North India
  - ✓ **Inter-Tropical Convergence Zone:**
    - Role: monsoon onset driver (June)
    - Not responsible for pre-monsoon thunderstorms
  - ✓ **Retreating Monsoon:**
    - Season: November–December
    - Region: Coromandel Coast (Tamil Nadu)

Bihar Linkage

- **Kalbaisakhi Impact:**
  - ✓ Region: Eastern and Northeastern Bihar
  - ✓ Districts: Purnea, Katihar, Kishanganj
- **Mechanism:**
  - ✓ Heating over Chotanagpur Plateau
  - ✓ Moisture source: Bay of Bengal
  - ✓ Clouds: cumulonimbus

Q 92. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Pumped Storage Plant (PSP): net consumer of electricity
  - ✓ Energy loss: pumping > generation
  - ✓ Efficiency: ~70–80%
  - ✓ Function: energy storage and time-shifting
- **Statement 2 is correct:**
  - ✓ PSP: water-based energy storage system
  - ✓ Off-peak: electricity used to pump water to upper reservoir

- ✓ Peak demand: water released to generate power
- **Statement 3 is correct:**
  - ✓ Srisailam Left Bank Power Station: Pumped Storage Plant
  - ✓ Capacity: 900 MW
  - ✓ River: Krishna
- **Additional Information:**
  - ✓ **Working Principle:**
    - Closed-loop system: stores and releases energy
    - Purpose: grid balancing, not net generation
  - ✓ **Grid Role:**
    - Supports renewable energy integration
    - Solar issue: generation drop during evening peak
    - PSP: supplies peak-time electricity
  - ✓ **Major PSPs in India:**
    - Srisailam: Telangana: Krishna River
    - Purulia: West Bengal: Kistobazar Nala
    - Kadamparai: Tamil Nadu: Anamalai Hills
    - Ghatghar: Maharashtra: Pravara basin
- **Bihar Linkage:**
  - ✓ Region: Kaimur Plateau
  - ✓ Feature: steep escarpment suitable for PSP
  - ✓ Districts: Rohtas, Kaimur

Q 93. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Monsoon Break: northward shift of monsoon trough
  - ✓ Normal position: across the Indo-Gangetic plains
  - ✓ Break phase: shifts to the Himalayan foothills
  - ✓ Effect: rainfall stops over Central and Peninsular India
- **Statement 2 is correct:**
  - ✓ Trough near the Himalayas: strong orographic rainfall
  - ✓ Region: Himalayan slopes and Northeast India
  - ✓ Plains (south of foothills): dry, humid, sunny conditions
- **Statement 3 is incorrect:**
  - ✓ Bay of Bengal depressions cause an active monsoon
  - ✓ Movement: west-northwest along the trough
  - ✓ Effect: pull through southwards
  - ✓ Result: rainfall over Central India
  - ✓ Break phase: absence of depressions

Additional Information

- **Monsoon Trough:**
  - ✓ Low-pressure axis over North India
  - ✓ Linked to: Inter-Tropical Convergence Zone
  - ✓ Nature: north-south oscillation
  - ✓ Result: active-break cycle
- **Active vs Break Phase:**
  - ✓ **Active phase:**
    - Trough: normal position
    - Depressions: frequent
    - Rainfall: Central India, Western Ghats, plains
    - Weather: overcast, widespread rain
  - ✓ **Break phase:**
    - Trough: Himalayan foothills
    - Depressions: absent
    - Rainfall: Himalayas, Northeast, Tamil Nadu coast
    - Weather: dry plains, high humidity

Q 94. Ans. C

Explanation:

- **Zawar: Lead and Zinc**
  - ✓ Oldest zinc smelting site: >2000 years
  - ✓ Current importance: major base metal production
- **Additional Information**
  - ✓ Khetri: Copper
  - ✓ Khetri Copper Belt: since Indus Valley Civilization
  - ✓ Operator: Hindustan Copper Limited
- **Makrana: Marble**
  - ✓ Type: white marble
  - ✓ Used in: Taj Mahal, Victoria Memorial
- **Jhamarkotra: Rock Phosphate**
  - ✓ Status: the largest deposit in India
  - ✓ Use: fertiliser industry
- **Key Facts**
  - ✓ **Lead-Zinc Association:**
    - Polymetallic: occur together
    - Lead ore: Galena
    - Zinc ore: Sphalerite
  - ✓ **Geographical Concentration:**

- Region: Aravalli Range
- State share: >90% of India's lead and zinc resources

✓ **Commercial Operations:**
- Operator: Hindustan Zinc Limited
- Major site: Rampura Agucha Mine

Q 95. Ans. B

Explanation:

- **Statement 1 is correct:**
  - ✓ Agasthyamala Biosphere Reserve: southernmost Western Ghats
  - ✓ Location: Kerala and Tamil Nadu
  - ✓ Kerala districts: Pathanamthitta, Kollam, Thiruvananthapuram
  - ✓ Tamil Nadu districts: Tirunelveli, Kanyakumari
- **Statement 2 is correct:**
  - ✓ Kerala components: Neyyar WLS, Peppara WLS, Shendurney WLS
  - ✓ Also includes: Achencoil, Thenmala, Konni, Punalur, Thiruvananthapuram divisions
  - ✓ Tamil Nadu component: Kalakad Mundanthurai Tiger Reserve
- **Statement 3 is incorrect:**
  - ✓ Latitude: ~8°N: tropical climate
  - ✓ Vegetation: Tropical Wet Evergreen, Moist Deciduous, Shola forests
  - ✓ Alpine meadows: absent
  - ✓ Coniferous taiga: absent
- **Key Facts:**
  - ✓ **Climate and Ecology:**
    - High altitude (~1868 m): no alpine ecosystem due to tropical latitude
    - Unique system: Shola–Grassland complex
  - ✓ **UNESCO Status:**
    - Year: 2016
    - Programme: Man and Biosphere (MAB)
    - India: among 12 biosphere reserves in MAB network
  - ✓ **Tribal Association:**
    - **Tribe: Kani tribe**
- **Medicinal Plant:**
  - ✓ Species: Trichopus zeylanicus
  - ✓ Use: anti-fatigue, energy source
  - ✓ Knowledge source: Kani tribe
- **Cultural Significance:**
  - ✓ Peak: Agasthyamala Peak
  - ✓ Linked to: Agastya

Q 96. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ 1686: Edmund Halley: Classical Thermal Concept of monsoon
  - ✓ Monsoon: modified large-scale land–sea breeze
  - ✓ Summer: land heats faster than ocean: low pressure over land
  - ✓ Ocean: high pressure: winds move toward land
  - ✓ Result: moisture-laden Southwest Monsoon
- **Statement 2 is incorrect:**
  - ✓ Jet streams: 20th century discovery: World War II pilots
  - ✓ Not part of Halley's theory
  - ✓ Halley explained Northeast Monsoon using thermal reversal:
  - ✓ Winter: land cools faster: high pressure over land
  - ✓ Sea: relatively low pressure
  - ✓ Result: dry land-to-sea winds
- **Key Facts:**
  - ✓ Limitations of Thermal Theory:
    - Fails to explain: monsoon burst
    - Fails to explain: break (pulsatory nature)
    - Fails to explain: upper-air circulation
    - Fails to explain: uneven global monsoon distribution
  - ✓ Evolution of Monsoon Theories:
    - Classical Thermal Concept:
      - ◊ Proponent: Edmund Halley: 1686
      - ◊ Mechanism: surface differential heating: land–sea breeze

Q 97. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ The Cold Desert Biosphere Reserve includes:
    - Core zone: Pin Valley National Park
  - ✓ Other components:
    - Chandratal Wildlife Sanctuary

- Sarchu Wildlife Sanctuary
- Kibber Wildlife Sanctuary

- **Statement 2 is incorrect:**
  - ✓ Cold deserts are extremely arid due to rain-shadow effect of the Great Himalayas.
  - ✓ Southwest Monsoon cannot reach this region.
  - ✓ Moisture source: winter snowfall (Western Disturbances)
  - ✓ Vegetation: sparse alpine scrub / dry steppe, NOT rhododendron forests

Additional Information

- Location: Lahaul & Spiti (Himachal Pradesh)
- Flagship species:
  - ✓ Snow Leopard ("Ghost of the Mountains")
  - ✓ Himalayan Ibex
  - ✓ Blue Sheep (Bharal)

Q 98. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Descending air does not cool, it warms due to compression.
  - ✓ The process is dry adiabatic warming, not moist adiabatic cooling.
  - ✓ Moist adiabatic cooling occurs only on the windward side of the Western Ghats, where air rises, expands, and condenses to produce rainfall.
- **Statement 2 is correct:**
  - ✓ After crossing the Western Ghats, winds descend on the leeward side (Deccan Plateau).
  - ✓ Descending air undergoes compression → temperature increases.
  - ✓ Warmer air → higher moisture-holding capacity → relative humidity drops.
  - ✓ Result: Suppressed cloud formation and low rainfall (rain-shadow region of interior Maharashtra).

Q 99. Ans. B

Explanation:

- Phumdis are floating masses of vegetation, soil, and organic matter. They are the defining feature of Loktak Lake.
- The core area of these phumdis forms Keibul Lamjao National Park, which is the world's only floating national park.
- **Additional Information**
  - ✓ **Bhitarkanika National Park**
    - Water-based ecosystem → assumed similar
    - Reality: Mangrove estuarine ecosystem, famous for saltwater crocodiles, not phumdis
  - ✓ **Chilika Lake**
    - Large lake → assumed floating vegetation
    - Reality: Brackish lagoon, known for Irrawaddy dolphins and birds, not phumdis
  - ✓ **Point Calimere Wildlife Sanctuary**
    - Coastal ecosystem with dry evergreen forests and salt swamps, not floating meadows
    - High-Yield Linkages
    - Sangai (Dancing Deer): Endemic to Keibul Lamjao; appears to "dance" due to floating phumdis
    - 20% Rule: ~20% above water, ~80% submerged
  - ✓ **Montreux Record:**
    - Loktak Lake is listed due to ecological threats (e.g., Ithai Barrage)
    - Keoladeo National Park (Rajasthan)
  - ✓ **Bihar Contrast**
    - No natural phumdis in Bihar wetlands (Chaurs, Maans, oxbow lakes like Kanwar)
    - Dominant aquatic system: Makhana (Euryale ferox) cultivation
    - Only National Park: Valmiki National Park
    - Terrestrial Terai forest ecosystem, unlike floating Loktak system

Q 100. Ans. B

Explanation:

- **India's Nuclear Reactor Base**
  - ✓ **Dominant reactor type:**
    - Pressurised Heavy Water Reactors (PHWRs)
  - ✓ **Share:**
    - Total reactors: 23
    - PHWRs: 19
  - ✓ **Key Features of PHWR:**
    - Fuel: Natural uranium (unenriched)

- Moderator + Coolant: Heavy water (D_2O)
- Role: Stage I of nuclear programme

- **Additional Information**
  - ✓ **Boiling Water Reactors (BWRs):**
    - Location: Tarapur
    - Units: 2
    - Fuel: Enriched uranium
    - Nature: Imported technology
  - ✓ **Light Water Reactors (LWRs):**
    - Location: Kudankulam
    - Type: VVER (Russian)
    - Fuel: Enriched uranium
    - Role: Not indigenous base
  - ✓ **Fast Breeder Reactors (FBRs):**
    - Stage: II (future)
    - Example: Kalpakkam (PFBR)
    - Status: Under commissioning
    - **Three-Stage Nuclear Programme:**
- Architect: Homi Jehangir Bhabha
- Stage I:
  - ✓ PHWR → Uranium → Plutonium-239
- Stage II:
  - ✓ FBR → Plutonium → Uranium-233 (from Thorium)
- Stage III:
  - ✓ AHWR → Thorium-232 + Uranium-233
- **Resource Base:**
  - ✓ Uranium: Limited
  - ✓ Locations: Jaduguda (Jharkhand), Tummalapalle (Andhra Pradesh)
  - ✓ Thorium: Abundant
  - ✓ Locations: Kerala, Odisha (monazite sands)
- **Supporting Infrastructure:**
  - ✓ Heavy Water Board: Produces D_2O
  - ✓ Plants: Kota, Manuguru, Thal, Hazira
- **Bihar Link:**
  - ✓ Proposed site: Rajauli
  - ✓ Planned reactors: 700 MWe PHWR units
  - ✓ Current power dependence:
  - ✓ Coal: Barh, Kahalgaon, Nabinagar
  - ✓ Nuclear supply:
  - ✓ Narora (PHWR units via grid)

Q 101. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Dugong: Only fully herbivorous marine mammal
  - ✓ Diet: Seagrass (benthic)
- **Statement 2 is correct:**
  - ✓ Distribution in India: Restricted
  - ✓ Locations:
    - Gulf of Mannar
    - Palk Bay
    - Gulf of Kutch
    - Andaman and Nicobar Islands
- **Statement 3 is incorrect:**
  - ✓ First Dugong Conservation Reserve:
  - ✓ Location: Palk Bay (Tamil Nadu, 2022)
  - ✓ Not in Andaman & Nicobar
- **Key Facts:**
  - ✓ **Conservation status:**
    - IUCN: Vulnerable
    - Legal protection: Schedule I, Wildlife Protection Act 1972
  - ✓ **Ecological role:**
    - Ecosystem engineer: Maintains seagrass health
    - Seagrass: Major Blue Carbon sink
  - ✓ **State animal:**
    - Andaman & Nicobar: Dugong
  - ✓ **Marine mammals (India):**
    - Dugong: Herbivore
    - Irrawaddy dolphin: Carnivore (Chilika, Sundarbans)
- **Bihar Link:**
  - ✓ **Equivalent species:**
    - Ganges river dolphin
    - Habitat: Freshwater (Ganga system)
  - ✓ **Protected area:**
    - Vikramshila Gangetic Dolphin Sanctuary
    - Role: Indicator of river health

Q 102. Ans. B

Explanation:

- **Statement 1 is correct:**
  - ✓ Withdrawal of the Southwest Monsoon begins from northwestern India.
  - ✓ Usual starting region:
    - Western Rajasthan
    - Parts of Punjab and Haryana.
  - ✓ Usual starting time:
    - ◊ Early September.
  - ✓ Withdrawal then progresses south-eastwards across India.
- **Statement 2 is incorrect:**

- ✓ Retreating monsoon winds:
  - Blow from northeast to southwest.
  - Pick up moisture from the Bay of Bengal, not the Arabian Sea.
- ✓ They strike the Tamil Nadu coast as:
  - Northeasterly winds, not westerlies.
- ✓ Therefore, this statement contains multiple factual errors.

Additional facts:

- Retreating monsoon period
  - ✓ October to November.
- Main rainfall region
  - ✓ Tamil Nadu coast receives most rainfall during this phase.
  - ✓ This is called the Northeast Monsoon.
- Direction of winds
  - ✓ Southwest Monsoon: Sea to land
  - ✓ Northeast Monsoon: Land to sea, gaining moisture over Bay of Bengal.
- Withdrawal sequence
  - ✓ Begins: Northwestern India
  - ✓ Ends: Southern peninsula
  - ✓ Complete withdrawal: Early December.

Q 103. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Red Panda is not found in the Western Ghats.
  - ✓ It does not live in tropical rainforests.
  - ✓ In India, it occurs in temperate and sub-alpine forests of the Eastern Himalayas.
  - ✓ Major Indian distribution areas:
  - ✓ Sikkim
  - ✓ Arunachal Pradesh
  - ✓ Northern West Bengal (Darjeeling and Kalimpong)
  - ✓ Parts of Meghalaya.
- **Statement 2 is correct:**
  - ✓ Singalila National Park and Neora Valley National Park in West Bengal are key habitats.
  - ✓ These parks provide:
    - High-altitude forests
    - Dense bamboo vegetation, which is essential food for Red Panda.
  - ✓ Singalila is known for Red Panda conservation and reintroduction programmes.

Additional facts:

- Habitat type
  - ✓ Temperate broadleaf forests
  - ✓ Sub-alpine forests
  - ✓ Bamboo-rich mountain regions.
- Major protected areas in India
  - ✓ Singalila National Park (West Bengal)
  - ✓ Neora Valley National Park (West Bengal)
  - ✓ Khangchendzonga National Park (Sikkim)
  - ✓ Namdapha National Park (Arunachal Pradesh).
- Conservation status
  - ✓ IUCN status: Endangered
  - ✓ Wildlife Protection Act, 1972: Schedule I.
- State animal
  - ✓ Red Panda is the state animal of Sikkim.

Q 104. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ The description given refers to a clustered (nucleated) settlement, not a semi-clustered one.
  - ✓ Features described:
    - Houses closely built together
    - Residential area clearly separated from fields and pastures
  - ✓ These are the defining characteristics of a clustered settlement.
- **Statement 2 is correct:**
  - ✓ Semi-clustered settlements form due to fragmentation of a main village.
  - ✓ Often occurs when:
    - Certain social groups live slightly away from the main settlement.
  - ✓ This pattern is commonly found in Gujarat plains and parts of Rajasthan.

Additional facts:

- Clustered (nucleated) settlements
  - ✓ Houses closely packed together.
  - ✓ Common in fertile alluvial plains such as:
    - Indo-Gangetic plains.
- Semi-clustered settlements
  - ✓ Main settlement with small fragmented groups nearby.
  - ✓ Common in:
    - Gujarat
    - Rajasthan.
- Hamleted settlements

- Village divided into several small units.
- Common in:
  - Middle and lower Ganga plains
  - Chhattisgarh.
- Dispersed settlements
  - Houses widely scattered.
  - Common in:
    - Hilly and forested regions
    - Meghalaya, Himachal Pradesh, Uttarakhand, Kerala.

Q 105. Ans. C

Explanation:

- **Vivipary:**
  - Vivipary means seeds germinate while still attached to the parent plant.
  - The young seedling formed is called a propagule.
  - After growing sufficiently, the propagule:
    - Falls into soft mud and anchors quickly, or
    - Floats in water until reaching a suitable site.
  - Why vivipary is necessary in mangroves
    - Mangroves grow in:
      - Saline conditions
      - Waterlogged soils
      - Oxygen-deficient mud.
    - Ordinary seeds would:
      - Rot
      - Suffocate
      - Be washed away by tides.
    - Vivipary increases survival chances.

Additional facts:

- Vivipary
  - Seed germinates on parent plant.
- Pneumatophores
  - Breathing roots grow upward to absorb oxygen.
- Salt regulation
  - Salt exclusion or salt secretion through leaves.
- Stilt roots
  - Provide support in soft coastal mud.
- Common viviparous mangrove genus:
  - Rhizophora.
- Major mangrove regions in India:
  - Sundarbans (largest mangrove forest in India)
  - Bhitarkanika (Odisha)
  - Pichavaram (Tamil Nadu)
  - Coringa (Andhra Pradesh)
  - Gulf of Kutch (Gujarat).

Q 106. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - The Deccan Traps were not formed by explosive eruptions.
  - They were formed by fissure eruptions, where lava flowed quietly from long cracks.
  - Explosive eruptions usually form:
    - Volcanic cones
    - Ash deposits
  - Deccan Traps instead formed broad lava plateaus, not cone-shaped volcanoes.
- **Statement 2 is correct:**
  - The Deccan Traps consist mainly of layered basaltic lava flows.
  - These eruptions occurred around the Cretaceous–Paleogene boundary, about 66 million years ago.
  - Repeated lava flows formed:
    - Thick horizontal basalt layers
    - Step-like plateau structure.

Additional facts:

- Type of eruption
  - Fissure eruption, not central volcanic eruption.
- Rock type
  - Mainly basalt, an igneous rock.
- Meaning of "Traps"
  - Derived from Swedish word trappa, meaning stairs, referring to step-like topography.
- Geological age
  - Formed near the end of the Mesozoic Era, at the Cretaceous–Paleogene boundary.
- Soil formed
  - Weathering of basalt formed black soil (regur).
  - Suitable for cotton cultivation.
- Geographical spread
  - Major areas:
    - Maharashtra
    - Gujarat
    - Madhya Pradesh
    - Parts of Karnataka and Telangana.

- Cause
  - ✓ Linked to Réunion hotspot activity beneath the Indian plate.

Q 107. Ans. C

Explanation:

- **Correct statement: Its rapid southward shift and weakening help in the development of anti-cyclonic conditions over northwestern India.**
  - ✓ During late monsoon season (September onwards):
    - The Tropical Easterly Jet (TEJ) weakens.
    - It shifts southward as the Tibetan Plateau cools.
  - ✓ This leads to:
    - Reduced upper-air divergence
    - Air subsidence over northwestern India
    - Formation of high-pressure (anti-cyclonic) conditions.
  - ✓ These conditions initiate the withdrawal of the Southwest Monsoon, beginning from northwestern India.

Additional facts:

- Tropical Easterly Jet (TEJ)
  - ✓ Active during summer monsoon.
  - ✓ Weakening helps monsoon withdrawal.
- Sub-Tropical Westerly Jet (STWJ)
  - ✓ Present over India in winter.
  - ✓ Moves north of Himalayas during monsoon onset.
- Somali Jet
  - ✓ A low-level jet.
  - ✓ Strengthens Arabian Sea branch of the monsoon.

Q 108. Ans. A

Explanation:

- **Assertion (A) is true:**
  - ✓ The Satpura Range is a classic example of a block mountain (horst).
  - ✓ Block mountains form due to faulting and vertical movement of crustal blocks.
- **Reason (R) is true and correct explanation:**
  - ✓ The Satpura Range formed between two major rift valleys:
    - Narmada rift valley to the north
    - Tapi rift valley to the south.
  - ✓ During tectonic activity:
    - Adjacent blocks subsided to form grabens (rift valleys).
    - The land between them remained elevated, forming a horst, which is the Satpura Range.
  - ✓ Thus, the mechanism described in Reason directly explains why the Satpura is a block mountain.

Additional facts:

- Horst
  - ✓ Elevated block of land between two faults.
  - ✓ Example: Satpura Range.
- Graben
  - ✓ Sunken block between faults.
  - ✓ Examples:
    - Narmada valley
    - Tapi valley.
- Parallel mountain system
  - ✓ Vindhya Range lies north of the Satpura Range.
  - ✓ Narmada river flows between Vindhya and Satpura.
- Highest peak
  - ✓ Dhupgarh
  - ✓ Located near Pachmarhi in Madhya Pradesh.
- River flow pattern
  - ✓ Narmada and Tapi flow westward into the Arabian Sea, unlike most peninsular rivers.

Q 109. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Narmada and Tapi do not form deltas.
  - ✓ They form estuaries before entering the Arabian Sea.
  - ✓ Reason:
    - They carry less sediment.
    - Strong tidal action and deep channels prevent deposition.
- **Statement 2 is correct**
  - ✓ Many west-flowing rivers like Narmada and Tapi flow through:
    - Hard rocky terrain
    - Rift valleys (fault lines).
  - ✓ Because of hard rocks:
    - Less erosion occurs
    - Rivers carry very little alluvial sediment.
  - ✓ Lack of sediment prevents delta formation.

- **Statement 3 is correct:**
  - ✓ Rivers originating from the Western Ghats have:
    - Short course
    - Steep gradient
    - High velocity flow.
  - ✓ High velocity carries sediments directly into the sea.
  - ✓ This prevents slow sediment deposition, so deltas do not form.

Additional facts:

- Delta
  - ✓ Formed by deposition of sediments.
  - ✓ Requires:
    - Gentle slope
    - Slow river flow
    - Heavy sediment load.
  - ✓ Examples:
    - Godavari
    - Krishna
    - Mahanadi
    - Ganga-Brahmaputra.
- Estuary
  - ✓ Funnel-shaped river mouth.
  - ✓ Little or no sediment deposition.
  - ✓ Common in west-flowing rivers.
- Major west-flowing rivers forming estuaries
  - ✓ Narmada
  - ✓ Tapi
  - ✓ Mahi
  - ✓ Sabarmati
  - ✓ Periyar
  - ✓ Mandovi.
- East-flowing rivers
  - ✓ Longer courses.
  - ✓ Gentle slopes.
  - ✓ Form large deltas on the East Coast.

Q 110. Ans. D

Explanation:

- **Statement 1 is incorrect**
  - ✓ Fringing reefs are widely found in the Andaman and Nicobar Islands.
  - ✓ India does not have major barrier reefs like those of Australia.
  - ✓ Major reef types in India:
    - Fringing reefs →Andaman and Nicobar, Gulf of Mannar, Gulf of Kutch
    - Atolls →Lakshadweep.
  - ✓ Therefore, saying fringing reefs are absent in Andaman and Nicobar is incorrect.
- **Statement 2 is incorrect:**
  - ✓ The main cause of coral bleaching is increase in sea surface temperature (SST).
  - ✓ Coral bleaching occurs when:
    - High temperature stresses corals
    - They expel zooxanthellae algae
    - Coral turns white and may die.
  - ✓ Reduced salinity due to freshwater inflow can cause local stress, but it is not the main cause of bleaching in the Gulf of Mannar.

Additional facts:

Major coral reef regions in India

- Andaman and Nicobar Islands
  - ✓ Mainly fringing reefs.
- Lakshadweep Islands
  - ✓ Mostly atolls.
- Gulf of Mannar
  - ✓ Fringing reefs along Tamil Nadu coast.
- Gulf of Kutch
  - ✓ Fringing reefs in extreme tidal conditions.
- Types of coral reefs
  - ✓ Fringing reefs
    - Grow close to shore.
    - Most common in India.
  - ✓ Barrier reefs
    - Parallel to coast but separated by wide lagoon.
    - Rare in India.
  - ✓ Atolls
    - Circular reef enclosing lagoon.
    - Example: Lakshadweep.
- Ideal temperature for coral growth:
  - ✓ About 20°C to 28°C.
- Major global trigger of bleaching:
  - ✓ El Niño years often raise ocean temperatures.
- Zooxanthellae:
  - ✓ Symbiotic algae that give coral:
    - Food
    - Colour.

Q 111. Ans. C

Explanation:

- **Correct statement: Strong summer upwelling cools the Somali coast, which strengthens the cross-equatorial pressure gradient and increases the strength of monsoon winds.**

- ✓ During summer, the Somali Current flows northward along the Somali coast.
- ✓ Strong southwest monsoon winds cause:
  - Coastal upwelling
  - Cold, nutrient-rich water rising to the surface.
- ✓ This cooling creates:
  - High pressure over western Arabian Sea
  - Strong pressure gradient toward low pressure over India.
- ✓ This strengthens the Somali Jet, which intensifies the Southwest Monsoon.

Additional facts:

- Somali Current
  - ✓ Unique current that reverses direction seasonally.
  - ✓ Summer: Cold current, flows northward.
  - ✓ Winter: Warm current, flows southward.
- Somali Jet (Findlater Jet)
  - ✓ Low-level jet stream across Arabian Sea.
  - ✓ Helps push moisture-laden winds toward India.
- Upwelling effect
  - ✓ Brings cold water to surface.
  - ✓ Strengthens pressure gradient and monsoon circulation.
- Indian Ocean uniqueness
  - ✓ Only ocean where major currents reverse seasonally due to monsoon winds.
- Impact on fisheries
  - ✓ Upwelling increases nutrients.
  - ✓ Leads to rich fishing zones along the Somali coast.

Q 112. Ans. B

Explanation:

Correct option: Negrito

- ✓ The Jarawa, Sentinelese, and Onge tribes of the Andaman Islands belong to the Negrito ethnolinguistic group.
- ✓ Key physical characteristics:
  - Short stature
  - Dark complexion
  - Frizzy or curly hair.
- ✓ These tribes are among the earliest human settlers believed to have migrated from Africa about 60,000 years ago.

- **Mongoloid**
  - ✓ Found mainly in:
    - Northeast India
    - Himalayan region.
  - ✓ In the Andaman and Nicobar region:
    - Nicobarese and Shompen tribes belong to the Mongoloid group, not Andaman tribes.
- **Australoid**
  - ✓ Common among tribes of:
    - Central and peninsular India.
  - ✓ Examples:
    - Santhal
    - Munda
    - Gond
    - Bhil.
- **Caucasoid**
  - ✓ Associated mainly with:
    - Many populations of North and West India.
  - ✓ Not linked to indigenous Andaman tribes.

Additional facts:

- Andaman tribes (Negrito group)
  - ✓ Jarawa
  - ✓ Sentinelese
  - ✓ Onge
  - ✓ Great Andamanese.
- Nicobar tribes (Mongoloid group)
  - ✓ Nicobarese
  - ✓ Shompen.
- Particularly Vulnerable Tribal Groups (PVTGs) in this region
  - ✓ Great Andamanese
  - ✓ Jarawa
  - ✓ Onge
  - ✓ Sentinelese
  - ✓ Shompen.

Q 113. Ans. C

Explanation:

- **Incorrect statement: The strong thermal low over the land pulls the Arabian Sea branch inland and causes the monsoon onset in Bihar.**
  - ✓ Monsoon onset in Bihar is not caused by the Arabian Sea branch.
  - ✓ Bihar mainly receives monsoon from the Bay of Bengal branch.
  - ✓ The Bay of Bengal branch:
    - Moves north-eastward
    - Gets deflected by the Himalayas and Arakan Yoma

    - Then flows westward across the Gangetic plains, bringing rain to Bihar.
- Arabian Sea branch brings monsoon to Kerala around 1 June
  - ✓ This marks the official onset of monsoon in India.
  - ✓ Kerala is the first mainland region to receive monsoon rainfall.
- Deflection by eastern Himalayas directs winds toward Bihar
  - ✓ The Himalayas and Arakan Hills force winds westward.
  - ✓ This allows the Bay of Bengal branch to reach Bihar by mid-June.
- Monsoon reaches Bihar later than Kerala
  - ✓ Kerala lies directly in the monsoon path.
  - ✓ Bihar receives rain after winds:
    - Travel across Bay of Bengal
    - Turn westward into northern plains.

Key timeline:

- Kerala onset: Around 1 June
- West Bengal: Around 7–10 June
- Bihar: Around 10–15 June
- Entire North India: By late June to early July

Additional facts:

- Two main branches of Southwest Monsoon
  - ✓ Arabian Sea branch →Western Ghats and western India
  - ✓ Bay of Bengal branch →Eastern India and Gangetic plains.
- Rainfall pattern in Bihar
  - ✓ Decreases from east to west due to gradual loss of moisture.

Q 114. Ans. B

Explanation:

- **Bengaluru is known as the "Silicon Valley of India."**
  - ✓ Reason:
    - It is India's largest Information Technology (IT) hub.
    - Hosts major IT companies, startups, and research institutions.
    - Major IT exports originate from this city.

Additional facts:

- Hyderabad
  - ✓ Major IT hub with HITEC City.
  - ✓ Often called "Cyberabad", not Silicon Valley of India.
- Pune
  - ✓ Known for:
    - Education institutions
    - Automobile industry.
  - ✓ Nickname: "Oxford of the East."
- Kolkata
  - ✓ Known as:
    - "Cultural Capital of India"
    - "City of Joy."
- Major IT hubs in India
  - ✓ Bengaluru → Silicon Valley of India
  - ✓ Hyderabad → Cyberabad
  - ✓ Pune → Oxford of the East
  - ✓ Chennai → Detroit of India (automobile hub)
  - ✓ Gurugram and Noida → Major NCR IT centres.

Q 115. Ans. B

Explanation:

- **Assertion (A) is correct:**
  - ✓ In 2023, the Ministry of Mines released India's first official list of 30 critical minerals.
  - ✓ The list includes minerals such as:
    - Phosphorus
    - Potash
    - Silicon
    - Tin.
  - ✓ Therefore, the statement that these minerals are included in the list is factually correct.
- **Reason (R) is correct:**
  - ✓ Many critical minerals are essential for clean energy technologies, such as:
    - Solar photovoltaic panels
    - Wind turbines
    - Electric vehicle batteries.
  - ✓ Examples:
    - Lithium, cobalt, nickel → battery technology
    - Silicon → solar cells
    - Rare earth elements → magnets in wind turbines and EV motors.
  - ✓ Hence, the reason is scientifically correct.
- **Why R is not the correct explanation of A**
  - ✓ The inclusion of phosphorus and potash in India's critical minerals list is mainly due to agricultural importance, not clean energy needs.

- ✓ These minerals are:
  - Essential components of fertilizers
  - Critical for food security
  - Largely import-dependent for India.
- ✓ Therefore:
  - R explains the energy-related importance of some critical minerals.
  - But it does not explain the agricultural rationale behind including phosphorus and potash.
- ✓ Hence, R does not fully explain A.

Additional facts:

- India's 30 critical minerals:
  - ✓ Battery and clean energy minerals
    - Lithium, Cobalt, Nickel, Graphite, Vanadium, Rare Earth Elements (REE), Silicon, Gallium, Germanium, Tellurium
  - ✓ Fertilizer-related minerals (agriculture)
    - Phosphorus, Potash
  - ✓ Electronics and high-tech minerals
    - Indium, Hafnium, Niobium, Tantalum, Beryllium, Selenium, Strontium
  - ✓ Alloy, defence and industrial metals
    - Antimony, Bismuth, Cadmium, Copper, Molybdenum, Rhenium, Tin, Titanium, Tungsten, Zirconium
- Definition of critical minerals:
  - ✓ Minerals that are:
    - Essential for economic development
    - Important for national security
    - Vulnerable to supply disruption.
- Two major reasons minerals are declared critical:
  - ✓ Clean energy and technology dependence
  - ✓ Agricultural and industrial dependence.

Q 116. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Transhumance is not a permanent one-way movement.
  - ✓ It refers to seasonal and cyclical movement of pastoral groups.
  - ✓ Movement pattern:
    - Summer: Toward high-altitude pastures
    - Winter: Toward lower valleys or plains.
  - ✓ Therefore, describing it as a permanent one-way migration is incorrect.
- **Statement 2 is correct:**
  - ✓ The Bakarwal community of Jammu and Kashmir practices transhumance.
  - ✓ Seasonal movement pattern:
    - Summer: Move to high mountain pastures (alpine meadows).
    - Winter: Move to lower valleys and foothills due to snowfall in high regions.
  - ✓ Hence, this statement correctly describes their lifestyle.

Additional facts:

- Meaning of transhumance
  - ✓ Seasonal migration of pastoral communities.
  - ✓ Movement between fixed summer and winter grazing grounds.
  - ✓ Common in mountain regions.
- Important transhumant communities in India
  - ✓ Bakarwals → Jammu and Kashmir
  - ✓ Gujjars → Jammu and Kashmir and Himachal Pradesh
  - ✓ Gaddis → Himachal Pradesh
  - ✓ Bhotiyas → Uttarakhand
  - ✓ Changpas → Ladakh (known for Pashmina goats).
- Important related terms
  - ✓ Bugyal → High-altitude alpine meadow in Uttarakhand
  - ✓ Marg → Alpine pasture in Kashmir
    - Example: Gulmarg, Sonamarg.

Q 117. Ans. D

Explanation:

- **Bewar:**
  - ✓ Local name of shifting cultivation in Madhya Pradesh.
  - ✓ Practised by tribal groups such as the Baiga community.
- **Podu:**
  - ✓ Local name used in Andhra Pradesh and Telangana (also parts of Odisha).
  - ✓ A well-known term in eastern ghats tribal regions.
- **Ponam (or Ponum):**
  - ✓ Local term used in Kerala / Western Ghats hill regions.
  - ✓ Often associated with the same regional group as Kumari cultivation.
- **Other Indian regional names:**

- ✓ North-East India: Jhumming (Jhum)
- ✓ Madhya Pradesh: Bewar or Dahiya
- ✓ Andhra Pradesh / Telangana: Podu or Penda
- ✓ Odisha: Koman, Bringa
- ✓ Western Ghats: Kumari
- ✓ Kerala hills: Ponam (Ponum)
- ✓ Jharkhand: Kuruwa
- ✓ Manipur: Pamlou
- ✓ Bastar (Chhattisgarh) & Andaman Islands: Dipa
- ✓ South-eastern Rajasthan: Valre or Waltre
- ✓ Himalayan region: Khil

Q 118. Ans. B

Explanation:

- **Statement 1 is correct:**
  - ✓ During El Niño, Pacific Walker circulation weakens.
  - ✓ Warm water and convection shift eastward toward central/eastern Pacific (near South America).
  - ✓ This is the core defining feature of El Niño.
- **Statement 2 is correct:**
  - ✓ Warming near the Peruvian coast shifts convection eastward.
  - ✓ This produces subsiding air and relatively higher pressure over Indonesia and nearby regions, weakening the monsoon circulation over India.
  - ✓ Hence, El Niño years often correlate with weak monsoon or drought conditions.
- **Statement 3 is incorrect:**
  - ✓ During La Niña, low pressure strengthens over western Pacific (Indonesia), not central Pacific.
  - ✓ This enhances monsoon circulation, generally leading to above-normal rainfall in India, not reduced rainfall.

Additional facts:

- El Niño
  - ✓ Warm eastern Pacific
  - ✓ Weak Walker circulation
  - ✓ Weak Indian monsoon (often drought risk)
- La Niña
  - ✓ Cold eastern Pacific
  - ✓ Strong Walker circulation
  - ✓ Strong Indian monsoon (often heavy rainfall)
- Key trigger
  - ✓ Shift of convection from west Pacific → east Pacific = El Niño
  - ✓ Strengthening over west Pacific = La Niña

Q 119. Ans. A

Explanation:

- Mahawat (or Mawat)
  - ✓ Refers to the light winter rainfall received during December to February.
  - ✓ Occurs mainly in:
    - Punjab
    - Haryana
    - Delhi
    - Western Uttar Pradesh
    - Parts of Rajasthan

Q 120. Ans. C

Explanation:

- **Khasi language → Austroasiatic (Mon–Khmer branch)**
  - ✓ Khasi belongs to the Austroasiatic language family, specifically the Mon–Khmer branch.
  - ✓ This makes it linguistically distinct from most other northeastern tribal languages, which are largely Sino-Tibetan.
  - ✓ Khasi shares distant linguistic links with languages of Southeast Asia.
- **Sino-Tibetan:**
  - ✓ Most northeastern tribal languages belong to this family.
  - ✓ However, Khasi is an exception.
  - ✓ Example: Garo language in Meghalaya belongs to Sino-Tibetan, but Khasi does not.
- **Dravidian:**
  - ✓ Dravidian languages dominate South India.
  - ✓ Examples: Tamil, Telugu, Kannada, Malayalam.
  - ✓ No major Dravidian linguistic presence in Meghalaya.
- **Indo-Aryan:**
  - ✓ Dominant across North and East India.
  - ✓ Examples: Hindi, Bengali, Assamese.
  - ✓ Khasi is structurally unrelated to Indo-Aryan languages.

Additional facts:

Austroasiatic languages in India:
Two major branches:

- Mon–Khmer branch
  - ✓ Khasi → Meghalaya
  - ✓ Nicobarese → Andaman and Nicobar Islands
- Munda branch
  - ✓ Santhali → Jharkhand, Odisha, West Bengal
  - ✓ Mundari → Jharkhand region
  - ✓ Ho → Jharkhand and Odisha

Important cultural fact:

- Khasi society is matrilineal
  - ✓ Lineage follows the mother's line.
  - ✓ The youngest daughter (Khadduh) inherits ancestral property.

Q 121. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ The Western Ghats and Meghalaya Plateau act as orographic barriers to southwest monsoon winds.
  - ✓ Moist air is forced to rise, cools adiabatically, condenses, and produces heavy rainfall on windward slopes.
  - ✓ This explains very heavy rainfall in regions like:
    - ▪ Windward side of Western Ghats
    - ▪ Southern slopes of Khasi hills (e.g., Mawsynram, Cherrapunji).
- **Statement 2 is incorrect:**
  - ✓ Rainfall does not always increase up to the highest peaks.
  - ✓ Rainfall increases only up to a level of maximum precipitation, after which it declines due to moisture depletion.
  - ✓ Highest peaks often receive less rainfall than mid-elevation slopes.

Additional facts:

- Level of maximum precipitation
  - ✓ Rainfall peaks at mid-elevations, not at the highest summit.
- Classic Indian examples
  - ✓ Mawsynram and Cherrapunji (~1300–1400 m) receive extremely high rainfall.
  - ✓ Shillong Peak (~1960 m) receives comparatively less rainfall.
- Rain shadow effect
  - ✓ Leeward slopes become dry due to descending warm air.
  - ✓ Example:
    - ▪ Mumbai (windward) → heavy rainfall
    - ▪ Pune (leeward) → much lower rainfall

Q 122. Ans. A

Explanation:

- **Correct answer: Panna, Para, Palli, Nagla, Dhani**
  - ✓ These are the local names for hamleted settlement units in different regions of India.
  - ✓ Regional mapping:
    - ▪ Panna / Para / Palli → West Bengal, Odisha, Jharkhand, parts of Bihar
    - ▪ Nagla → Uttar Pradesh
    - ▪ Dhani → Rajasthan (especially arid regions)

Type of Settlement	Defining Characteristic	Primary Geographical Regions	Main Driving Factor
Clustered (Nucleated)	Houses tightly packed together; distinct separation between residential and farm areas.	Punjab, Haryana, Upper Ganga Plain, Northeastern valleys.	Defense/Security, shared water sources (ponds/wells), highly fertile land.
Semi-Clustered	A large clustered village where a specific group has been forced to live slightly outside the main boundary.	Gujarat plains, parts of Rajasthan.	Segregation from an originally compact village.
Hamleted	One village fragmented into multiple independent clusters spread across the land.	Middle/Lower Ganga Plains (Bihar, WB), Chhattisgarh, Lower Himalayas.	Social, ethnic, and caste-based stratification.
Dispersed (Isolated)	Single isolated huts or very small hamlets separated by large distances.	Meghalaya, Uttarakhand, Himachal Pradesh, Kerala.	Highly fragmented terrain, dense forests, or individual hillside farms.

Q 123. Ans. A

Explanation:

- **Umiam Lake**
  - ✓ Also Known As:
    - Barapani ("big water")
  - ✓ Type:
    - Artificial reservoir
  - ✓ **Formation:**
    - River: Umiam River
    - Period: early 1960s
  - ✓ **Purpose:**
    - Hydroelectric power generation
  - ✓ **Project:**
    - **Umiam-Umtru Hydroelectric Project**
    - **Significance: first hydroelectric project in Northeast India**
- **Key Facts**
  - ✓ **Physiography:**
    - Located on Shillong Plateau
  - ✓ **Regional Comparisons:**
    - Loktak Lake: largest freshwater lake NE India
    - Feature: phumdis, floating national park
- **Bihar Linkage**
  - ✓ Natural Lakes:
    - Type: oxbow lakes (maans/chaurs)
  - ✓ Example:
    - Kanwar Jheel: largest freshwater oxbow lake Asia, Ramsar site
  - ✓ Artificial Lakes:
    - Ghora Katora: Rajgir (Nalanda)

Q 124. Ans. D

Explanation:

- **Statement 1 is correct:**
  - ✓ Formation: intense tropical weathering (leaching)
  - ✓ Process: removal of silica + bases
  - ✓ Residue: aluminium hydroxides (gibbsite, boehmite, diaspore)
- **Statement 2 is correct:**
  - ✓ Laterite: iron-rich residual soil/rock
  - ✓ Bauxite: aluminous laterite
  - ✓ Common origin: same weathering process
- **Statement 3 is correct:**
  - ✓ Types: lateritic and karstic bauxite
  - ✓ Lateritic share: ~86–88% (dominant globally)
- **Key Facts**
  - ✓ **Nature:**
    - Bauxite: secondary (residual) ore
  - ✓ **Processing:**
    - Bayer process: bauxite → alumina (Al_2O_3)
  - ✓ **Parent Rocks (India):**
    - Deccan Trap basalts
    - Khondalites (Eastern Ghats)
- **Distribution (India):**
  - ✓ Odisha: >50% reserves
  - ✓ Major deposit: Panchpatmali (Koraput)

Q 125. Ans. A

Explanation:

- **Renewable Energy Installed Capacity (India)**
  - ✓ Solar Power: >80 GW
  - ✓ Wind Power: ~45–48 GW
  - ✓ Bio Power: ~11 GW
  - ✓ Small Hydro (<25 MW): ~5 GW
- **Key Facts**
  - ✓ **Classification:**
    - Small Hydro: <25 MW (MNRE classification)
    - Large Hydro: counted as renewable since 2019
  - ✓ **Trend Shift:**
    - Solar overtook Wind: ~2021
    - Reason: falling solar costs + policy push
  - ✓ **Geographic Pattern:**
    - Solar: widespread (Rajasthan, Gujarat, Karnataka)
    - Wind: localized (Tamil Nadu, Gujarat, Maharashtra)
    - Bio Power: agri-states (UP, Maharashtra, Punjab)
    - Small Hydro: Himalayan states

Q 126. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Describes Western Disturbances
  - ✓ Season: winter
  - ✓ Region: NW India
  - ✓ Not related to Kalbaisakhi
- **Statement 2 is correct:**
  - ✓ Origin: Chotanagpur Plateau
  - ✓ Cause: intense surface heating → low pressure
  - ✓ Moisture source: Bay of Bengal
  - ✓ Type: convective thunderstorms

- **Key Facts**
  - ✓ Nature:
    - Kalbaisakhi: pre-monsoon (April–May)
    - Type: violent squalls, heavy rain, hail
  - ✓ Direction:
    - Cloud movement: northwest → southeast
    - Name: Nor'westers (from Bengal perspective)
  - ✓ Agricultural Impact:
    - Beneficial: jute, tea, aus rice
    - Destructive: hail, strong winds

Bihar Linkage

  - ✓ Affected Region:
    - Seemanchal: Purnea, Katihar, Kishanganj, Araria
  - ✓ Positive Impact:
    - Supports jute cultivation
  - ✓ Negative Impact:
    - Damages maize, mango orchards

Q 127. Ans. C

Explanation:

- **Zaid: short intermediate cropping season**
  - ✓ Between: Rabi and Kharif
  - ✓ **Time Period:**
    - Sowing: March–May
    - Harvest: May–June
  - ✓ **Climatic Features:**
    - Season: peak summer
    - Condition: hot, dry
    - Day length: long
  - ✓ **Water Requirement:**
    - Needs: assured irrigation or residual moisture
    - Sources: tube wells, canals, riverbeds
  - ✓ **Major Crops:**
    - Cucurbits: watermelon, muskmelon, cucumber
    - Vegetables: short-duration crops
    - Fodder: summer fodder crops
- **Key Facts**
  - ✓ **Cropping Seasons:**
    - **Kharif:**
      - ◊ Sowing: June–July
      - ◊ Harvest: Sept–Oct
      - ◊ Crops: rice, jute, cotton, maize
    - **Rabi:**
      - ◊ Sowing: Oct–Dec
      - ◊ Harvest: April–May
      - ◊ Crops: wheat, gram, mustard
    - **Zaid:**
      - ◊ Sowing: March–May
      - ◊ Harvest: May–June
      - ◊ Crops: melons, vegetables, fodder
  - ✓ **Economic Role:**
    - Short-duration cash crops
    - Provides interim income before Kharif
- **Bihar Linkage**
  - ✓ Local Name: Garma season
  - ✓ Major Crops:
    - Garma maize
    - Moong
    - Summer vegetables
  - ✓ Region:
    - Diara lands: Ganga, Gandak, Kosi floodplains
  - ✓ Bihar Cropping Cycle:
    - Bhadai: May–June → Aug–Sept (autumn crops)
    - Aghani: June–July → Nov–Dec (main paddy)
    - Rabi: Oct–Nov → Mar–Apr (wheat, pulses)
    - Garma: March → May–June (Zaid crops)

Q 128. Ans. C

Explanation:

- **Puga Valley**
  - ✓ India's most promising geothermal field
  - ✓ Type: high-enthalpy geothermal system
  - ✓ **Project:**
    - Agency: Oil and Natural Gas Corporation
    - Project: India's first geothermal power plant
    - Capacity: 1 MW (pilot)
    - Output: baseload electricity (24×7)
  - ✓ **Mechanism:**
    - Source: subsurface steam
    - Process: steam → turbines → electricity
- **Key Facts**
  - ✓ **Geological Basis:**
    - Location: near Indus-Tsangpo Suture Zone
    - Cause: Indian–Eurasian plate collision
    - Effect: magma heating + fractures → hot springs, steam
  - ✓ **Energy Advantage:**

- Geothermal: continuous (baseload)
- Solar/Wind: intermittent

✓ **Other Renewable Context (Ladakh):**
- Solar: high insolation (Hanle, Pangong region)
- Wind: strong but variable
- Tidal: not possible (landlocked region)

✓ **Major Geothermal Sites (India):**
- Puga Valley: Ladakh: high enthalpy
- Chumathang: Ladakh: medium-high
- Manikaran: Himachal Pradesh: medium
- Tapovan: Uttarakhand: medium

- **Bihar Linkage**
  - ✓ Type: low-enthalpy geothermal
  - ✓ Examples:
    - Rajgir: Nalanda district
    - Munger: Kharagpur Hills
  - ✓ Temperature: ~60°C–87°C
  - ✓ Use: tourism, balneotherapy
  - ✓ Not suitable: commercial power generation
- Bihar springs: stable crust, low heat → no power generation

Q 129. Ans. A

Explanation:

- **Assertion (A) is true:**
  - ✓ India is currently experiencing a demographic dividend phase.
  - ✓ A demographic dividend occurs when:
    - Birth rates decline
    - The share of the working-age population increases
    - The dependent population share decreases
  - ✓ This creates favourable conditions for accelerated economic growth.
- **Reason (R) is true:**
  - ✓ Population structure is divided into:
    - Working-age population: 15–59 years (sometimes 15–64 globally)
  - ✓ Dependent population:
    - ◊ Children (0–14 years)
    - ◊ Elderly (60+ years)
- **Why R explains A:**
  - ✓ When working-age population increases relative to dependents:
    - More people earn income
    - Fewer dependents rely on them
    - Savings, investment, and productivity increase
  - ✓ This structural shift creates the economic advantage called demographic dividend.

Additional information:

- India's demographic dividend window
  - ✓ Started: Around 2005–06
  - ✓ Expected to last until: Around 2055–56
  - ✓ Current stage: Near peak phase
  - ✓ Working-age share: Over 64% of total population
- NFHS-5 demographic indicator:
  - ✓ Total Fertility Rate (TFR): 2.0
  - ✓ Replacement level fertility: 2.1
  - ✓ Falling TFR leads to:
    - Smaller child population
    - Increasing working-age share
    - Expansion of dividend phase

Q 130. Ans. B

Explanation:

- **Loo: hot, dry, westerly/north-westerly wind**
  - ✓ Origin: Thar Desert and Balochistan
  - ✓ Nature: strong, dusty, high temperature
  - ✓ **Movement:**
    - Direction: west → east
    - Region: Indo-Gangetic Plains
  - ✓ **Modification Eastward:**
    - Humidity increases (Bay of Bengal influence)
  - ✓ **Effect:**
    - Temperature intensity ↓
    - Dryness ↓
    - Impact weaker in eastern India
  - ✓ **Key Characteristics:**
    - Type: advection (horizontal heat transfer)
    - Season: pre-monsoon (April–June)
    - Temperature: often > 45°C
  - ✓ **Ecological Impact:**
    - Deciduous trees: leaf shedding
    - Reason: reduce transpiration
  - ✓ **Atmospheric Interaction:**
    - Loo + moist easterlies → instability
    - Result: thunderstorms (Kalbaisakhi)
- **Bihar-Specific Pattern**
  - ✓ **Western Bihar:**
    - Districts: Buxar, Bhojpur, Rohtas, Kaimur, Aurangabad

- Condition: strong Loo
    - Temp: 44–45°C
    - Humidity: very low
- ✓ **Eastern Bihar (Seemanchal):**
    - Districts: Purnea, Katihar, Kishanganj
    - Condition: Loo weak/absent
    - Dominant: humid winds + Kalbaisakhi
    - Humidity: high

Q 131. Ans. D

Explanation:

- **Statement 1 is correct:**
  - ✓ Luni: ephemeral inland drainage river
  - ✓ Termination: Rann of Kutch
  - ✓ Does not reach Arabian Sea
- **Statement 2 is correct:**
  - ✓ Origin: Aravalli Range near Pushkar valley (Rajasthan)
  - ✓ Initial name: Sagarmati
  - ✓ After confluence with Saraswati at Govindgarh: becomes Luni
  - ✓ Upper course water: fresh
- **Statement 3 is correct:**
  - ✓ Name origin: Lavanavati (salt river)
  - ✓ Salinity: develops downstream
  - ✓ Point: after Balotra (Barmer district)
  - ✓ Reason: flow over salt-rich soils
  - ✓ Final water: highly saline till Rann of Kutch
- **Key Facts**
  - ✓ **Drainage Type:**
    - Inland drainage: river disappears in marsh/desert
    - Example: Luni → Rann of Kutch
  - ✓ **Tributaries:**
    - Left-bank: Sukri, Mithri, Bandi, Jawai
    - Right-bank: Jojari (only major one)
  - ✓ **Aravalli Divide:**
    - East-flowing rivers: Banas → Chambal → Bay of Bengal
    - West-flowing rivers: Luni → inland **basin**
  - ✓ **Inland Drainage Examples:**
    - Luni: Rajasthan–Gujarat → Rann of Kutch
    - Ghaggar-Hakra: Haryana–Rajasthan → desert sands
    - West Banas/Rupen: Gujarat → Little Rann

Q 132. Ans. C

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Renewable generation: not continuously increasing
  - ✓ Reason: weather dependence
  - ✓ Example: FY 2023–24: weak monsoon (El Niño)
  - ✓ Impact: large hydro generation → (~17%)
- **Statement 2 is correct:**
  - ✓ Renewable additions are dominated by solar
  - ✓ Share: >80% of new capacity (2020–2026)
  - ✓ FY 2025–26: ~43 GW added: ~35 GW solar
- **Statement 3 is correct:**
  - ✓ FY 2025–26: renewable share in generation: 25.86%
  - ✓ Includes: large hydro
  - ✓ Trend: rising RE share, declining thermal share
- **Key Facts**
  - ✓ National Target:
    - Policy: Paris Agreement (NDC)
    - Target: 500 GW non-fossil capacity by 2030
    - Current (2026): ~271 GW achieved
  - ✓ **Thermal vs Renewable:**
    - RE capacity share: >50%
    - RE generation share: ~25.8%
    - Thermal generation: ~70%
    - Reason: low PLF of solar/wind (15–25%)
- **Bihar Linkage**
  - ✓ **Solar Strategy:**
    - Floating solar plants: Darbhanga, Supaul
    - Advantage: no land requirement
  - ✓ **Bio-energy Focus:**
    - Feedstock: agricultural residue
    - Shift: 1G → 2G, 3G, 4G biofuels

Q 133. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Inter-Tropical Convergence Zone: equatorial low-pressure belt
  - ✓ Cause: intense heating → strong convection
  - ✓ Clouds: cumulonimbus
  - ✓ Weather: heavy rain, thunderstorms

- **Statement 2 is correct:**
  - ✓ ITCZ: convergence zone of trade winds
  - ✓ Winds: Northeast Trades + Southeast Trades
  - ✓ Result: rising air, low pressure
- **Statement 3 is incorrect:**
  - ✓ ITCZ shift: follows sun
  - ✓ NH summer: shifts north
  - ✓ NH winter: shifts south
- **Key Facts**
  - ✓ **Mechanism:**
    - Rising branch of Hadley Cell
    - Air rises at ITCZ → moves poleward → sinks at subtropics → returns as trade winds
  - ✓ **Thermal vs Geographic Equator:**
    - Geographic: fixed (0°)
    - Thermal (ITCZ): shifts seasonally
    - Over land: large shift (up to ~25°N over Asia)
    - Over ocean: limited shift

Q 134. Ans. C

Explanation:

- **Tattapani:**
  - ✓ Location: Chhattisgarh
  - ✓ Geology: SONATA (Son-Narmada-Tapi) lineament
  - ✓ Significance: major geothermal field
- **Tapovan:**
  - ✓ Location: Uttarakhand
  - ✓ River: Dhauliganga
  - ✓ Region: Uttarakhand Himalayas
- **Manikaran:**
  - ✓ Location: Parvati Valley, Kullu
  - ✓ Region: Himachal Pradesh
  - ✓ Feature: geothermal hot springs
- **Key Facts:**
  - ✓ Geothermal Energy:
    - Type: baseload renewable energy
    - Feature: continuous power supply
  - ✓ **Major Geothermal Sites:**
    - Puga Valley: highest potential (high enthalpy)
    - Chumathang: Himalayan province
    - Unai, Tuwa: Cambay Graben
    - Bakreshwar: Singhbhum shear zone

Q 135. Ans. C

Explanation:

- **Andhi: pre-monsoon convective dust storm**
  - ✓ Region: Northwest India
  - ✓ Areas: Punjab, Haryana, Rajasthan, Western Uttar Pradesh
  - ✓ Season: May–June
  - ✓ **Characteristics:**
    - Cause: intense surface heating → convection
    - Nature: sudden, violent squall
    - Feature: dust walls, low visibility
    - Effect: temporary temperature drop
    - Sometimes followed by light rain
- **Additional Information**
  - ✓ Mahawat:
    - Region: NW India
    - Season: winter
    - Cause: Western Disturbances
    - Role: beneficial for Rabi crops
- **Key Facts**
  - ✓ **Mechanism:**
    - Extreme heating → steep lapse rate
    - Rapid updraft → lifts loose dry soil
    - Dust carried into lower troposphere
  - ✓ **Global Analogy:**
    - Andhi ≈ Haboob (desert dust storms)
    - Pre-Monsoon Matrix:
      - ◊ Loo: hot dry wind
      - ◊ Andhi: dust storm

Q 136. Ans. B

Explanation:

- **Location: Reasi district Jammu and Kashmir**
  - ✓ Area: Salal–Haimana
  - ✓ Resource: Lithium
  - ✓ Quantity: 5.9 million tonnes
- **Significance:**
  - ✓ First major lithium deposit in India
  - ✓ Boost to green energy self-reliance
- **Additional Information**
  - ✓ Sukinda Valley: Chromite (>90% reserves)
  - ✓ Khetri Belt: Copper
  - ✓ Singhbhum region: Iron ore, copper, uranium
- **Key Facts**
  - ✓ **Lithium Importance:**
    - Term: “White Gold”

- Use: Li-ion batteries
- Applications: Electric Vehicles, energy storage

✓ **Global Dependence:**
- Imports from: Chile, Argentina, Bolivia, China

✓ **Policy Link:**
- Mission: National Mission on Transformative Mobility and Battery Storage

- **Bihar Linkage**
  - ✓ Exploration Areas:
    - Gaya, Nawada: lithium, REE potential
    - Region: Bihar Mica Belt

Q 137. Ans. A

Explanation:

- **Assertion (A) is correct:**
  - ✓ Traditionally, petroleum refineries were located at:
  - ✓ Field-based locations near oil wells (example: Digboi, Assam).
  - ✓ Port-based locations to process imported crude (examples: Mumbai, Kochi, Visakhapatnam).
  - ✓ Later, India developed market-based (inland) refineries such as:
    - Mathura (Uttar Pradesh)
    - Panipat (Haryana)
    - Barauni (Bihar)
  - ✓ These refineries are located deep inside consumption zones, far from both oilfields and ports.
- **Reason (R) is correct:**
  - ✓ Transporting crude oil through rail or road is:
    - Expensive
    - Risk-prone
    - Logistically inefficient at large scale.
  - ✓ Subterranean pipelines solved this constraint by providing:
    - Continuous flow transport
    - Lower operational cost after installation
    - Greater safety and efficiency.
- **Why R explains A**
  - ✓ Petroleum refining involves minimal weight loss between crude and finished products.
  - ✓ It is more efficient to:
    - Transport one bulk raw material (crude) via pipeline to interior markets.
    - Then distribute multiple refined products locally.
  - ✓ Therefore, pipeline technology directly enabled inland refinery locations, explaining the shift described in Assertion (A).

Additional information:

- Mapping pipelines to major inland refineries
- Mathura and Panipat refineries
  - ✓ Supplied by Salaya–Koyali–Mathura (SKM) pipeline.
  - ✓ Crude received from Salaya terminal (Gujarat).
- Barauni refinery
  - ✓ Initially linked with Naharkatiya–Nunmati–Barauni pipeline (Assam crude).
  - ✓ Now also receives imported crude via Haldia–Barauni pipeline.
- Bina refinery (Madhya Pradesh)
  - ✓ Receives crude from Vadinar terminal (Gujarat).
- Barauni refinery:
  - ✓ Located in Begusarai district.
  - ✓ Established with Soviet (USSR) collaboration.
  - ✓ Commissioned in 1964.
  - ✓ First major inland market-based refinery in India.
  - ✓ Built to serve the middle and lower Ganga plains market.
- Major pipeline corridors to remember
- HBJ (Hazira–Vijaipur–Jagdishpur) pipeline
  - ✓ India's first major cross-country gas pipeline.
  - ✓ Supplies fertiliser and power plants across Madhya Pradesh, Rajasthan, and Uttar Pradesh.
- Jamnagar–Loni LPG pipeline
  - ✓ One of the longest LPG pipelines in the world.
  - ✓ Transports LPG from the western coast to the northern plains.

Q 138. Ans. D

Explanation:

- **Statement 1 is correct:**
  - ✓ Siddi community: origin from Bantu people of Southeast Africa
  - ✓ Migration: 7th–19th century

- ✓ Roles: slaves, sailors, mercenaries
- **Statement 2 is correct:**
  - ✓ Major concentration: Gujarat
  - ✓ Region: around Gir Forest
  - ✓ Also present: Karnataka, Maharashtra, Andhra Pradesh
- **Key Supporting Facts**
  - ✓ **Origin:**
    - Region: Southeast Africa (Tanzania, Kenya, Mozambique)
    - Ethnicity: Bantu
  - ✓ **Status:**
    - Recognised as a Scheduled Tribe (ST) in Gujarat, Karnataka, and Andhra Pradesh
  - ✓ **Cultural Aspect:**
    - Dance: Sidi Dhamal
    - Features: drumming, Sufi influence
  - ✓ **Regional Distribution:**
    - Gujarat: Gir region
    - Karnataka: Uttara Kannada (Yellapur, Haliyal)

Q 139. Ans. A

Explanation:

- **Statement 1 is correct:**
  - ✓ Bhitarkanika Mangroves: located in Brahmani–Baitarani delta
  - ✓ Status: Ramsar site
  - ✓ Fauna: largest habitat of Saltwater Crocodile
- **Statement 2 is correct:**
  - ✓ Pichavaram Mangroves: located in Vellar–Coleroon estuary
  - ✓ Rank: second-largest mangrove forest in India
  - ✓ Feature: network of islands and backwaters
- **Statement 3 is incorrect:**
  - ✓ Mangroves: not restricted to East Coast
  - ✓ Also present on West Coast
- **Key Facts**
  - ✓ **Distribution:**
    - East Coast: Sundarbans, Bhitarkanika, Godavari–Krishna delta
    - West Coast: Gujarat, Maharashtra, Goa, Karnataka
  - ✓ **Gujarat:**
    - Second-largest mangrove cover in India
    - Regions: Gulf of Kutch, Gulf of Khambhat
  - ✓ **Types of Mangroves:**
    - Deltaic: large river deltas (East Coast)
    - Estuarine: estuaries, creeks (West Coast)
  - ✓ **Global Fact:**
    - India: ~3% of world mangroves
    - Sundarbans: largest mangrove forest

Q 140. Ans. B

Explanation:

- **Assertion (A) is true:**
  - ✓ The Dynamic concept of monsoon was proposed by Hermann Flohn (1951).
  - ✓ It states that the monsoon results from the seasonal shift of planetary wind belts and the ITCZ, not merely land–sea heating.
  - ✓ In summer, the ITCZ shifts north, allowing cross-equatorial winds to enter India as the Southwest Monsoon.
- **Reason (R) is true:**
  - ✓ During winter, Northeast Trade Winds blow over India.
  - ✓ These winds originate from continental interiors, making them dry, so most parts of India receive little rainfall in winter.
- **Why R does not explain A**
  - ✓ Assertion (A) deals with the theoretical basis of the Dynamic concept, specifically the migration of ITCZ and wind belts.
  - ✓ Reason (R) only describes a winter wind characteristic, not the mechanism that defines Flohn's Dynamic concept.
  - ✓ Therefore, both are correct but not causally linked.

Additional facts:

- Key monsoon theories:
  - ✓ Thermal theory
    - Edmund Halley (1686)
    - Monsoon caused by differential heating of land and sea
  - ✓ Dynamic concept
    - Hermann Flohn (1951)
    - Monsoon caused by shift of ITCZ and planetary winds
  - ✓ Jet stream theory
    - P. Koteswaram
    - Role of Sub-Tropical Westerly Jet and Tropical Easterly Jet

Q 141. Ans. A

Explanation:

- **Assertion (A) is true:**
  - ✓ Regur (Black soil) has high clay content and excellent moisture-retention capacity.
  - ✓ It develops wide cracks during dry periods, giving it the well-known "self-ploughing" character.
- **Reason (R) is true:**
  - ✓ The dominant clay mineral in black soil is montmorillonite.
  - ✓ This mineral swells when wet and shrinks when dry, producing deep cracks.
- **Why R explains A:**
  - ✓ Shrink–swell behaviour of montmorillonite forms deep cracks in dry seasons.
  - ✓ Soil from upper layers falls into cracks and mixes naturally.
  - ✓ This repeated swelling and shrinking naturally churns the soil, creating the self-ploughing property described in A.

Additional facts:

- Parent rock: Basalt (Deccan Traps origin)
- Main states: Maharashtra, Gujarat, Madhya Pradesh, Karnataka, Andhra Pradesh, Tamil Nadu
- Key clay mineral: Montmorillonite
- Major crop: Cotton (hence called Black cotton soil)
- Chemical composition:
  - ✓ Rich in:
    - Lime, iron, magnesia, alumina, potash
  - ✓ Deficient in:
    - Nitrogen, phosphorus, humus
- Key physical properties
  - ✓ Very fine and clayey texture
  - ✓ Sticky when wet
  - ✓ Cracks deeply when dry
  - ✓ Excellent moisture retention, suitable for rain-fed agriculture

Q 142. Ans. C

Explanation:

- **Cotton: tropical and sub-tropical Kharif crop**
  - ✓ **Key Requirement:**
    - Frost-free period: ≥ 210 days
    - Frost effect: fatal during flowering and boll formation
  - ✓ **Climate Needs:**
    - Temperature: 21°C–30°C
    - Rainfall: 50–100 cm
    - Sunshine: bright, clear during harvesting
  - ✓ **Harvest Condition:**
    - Dry weather essential
    - Cloud/rain: causes boll rot, fiber staining
  - ✓ **Soil Requirement:**
    - Best: Black soil (Regur)
    - Also: well-drained alluvial soil
  - ✓ Additional Information
    - Sugarcane:
      - ◊ Requirement: high humidity, long duration (10–12 months)
      - ◊ Not defined by frost-free condition
    - Jute:
      - ◊ Requirement: high temperature (25–35°C), rainfall >150 cm
      - ◊ Region: humid, flood-prone areas
    - Tea:
      - ◊ Requirement: cool, humid climate
      - ◊ Terrain: sloping land
      - ◊ Region: high rainfall areas

Q 143. Ans. B

Explanation:

- **Blossom Showers: pre-monsoon thunderstorms**
  - ✓ Season: March–April
  - ✓ Region: Karnataka, Kerala
  - ✓ Key Region:
    - Kodagu (Coorg), Chikmagalur
    - Mechanism:
    - Pre-monsoon convection: thermal heating
    - Sudden rainfall: triggers flowering
    - Agricultural Role:
    - Crop: Coffee (Coffea arabica, Coffea robusta)
    - Effect: synchronized blossoming of buds
    - Without showers: reduced yield
- **Additional Information**
  - ✓ **Chheerha:**
    - Region: Assam (Brahmaputra valley)
    - Nature: pre-monsoon squalls
    - Key Supporting Facts
    - Pre-Monsoon Season:
    - Period: March–May

- Cause: thermal low over land
- Nature: localized, short-duration, high-intensity storms

Q 144. Ans. B

Explanation:

- **Deendayal Port: first major port after independence**
  - ✓ Commissioned: 1950s
  - ✓ Purpose: replace loss of Karachi Port after Partition
  - ✓ Hinterland: Punjab, Haryana, Rajasthan, Delhi
  - ✓ Type: tidal port
  - ✓ Location: Gulf of Kachchh
  - ✓ Tidal Port Concept:
    - Depth depends on tides
    - Ships enter/exit during high tide
- **Additional Information:**
  - ✓ Jawaharlal Nehru Port:
    - Type: largest container port
    - Year: 1989
    - Purpose: decongest Mumbai Port
  - ✓ Visakhapatnam Port:
    - Type: natural harbor
    - Status: deepest landlocked port
    - Operational: 1933
  - ✓ Mormugao Port:
    - Type: iron ore export hub
    - Status: major port after 1961
- **Key Facts:**
  - ✓ Administrative Change:
    - Year: 2017
    - Renamed from Kandla Port Trust
    - Named after: Deendayal Upadhyaya
- Industrial Significance:
  - ✓ First Export Processing Zone (EPZ): 1965
  - ✓ Later evolved into SEZ model

Q 145. Ans. D

Explanation:

- **Statement 1 is correct:**
  - ✓ Nilgiri Biosphere Reserve: first biosphere reserve of India
  - ✓ Year: 1986
  - ✓ Programme: Man and Biosphere (MAB)
- **Statement 2 is correct:**
  - ✓ Location: Western Ghats
  - ✓ States: Tamil Nadu, Kerala, Karnataka
  - ✓ Tri-state ecological corridor: high biodiversity connectivity
- **Statement 3 is correct:**
  - ✓ Constituent Protected Areas:
    - Karnataka:
      - ◊ Bandipur National Park
      - ◊ Nagarhole (Rajiv Gandhi) National Park
    - Kerala:
      - ◊ Silent Valley National Park
      - ◊ Wayanad Wildlife Sanctuary
    - Tamil Nadu:
      - ◊ Mudumalai Wildlife Sanctuary
      - ◊ Mukurthi National Park
- **Key Facts:**
  - ✓ UNESCO Status:
    - Year: 2000
    - Network: World Network of Biosphere Reserves
    - Also part of: Western Ghats World Heritage Site
  - ✓ Key Fauna:
    - Asian Elephant: largest population
    - Nilgiri Tahr: endemic
    - Lion-tailed Macaque: endemic
  - ✓ Vegetation Gradient:
    - Eastern side: dry deciduous forests
    - Western side: tropical evergreen forests
    - Cause: rainfall gradient

Q 146. Ans. A

Explanation:

- **Assertion (A) is true:**
  - ✓ The Greater Himalayas lie roughly between 28°N to 35°N, within the subtropical latitude belt.
  - ✓ However, despite being in these latitudes, the region shows cold, alpine and tundra-like vegetation.
  - ✓ This occurs because very high altitude reduces temperature, leading to snow-covered peaks and alpine vegetation.
- **Reason (R) is true:**
  - ✓ The normal lapse rate states that temperature decreases by about 6.5°C per 1000 m of altitude.
  - ✓ As elevation increases, air becomes cooler, even if latitude suggests warm conditions.
- **Why R explains A:**

- ✓ The extreme height of the Himalayas causes temperature to fall drastically with altitude.
- ✓ This cooling produces cold climatic conditions and alpine vegetation, despite the subtropical latitude.
- ✓ Therefore, altitude overrides latitude, which is exactly what R explains.

Additional facts:

- Normal lapse rate:
  - ✓ About 6.5°C decrease per 1000 m rise in altitude.
- Key concept:
  - ✓ Altitude controls temperature locally, often more strongly than latitude.
- Vertical zonation in the Himalayas (very important)
  - ✓ Foothills → Tropical forests
  - ✓ Middle heights → Temperate forests
  - ✓ Higher levels → Coniferous forests
  - ✓ Above tree line → Alpine grasslands
  - ✓ Above snow line → Permanent snow
- Important altitude limits
  - ✓ Tree line: ~3600 to 3800 m
  - ✓ Snow line (Himalayas): ~4500 m (south slopes), higher on northern slopes

Q 147. Ans. B

Explanation:

- **Statement 1 is incorrect:**
  - ✓ Kari soils: not alkaline
  - ✓ Nature: highly acidic
  - ✓ pH: often < 4.0
  - ✓ Cause: oxidation of iron pyrites → sulfuric acid formation
- **Statement 2 is correct:**
  - ✓ Kari soils: subtype of peaty (bog) soils
  - ✓ Region: Kerala backwaters
  - ✓ Characteristics: high organic matter, black, heavy, clayey
- **Key Facts:**
  - ✓ **Soil Chemistry:**
    - Type: Acid-sulfate soils
    - Salinity present: due to seawater
    - Dominant issue: acidity, not alkalinity
    - Management: liming (calcium carbonate)
  - ✓ **Distribution of Peaty Soils:**
    - Kerala: Kari soils (Kottayam, Alappuzha)
    - Odisha, Tamil Nadu: coastal tracts
    - West Bengal: Sundarbans
    - Bihar: Tarai and marshy depressions
  - ✓ **Physical Properties:**
    - Color: black
    - Texture: heavy
    - Organic matter: up to 40–50%

Q 148. Ans. A

Explanation:

- **Sivasamudram Waterfalls → Cauvery River**
  - ✓ Location: Mandya district, Karnataka
  - ✓ River splits: forms an island
  - ✓ Branches: Gagana Chukki (west), Bhara Chukki (east)
- **Additional Information:**
  - ✓ Krishna River: no major waterfall on the main stem
  - ✓ Godavari River: waterfalls mainly on tributaries
  - ✓ Tungabhadra River: known for the dam, not Sivasamudram
- **Key Facts:**
  - ✓ **Hydropower:**
    - Year: 1902
    - Significance: first major hydroelectric project in Asia
    - Purpose: power supply to Kolar Gold Fields
  - ✓ **Cauvery River Islands:**
    - Srirangapatna
    - Sivasamudram
    - Srirangam
  - ✓ **Major Waterfalls Mapping:**
    - Sivasamudram: Cauvery: Karnataka
    - Jog: Sharavathi: Karnataka
    - Dhuandhar: Narmada: Madhya Pradesh
    - Dudhsagar: Mandovi: Goa/Karnataka
    - Hundru: Subarnarekha: Jharkhand
- **Bihar Linkage:**
  - ✓ **Major Waterfalls:**
    - Kakolat Waterfall
    - Karkatgarh Waterfall
    - Telhar Kund
    - Manjhar Kund
    - Dhua Kund

Q 149. Ans. C

Explanation:

- **Statement 1 is correct:**
  - ✓ Dryland farming: rainfall < 75 cm
  - ✓ Condition: chronic moisture stress, frequent droughts
- **Statement 2 is correct:**
  - ✓ Crops: drought-resistant
  - ✓ Examples: ragi, bajra, moong, guar, gram
  - ✓ Traits: short-duration, deep-rooted, low water requirement
- **Additional Information:**
  - ✓ **Concept Classification:**
    - ▪ Rainfed farming: based on soil moisture
    - ▪ Dryland: < 75 cm rainfall
    - ▪ Rainfed (wetland): > 115 cm: adequate moisture: rice, jute, sugarcane
  - ✓ **Geographical Spread:**
    - ▪ Regions: Rajasthan, Gujarat
    - ▪ Rain-shadow zones of Western Ghats
    - ▪ Areas: Marathwada, North Karnataka, Rayalaseema
- **Bihar Linkage:**
  - ✓ Districts: Aurangabad, Gaya, Nawada, Jamui
  - ✓ Cropping Shift:
    - ▪ Millets: ragi, jowar
    - ▪ Pulses: gram, lentils

Q 150. Ans. C

Explanation:

- **Assertion (A) is true:**
  - ✓ Kerala has consistently recorded the highest sex ratio among major Indian states.
  - ✓ Census 2011: 1084 females per 1000 males.
  - ✓ This reflects better social development indicators, especially for women.
- **Reason (R) is false:**
  - ✓ The statement claims large-scale migration of women to Gulf countries explains the high sex ratio.
  - ✓ This is incorrect logic.
  - ✓ If women migrated in large numbers, the female population would decrease, lowering the sex ratio.

Actual reasons behind Kerala's high sex ratio

- Male out-migration
  - ✓ Large-scale migration of men to Gulf countries increases the proportion of women in the resident population.
- High female literacy
  - ✓ Kerala has among the highest female literacy rates in India.
- Better healthcare
  - ✓ Low maternal mortality and improved healthcare access support female survival.
- Social development
  - ✓ Some communities historically followed matrilineal traditions, improving women's social status.